Cheap to Keep: Insurance Pricing and the Retirement of Hazardous Capital

Kaleb K. Javier*

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Abstract

In the midst of the clean energy transition, efficient coordination of hazardous capital retirement has become increasingly important. Much of the policy response operates on the adoption margin, and research on retiring capital already in place has focused largely on subsidies and mandates. This paper studies a different decommissioning instrument: pricing the cost of retaining hazardous capital through mandatory environmental insurance. I examine this question at U.S. gas stations, whose buried fuel tanks are the most commonly reported source of groundwater contamination in the country, and where federal law requires operators to carry pollution coverage but states differ in whether premiums are uniformly priced or risk-rated. I identify the effects of risk-based pricing using Texas's 1998 transition from a uniformly priced state insurance fund to risk-rated private insurance, which repriced 21,658 facilities. Using a panel of 7,862,172 tank-years and a difference-in-differences design that compares Texas tanks with tanks of the same construction and vintage in 21 control states, I find that risk-rated pricing increased annual tank closures by 1.3 percentage points, a 47 percent increase. The effect rises with tank age, from 0.5 points for the youngest quarter of tanks to 4.1 points for tanks 21 years or older at the reform. Small facilities disproportionately exited, while larger facilities replaced tanks and remained in operation. I then estimate a dynamic model of facility portfolio choice and entry to compare risk-based pricing with replacement subsidies and age-based retirement mandates. Among these instruments, risk-rated pricing comes closest to a first-best premium that prices both cleanup costs and health damages. Pricing mandatory environmental insurance is therefore a decommissioning policy, and one that outperforms the alternatives considered here.

*University of California, Berkeley, Department of Agricultural and Resource Economics. Email: kalebkja@berkeley.edu. Website: kalebkjavier.com. I thank Meredith Fowlie, Billy Pizer, James Sallee, Nano Barahona, Ben Handel, Matt Backus, and Cailin Slattery for their invaluable comments and feedback. I also thank the participants and discussants at Camp Resources XXXI and the AERE Summer Conference 2025 for helpful discussions. All remaining errors are my own.

1 Introduction

In the midst of the clean energy transition, efficient coordination of hazardous capital retirement has become increasingly important. Much of the economic research has focused on the adoption margin for the clean energy transition: subsidies for cleaner technologies, standards on new sources, and incentives for firms to replace older equipment with newer capital (e.g., Anderson and Sallee 2011; De Groote and Verboven 2019). Much less attention has been given to the other side of the transition, getting hazardous capital that is already in place out. Policies designed to accelerate adoption of new technologies do not necessarily solve this exit problem: new capital can be subsidized while old capital remains grandfathered and continues operating. Research on capital retirement has consequently focused largely on subsidies that encourage owners to scrap old assets and mandates that require them to remove them (e.g., Jacobsen and van Benthem 2015; Barahona, Gallego, and Montero 2020; Jacobsen et al. 2023). This paper studies a different decommissioning instrument: risk-based pricing of mandatory pollution insurance.

The private value of continuing to operate an asset need not reflect the full social cost of keeping it in service. As capital ages, failure risk and expected environmental damages can rise even while the asset continues to generate revenue. An aging pipeline, fuel tank, oil well, or chemical storage system may therefore remain privately profitable after the point at which society would prefer it to be retired. Two failures separate the private value from the social cost. Third parties bear part of the damage from a release, and liability does not restore the owner's incentive when the damage exceeds the assets available to satisfy a claim (Shavell 1986). Mandatory insurance is the standard remedy, and its effect on retirement depends on how the coverage is priced. Because coverage is mandatory, a higher price cannot be avoided by leaving the insurance market, and the cost of continued operation can rise with the observable risk of the asset. This mechanism is not automatically efficient. The price must respond to the dimensions of risk that matter for expected harm, firms must change their decisions in response, and the price must not be so high that it retires capital whose social value still exceeds its risk. I therefore ask three questions. By how much does pricing risk change the retirement of hazardous capital? Does the response concentrate in the riskiest capital? And how does priced risk compare in welfare with replacement subsidies and retirement mandates?

I answer these questions at U.S. gas stations, whose underground fuel tanks are the most commonly reported source of groundwater contamination in the United States. The EPA counts about 583,000 confirmed releases from underground tanks since 1984, more than the 534,000 tanks now in service (EPA 2026a). Federal law requires every station to carry $1 million of mandatory pollution insurance, but states differ in how they price that coverage. Most states historically charged tanks a uniform fee through a state insurance fund, while others relied on private insurers whose premiums vary with observable measures of tank risk, including age and construction. Texas provides a sharp change in those incentives. In 1998, the state closed its uniformly priced insurance fund to new claims and moved facilities into risk-rated private insurance, repricing 21,658 facilities. The timing of the transition was legislated in advance rather than chosen in response to facility behavior.

I identify the effect of risk-based pricing using a difference-in-differences design that compares retirement behavior in Texas before and after the reform with facilities in 21 states that retained uniform pricing. The comparison is made within cells defined by tank type and vintage and across the same calendar years, and the identifying variation therefore comes from the change in insurance pricing rather than differences in the age or construction of the tanks being compared. I build the data from the universe of 361,817 registered tank facilities in 22 states, premium schedules collected by hand from insurer rate filings, and 12,168 claims obtained from 6 state funds. The estimates use a panel of 7,862,172 tank-years from 1990 to 2015. I then estimate a dynamic model of facility portfolio choice, in which owners maintain, downsize, replace, or exit and new facilities enter, to compare risk-based pricing with other retirement policies.

Risk-based pricing substantially increases retirement. The annual tank closure rate rises by 1.3 percentage points relative to a control mean of 2.71 percent, a 47 percent increase. The timing of the response is consistent with the reform: pre-reform coefficients are small, the series breaks at the transition, and the elevated retirement rate persists through the seventeen post-reform years of the sample. A change in the price of mandatory coverage therefore moves the retirement margin. The reform itself changed no technical standard a tank must meet. The response is concentrated in the capital facing the largest increase in the price of continued operation. The effect rises with the age of the tank at the reform, from 0.5 percentage points for the youngest quarter of tanks to 4.1 percentage points for tanks 21 years or older. A level shift in the cost of operating a station would tend to affect tanks more uniformly. The response instead follows the risk gradient embedded in the insurance price. A retirement policy is valuable when it causes the assets with the highest social cost of continued operation to leave. The reform moves the oldest tanks most, and those are the tanks the filed schedules charge most. The reform also changes how facilities adjust their portfolios. Facilities in the smallest size bin increase their exit rate by 5.5 percentage points, while the largest facilities are more likely to replace tanks and continue operating. A low-value facility facing a large increase in the cost of retaining hazardous capital can exit altogether, while a more productive facility can replace the tank and continue operating. The reform thus removes marginal capital without requiring every owner to retire at the same age. Releases fall as well. Confirmed release discoveries at the facility fall by 1.2 percentage points a year (s.e. 0.3).

The reduced-form evidence establishes that risk-based pricing changes retirement, but it does not by itself determine whether the resulting allocation is efficient or how the instrument compares with alternative ways of inducing exit. I therefore use the estimated dynamic model to evaluate three decommissioning instruments: risk-based pricing, replacement subsidies, and age-based retirement mandates. The model combines the observed price schedules with facility revenue, expected leak risk, and adjustment costs to characterize how operators choose whether to maintain, downsize, replace, or exit their tanks, and how many new facilities open. I compare each instrument with a first-best premium that prices both the cleanup cost and the health damage of a leak, counting incumbent facilities and entrants. Risk-rated pricing comes closest, with net welfare 57.8 percent below the first best. Retaining the uniform fee falls 62.2 percent below it, a replacement subsidy 66.7 percent, and a mandate requiring single-walled tanks to be removed at age 20 70.5 percent.

The paper makes four contributions. First, it provides facility-level evidence on how repricing environmental liability changes capital retirement, adding to research on liability and hazardous capital (Shavell 1986; Ringleb and Wiggins 1990; Yin, Kunreuther, and White 2007; Yin, Kunreuther, and White 2011; Yin, Pfaff, and Kunreuther 2011; Muehlenbachs 2015; Boomhower 2019; Marcus 2021; Boomhower et al. 2024). I observe a jurisdiction move its existing capital stock from a uniform fee to risk-rated pricing, and I measure which capital leaves. Second, it extends research on cost-based pricing in insurance markets (Einav, Finkelstein, and Cullen 2010; Bundorf, Levin, and Mahoney 2012; Hendren 2013; Einav, Finkelstein, and Mahoney 2021). That research studies how prices change who holds coverage. When coverage is compulsory, a cost-based premium changes which capital stays in service and acts as an imperfect corrective price. It charges owners for the expected insured loss and not for health damages. Both come from the same leaks, and a price on the first therefore reduces the second.

Third, the paper compares priced risk with scrappage subsidies and age mandates on a common welfare footing. Existing research on retiring durable capital has emphasized scrappage subsidies, retirement mandates, and vintage-differentiated regulation (Gruenspecht 1982; Adda and Cooper 2000; Stavins 2006; Mian and Sufi 2012; Jacobsen and van Benthem 2015; Barahona, Gallego, and Montero 2020; R. J. Davis, Holladay, and Sims 2022; Jacobsen et al. 2023; Deshmukh et al. 2023; L. Davis and Weber 2025). Fourth, it adds to research that uses dynamic models to study capital replacement and environmental regulation (Rust 1987; Ryan 2012; Kellogg 2014; Fowlie, Reguant, and Ryan 2016). Much of that work emphasizes sunk costs of entry and investment. I evaluate capital replacement under a liability cost that recurs in every year an asset stays in service.

The paper proceeds as follows. Section 2 describes the tanks and the rules that govern them, the data, leak risk and its cost, the market for coverage, and the Texas reform. Section 3 sets out the research design, and Section 4 reports the retirement response. Section 5 develops and estimates the dynamic model, and Section 6 compares the decommissioning policies. Section 7 concludes.

2 Setting and Data

I begin with the tanks and the rules that govern them and then describe the data. The rest of the section shows what the data say about leak risk and its cost, how coverage is priced, and what happened in Texas at the reform.

2.1 Underground storage tanks and their regulation

The risk of a tank is fixed at installation. Whether a tank will leak is largely decided before it ever holds fuel. Every retail gasoline station in the United States stores its fuel underground, and those systems stay in the ground for decades while soil and fuel corrode the piping and walls. An operator buying a new system picks between two constructions. A single-walled tank carries one wall of bare steel, which soil corrodes from outside and fuel from inside. A double-walled tank adds a second shell of corrosion-protected steel or fiberglass around the

first, with a monitored space between them, and a sensor in that space registers a breach of the inner wall before fuel reaches the soil. The construction an operator selects at installation, not the care taken afterward, governs how quickly the system becomes dangerous.

Operators chose the cheaper wall. A double-walled system costs substantially more to install, and lower risk is the only thing it returns. The second wall added roughly seventy percent to the cost of a tank in the late 1980s, about $41,000 in 2023 dollars.[1] A typical station runs three or four tanks. Equipping a whole site with double-walled systems therefore cost on the order of $120,000 to $160,000 more than the single-walled alternative, and that expenditure bought no additional storage, no additional throughput, and no additional revenue. Operators chose accordingly. Single-walled tanks made up 89.6 percent of the stock in the ground when Texas repriced, and single wall stayed the majority choice even among tanks installed after 1988.

Closure and replacement are the only actions that reduce risk. An operator who installs a single-walled system cannot later convert it into a double-walled one. Changing a wall means excavating the tank and installing a different one, and 0.003 percent of active Texas facilities ever did so before closing the tank. Retrofits in practice reach the piping, the release detection equipment, and the sumps. Each interrupts business and costs real money, and none of them addresses the wall that holds the fuel. That leaves two actions, closure and replacement.

The law grandfathers the stock. The 1988 EPA regulations set the technical standards a tank must meet and gave tanks already in the ground until December 22, 1998 to meet them, upgrade to them, or close. Every facility that met the standard by that date was grandfathered. The minimum has ratcheted upward in the decades since, most consequentially through the 2005 Energy Policy Act's secondary containment requirement and the 2015 revision's operating and maintenance rules, and each revision has continued to grandfather existing systems, on the condition that any retrofit applied to a tank immediately subjects it to the current technical standard. The regulation therefore produces a Gruenspecht effect (Gruenspecht 1982; Stavins 2006). It protects old capital and penalizes its improvement. Combined with the physical impossibility of converting a wall, it leaves closure and replacement as the only risk-reducing actions an operator has.

Monitoring catches active leaks, and closure can uncover slow ones. Federal rules require an owner to monitor a tank for releases at least once every thirty days, with limited exceptions, to report a suspected release to the state agency within twenty-four hours, and to investigate it within seven days, unless the agency sets other periods (EPA 2026c). Monitoring does not catch every leak. A monitoring method must detect a leak of 0.2 gallons an hour with a probability of at least 95 percent, and a slower leak falls outside what the standard requires a method to find (EPA 2020). Closure is a point at which such a release can surface. An owner must assess the site for contamination before a closure is permanent, and a tank out of service for more than twelve months must be closed permanently unless it meets current standards or the agency grants an extension. Section 4 separates the releases reported within

[1] In 1987 a ten-thousand-gallon single-walled steel tank cost $21,400 in the ground against $36,700 for the double-walled version, and in fiberglass $23,300 against $39,000 (U.S. General Accounting Office 1992).

sixty days of a closure from the rest for that reason.

An owner out of compliance pays for it in penalties and in coverage. Since the Energy Policy Act of 2005, federal law requires every tank to be inspected at least once every three years, and fuel may not be delivered to a tank the agency has found ineligible (United States Congress 2005b, 2005a). The statute caps the civil penalty for a violation at $10,000 per tank per day, $29,980 after adjustment for inflation as of 2025 (EPA 2025). Coverage is conditional on compliance as well. In a 2011 EPA review, 22 of 25 sampled insurance policies excluded losses tied to noncompliance (EPA 2011). State funds in Tennessee and Maine raise the owner's deductible when a release goes unreported or leak-detection requirements are not met, and North Carolina's fund pays a smaller share of the cleanup for a tank that did not meet the upgrade standards (Tennessee Department of Environment and Conservation 2021; State of Maine 2026; State of North Carolina 2026).

2.2 Data

The registry records exactly what an underwriter can verify. Every state keeps a registry of the tanks in its ground, and the registry records the traits an underwriter can check before writing a policy: the install date, the closure date, the wall construction, the fuel, and the capacity of each tank, and the identity and location of the facility that holds it. The risk gradient and the cost gradient of Section 2.3 are built on those traits. That is what makes them traits a price can be written on. Four sources supply the records: the state tank and release registries, insurer rate filings, trust fund claims, and the EPA's performance measures.

The UST universe comes from the EPA's UST Finder, which compiles every state registry into one national schema (EPA 2021). Each record describes one tank at one facility. The universe holds 361,817 facilities in 22 states. Harmonization leaves some states missing entire fields, and I therefore rebuild the registry from the state agency's own database where the national file is incomplete. Texas supplies the richest of these registries. Its records name the carrier each facility used to meet its financial responsibility obligation in each year, and that record lets me map a facility's tanks to the rate filing that priced them. Federal rules require every facility to report a suspected release to its state agency. States record the report date, the facility identifier, and a release identifier consistently. They do not record closure or No Further Action dates consistently. I collect those dates where a state supplies them but do not use them in the analysis.

A rate manual states the algorithm and the rules a carrier uses to price coverage in a given state. It fixes a base rate and the factors that multiply it. Carriers file these manuals with state insurance regulators, and the filings enter the public record. The same filings underlie recent work on how insurers price wildfire risk in homeowners coverage (Boomhower et al. 2024). I collected the full filing history of the six largest carriers in the Texas market through the Texas and Iowa filing systems. The filings span 2000 to 2025. I reimplemented each manual as an executable rating engine. Running a facility's observed characteristics through its carrier's engine returns the premium that facility faced that year. Running a hypothetical facility through the same engine returns the premium the market would have charged it. Section A.6 documents the engines and the contract terms I hold fixed, and

Section A.1 reports how much of the Texas market they price.

Public insurance programs hold facility-level claims records that no private carrier discloses. I assembled the first multi-state facility-level claims dataset for this market through records requests to state remediation trust funds, covering 6 states: Colorado, Louisiana, New Mexico, Pennsylvania, Tennessee, and Utah. The records hold 12,168 claims. Five of those states also serve as controls in the closure analysis, and New Mexico does not. Each record carries the facility identifier, the date the facility reported the claim, and the cleanup cost the fund paid on it, harmonized across states by the rule Appendix C states for each fund.

Every state reports to the EPA twice a year its aggregate inspection and compliance measures, among them the percentage of its recently inspected tank facilities found in compliance with the federal technical standards. I assemble these measures by state and year. Section B.10 builds the enforcement comparison from the complement of that percentage.

I observe the data at the facility-year level, building each record by rolling a facility's tanks into a portfolio. The facility is also the unit of treatment, because an operator decides for the site rather than tank by tank. The panel covers 22 states from 1990 through 2021 and holds the universe of registered facilities in those states rather than a sample. The reduced-form estimates use panel years 1990 to 2015. States that never operated a fund priced risk throughout the period and cannot serve as uniform-price controls, and I hold no rate filings for them outside Texas and Iowa.

The sample is the incumbent stock of the fund states with complete registries. Three fields decide whether a state can enter the sample at all: a tank's install date, its closure date, and its wall type. The two dates place the tank in event time. Wall type, together with age, defines the risk class the market prices. State registries do not report these three fields consistently. I therefore begin with the 42 states that created an assurance fund and keep the 22 whose registries report all three, which leaves Texas and 21 control states. I then restrict those registries to incumbents, the facilities holding at least one active tank on December 23, 1998. The reform repriced this stock. Tanks installed later entered a market that was already risk-rated and carry no before-and-after. Table 8 reports the sample remaining after each restriction, and Section A.2 lists the dropped states and the field each one lacks.

The tank-year panel carries an annual closure indicator, equal to one in the year a tank's permanent-removal date falls and zero in every other year the tank is in the ground. The facility-year panel carries exit, replacement, downsizing, and consolidation, each defined from the facility's tank count and capacity against the previous year, and the release records attach to it with a split by the days between a release report and the nearest closure at the same facility. Every covariate in the heterogeneity analysis is measured at the reform and held there, because the reform itself changes local market structure. Section A.3 states each rule exactly.

2.3 Leak risk and its cost

A release is a cost to third parties before it is a cost to the operator. A release from a tank does not stay on the site. Gasoline reaches groundwater, and groundwater reaches

drinking-water wells, neighboring parcels, and surface water. Underground tanks are the most commonly reported source of groundwater contamination in the United States (EPA 2024b). The people who bear the damage are third parties. Marcus (2021) finds that infants born near a leaking tank site are more likely to be born preterm and at low birth weight, and the cleanup that follows a confirmed release is paid by the state fund or the insurer rather than by the household exposed. The operator bears the deductible and the disruption. The risk this section measures is therefore a risk to society first, and the question the paper asks is whether the price the operator pays reflects it.

Holding a single-walled tank raises the release rate as tanks age. I regress an indicator that a facility reports a release in the year on the facility's average-tank-age bin, an indicator that it holds at least one single-walled tank, and their interaction, with state-by-year fixed effects and standard errors clustered by state. Panel A of Figure 1 plots the effect of holding a single-walled tank in each age bin. The effect is positive and its 95 percent confidence interval excludes zero in 5 of the nine age bins, peaking at +0.81 percentage points at ages 18-20. The effect grows with age through the late teens and is indistinguishable from zero in the first years of service. The comparison group holds no single-walled tank, and it includes tanks with no recorded wall type.

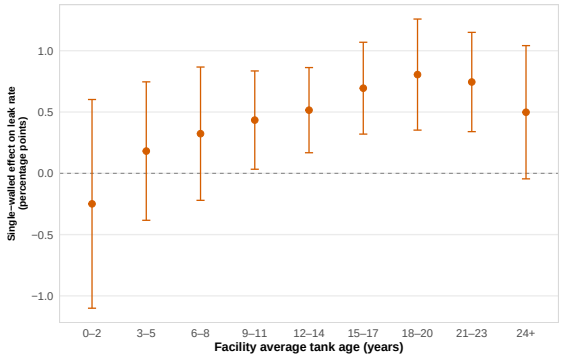
Release risk rises with age and is higher behind a single wall. The observed rate of first confirmed releases, tabulated by the facility's average tank age in three-year bins and by wall construction, shows the pattern directly. Across the sample, 2.37 percent of facility-years record one. The rates show three facts. First, release rates rise with age. Among facilities with no single-walled tank the rate climbs from 1.01 percent at ages three to five to 1.99 percent past twenty-four. Second, single-walled facilities sit above facilities with no single-walled tank at every age. Their rate reaches 2.8 percent at ages eighteen to twenty, and the gap between the constructions is widest at ages 21-23, at 1.15 points. Third, the two ends of the profile are not age effects. Rates are highest in the first years of service, 4.7 percent for single-walled facilities and 4.0 percent for facilities with no single-walled tank, reflecting failures at installation rather than the ageing risk the rest of the paper prices, and the oldest single-walled bin falls as the leakiest tanks exit the sample.

Facilities holding a single-walled tank file claims more often. In the 4 states whose cleanup-fund claims link to the registry, Colorado, Louisiana, Pennsylvania, and Tennessee, a claim arrives in 1.17 percent of the facility-years of facilities holding a single-walled tank and 0.83 percent of those holding none (panel B of Figure 1). A claim costs $353,099 at the mean in the first group and $271,068 in the second, with medians of $90,481 and $77,020, but the state-clustered intervals on the two means overlap. Within state and year the difference in mean claim cost is +61,338 dollars ($p = 0.066$). The comparison group includes tanks with no recorded wall type.

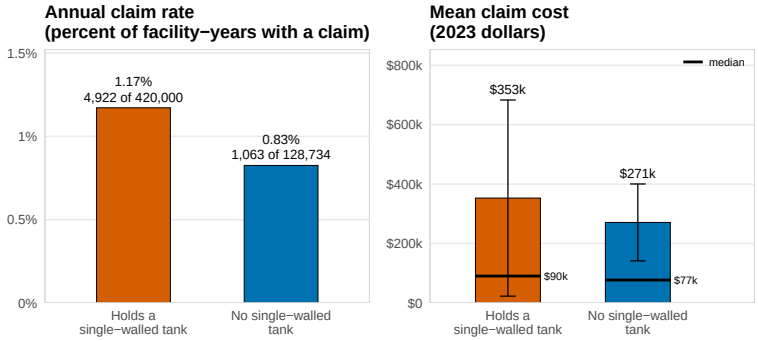
A release costs $106,000 at the median, and the tail is long. The claims data measure what a release costs. 12,168 claims paid by 6 state trust funds put a distribution on cleanup cost, plotted in Figure 17 of Section B.2. The median claim runs $106,000 against a mean of $348,000, and the gap between them reflects a right-skewed distribution in which a minority of releases accounts for most of the spending. The claims enter the analysis once, as the severity behind every expected-cost figure that follows.

Figure 1: Leak risk and claims by tank construction

A. Effect of a single-walled tank on the annual release rate, by tank age



B. Claim rate and claim cost, by whether a facility holds a single-walled tank



Notes: Panel A: the effect of holding at least one single-walled tank on the annual probability that a facility reports a release, by three-year bin of the facility's average tank age, in percentage points. Each point is the single-walled coefficient plus its interaction with the age bin, from a linear probability model with state-by-year fixed effects, estimated on 3,146,158 facility-years in 22 states, Texas from 1990 to 1998 and the control states from 1990 onward. Bars are 95 percent confidence intervals from standard errors clustered by state. Wall type and age are measured in December. Panel B: claims to the state cleanup funds of Colorado, Louisiana, Pennsylvania, and Tennessee, in the years each state's claim records cover, from 1990 onward (Pennsylvania from 1994). Left: the percent of facility-years with at least one claim, with the number of facility-years with a claim and the number of facility-years printed above each bar. Right: the mean cost per claim in 2023 dollars, printed above each bar, with a 95 percent confidence interval from standard errors clustered by state (four clusters) and the median marked by a horizontal tick. In panels A and B the comparison group holds no tank recorded as single-walled and includes tanks with no recorded wall type.

Risk is predictable from the registry. To measure the risk of an individual facility rather than a cell, I estimate the annual probability that facility i in state s records its first confirmed release in year t:

$$\Pr\left(\text{Release}_{ist}=1\right) \;=\; F(\alpha_t+\delta_s+\mathbf{X}'_{ist}\theta) \tag{1}$$

where F is the logistic function, the sample holds facility-years before any confirmed release, and the year and state effects α_t and δ_s absorb common shocks and fixed differences in state reporting and enforcement. $\mathbf{X}_{ist}$ is the registry record. It holds the three-year bins of average tank age used above, an indicator for single-walled tanks, and their interactions, and the interactions let the age profile differ by wall type. It also holds the tank count, total capacity, fuel type, the roster's composition by age, vintage, capacity, and fuel, and pairwise interactions among these traits. I estimate the equation as a lasso logit, a penalized logistic regression whose penalty is chosen by ten-fold cross-validation and whose selection shrinks uninformative terms to zero. The folds are grouped by facility, all of a facility's years are held out together, and every prediction reported in this paper is out of fold. The model predicts releases well out of sample. It reaches an AUC of 0.81, the highest-risk decile of facilities contains 55.6 percent of all releases, and the predicted cell rates recover the observed age-by-wall profile cell by cell (Figure 13, in Section B.1). An insurer or a regulator observing only the registry can therefore assign each facility a risk before coverage is written.

At every age, a single-walled facility carries more expected cleanup cost than a facility with no single-walled tank. Combining the two measures prices the risk. A facility's expected annual cleanup cost is its cell's release rate times the mean cleanup cost of the claims in that cell, the same two objects the structural model prices in Section 5. I compute that cost in each age-by-wall cell, and the levels are measured rather than modelled: each is an observed rate times an observed mean claim. The two cells I quote differ in both age and wall. A facility with no single-walled tank whose tanks are three to five years old carries $3,138 a year of expected cleanup cost. A single-walled facility whose tanks are twenty-four or older carries $7,263, a factor of 2.31. Between those two cells, a single-walled facility carries more expected cleanup cost than a facility of the same age with no single-walled tank in every age bin.

The expected cleanup cost above puts a value on the two actions that reduce risk, closure and replacement. Consider a single-walled facility whose tanks are 29 years old, the median in 2015. Retiring five years early avoids $24,511 of expected cleanup cost in present value, the annual cost above carried over five discounted years. Replacing with a new double-walled unit avoids $23,036 over the new tank's life. Replacement earns less than early retirement for a reason visible in the release rates above: a newly installed tank is not immediately safe, and those first years of the swap are the ones discounting weights most. Both figures are avoided expected cleanup cost rather than profit. Shedding tanks and operating on fewer is the third margin, and it removes risk only to the extent the tanks removed are the old ones.

2.4 The market for pollution coverage

Federal rules also require coverage. An operator whose liability stops at its own assets has little reason to insure against a loss it would never bear (Shavell 1986), and a cleanup averaging $348,000 exceeds the assets of most single-site firms. RCRA Subtitle I has therefore required every underground storage tank owner since 1988 to demonstrate one million dollars

of coverage per occurrence for cleanup and third-party liability. An owner may satisfy the requirement through commercial insurance, a surety bond, a letter of credit, a financial test of self-insurance, or participation in a state assurance fund. Coverage is mandatory everywhere. What a state chooses is how the coverage is priced, and that choice produces two contracts. The technical standards act on new capital and on a schedule keyed to install cohort, and the coverage requirement acts on who pays for a release. Neither touches whether a grandfathered tank stays in the ground for another year. The one instrument that does is the price of the coverage.

A fund charges every tank the same fee. 42 states met the requirement by creating an assurance fund. A fund reimburses cleanup costs, satisfies the federal requirement on its own, and charges an annual fee per tank that does not vary with the tank's age, its wall construction, or its release history. In the year of the Texas reform, the annual per-tank fees reported to the state-fund survey ran from $25 to $500 across the 26 states that levied one, with a median of $100, and the Texas fee of $50 sat near the bottom of that range (ASTSWMO 1999). Funds draw the balance of their obligations from state motor fuel taxes. Because the fund satisfies the requirement at that fee, an operator in a fund state has no reason to buy private coverage, and the four states answering the survey's coverage question report that the fund covers 98 to 100 percent of their facilities (ASTSWMO 1999).

A carrier charges for the traits the registry records. The remaining states leave the requirement to a private market. Carriers there write pollution liability specifically for tank operators and set premiums from the characteristics a state registry records, principally a tank's age and its wall construction. Every state maintains a registry covering the universe of these facilities, and a carrier verifies those characteristics against it before writing the policy. The operator fixes construction at installation, and age is therefore the only characteristic of a tank that changes. A premium rises as a tank ages. A fund fee does not.

With x_j the registry traits of tank j and $\bar{P}$ the fund's fee, the rate on tank j and the premium of facility i are:

$$p_j = \bar{P} \ \text{(fund)}, \qquad p_j = p(x_j) \ \text{(private insurance)}, \qquad P_i = \sum_{j \in i} p_j \tag{2}$$

The slope of the rate in a tank's age is positive under private insurance and zero under a fund.

The filed risk-rated schedule for a single-walled tank sits far above the fee it displaced in Texas. The fund charged $50 per tank at every age. The schedule carriers filed runs from $359 at age 1 to $796 at age 35 in 2006, roughly 7 times the old fee for a new tank and 16 times for the oldest. The schedule itself barely moved over the risk-rated era. Priced in four snapshot years across 14 years, it differs by at most 6.2 percent at any fixed age. The reform changed the schedule a facility faced, not the schedule's own drift over time.

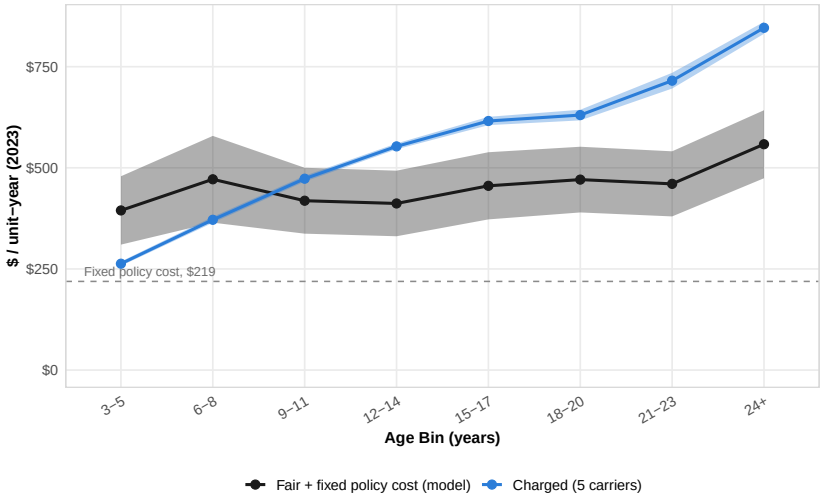
Applying each carrier's filed rating manual to a facility's insured tanks gives the premium it faces each year. That premium rises from $263 for a tank aged three to five to $846 past twenty-four in 2023 dollars, a factor of 3.22, and from $588 to $2,708 at the facility level.

Figure 32 follows that facility premium over the risk-rated era. The mean Texas facility faces several times the fund fee a control facility pays in every year from 2006 to 2020, and the Texas total falls over the period as facilities shed their oldest tanks, a composition change Appendix C documents.

The filed premium was about a fifth of a single store's profit in 2006. The mean Texas facility faced a filed premium of $3,035 in 2006, in 2023 dollars, against a fund fee of $673 for the mean control facility (Table 1). For the profit of a single independent store I use the mean profit per sole proprietorship return in gasoline stations that the Internal Revenue Service reports (IRS 2008). It was $15,551 in 2006, in 2023 dollars. The Texas premium is 19.5 percent of it and the control fee is 4.3 percent. The Census Bureau's retail data imply an operating profit of $248,876 for the average gasoline station in 2006 (U.S. Census Bureau 2024, 2008), and the Texas premium is 1.2 percent of that.

Set against the insurer's expected cost with the filed minimum policy premium added, the charged premium runs 1.42 times expected cost pooled, from 0.67 in the youngest age bins to 1.55 in the oldest, with 15.0 percent of Texas facilities priced exactly at the minimum. Carriers charge the young below cost and the old above it. The rate filings were set without reference to this model or these claims, and Figure 2 plots the comparison.

Figure 2: Filed Texas premiums against the insurer cost benchmark



Notes: Mean charged premium (blue line) and the estimated insurer cost benchmark (black line) per tank-year, by three-year bin of tank age, for Texas tanks in 2015, in 2023 dollars. The figure plots 17,718 tanks. The charged premium applies the filed rating manual of each of five carriers to the insured tanks of the facility. The benchmark is the tank's share of its facility's expected cleanup cost, a release rate times a fitted cleanup cost, plus the carrier-weighted filed minimum policy premium split equally across the facility's tanks. The dashed line marks that minimum at 219 dollars per tank-year. Shaded bands are 95 percent intervals. The benchmark band carries the estimation error of the release-rate and cleanup-cost models, and the charged-premium band carries the sampling error of the mean.

Table 1: Premium as a share of store profit

	Texas	Controls
Facility total, annual premium (2023 $)		
2006	$3,035	$673
	(120.4)	(3.4)
2010	$2,589	$620
	(29.3)	(3.1)
2015	$1,854	$534
	(26.6)	(2.7)
Per tank, annual premium (2023 $)		
2006	$1,035	$234
	(19.5)	(0.9)
2010	$977	$218
	(5.3)	(0.9)
2015	$715	$187
	(4.2)	(0.7)
Single-store benchmark, sole proprietorship returns, 2006		
Revenue per store	$1,756,800	
Net profit per store	$15,551	
2006 Texas premium as share of profit	19.5%	
2015 Texas premium as share of 2006 profit	11.9%	
All-establishment benchmark, Census Bureau data, 2006		
Revenue per station	$5,458,112	
Implied operating profit per station	$248,876	
Gross margin share of revenue	14.3%	
2006 Texas premium as share of operating profit	1.2%	
2015 Texas premium as share of 2006 operating profit	0.7%	

Notes: Annual premium in 2023 dollars, with the standard error of the mean in parentheses. Facility total is the premium of a facility and per tank is the premium of a tank. The Texas figures apply the filed rating manual of each of five carriers to the insured tanks. They are filed premiums. The control figures are the per-tank fund fee in force, and the facility total multiplies it by the facility's active tank count. The per-tank rows cover the matched sample of Texas tanks. In 2006, 2010, and 2015 the Texas facility means are based on 474, 9,696, and 8,125 facilities, the Texas per-tank means on 1,388, 25,615, and 20,973 tanks, and the control means on 81,385, 77,490, and 74,193 facilities. The single-store benchmark is the mean net income, less deficit, per sole proprietorship return in gasoline stations (NAICS 447) for tax year 2006, in 2023 dollars. The all-establishment benchmark is a rough proxy for operating profit per station in 2006, the difference between national gross margin and operating expenses, divided by the number of establishments. The 2015 shares divide the 2015 Texas premium by the 2006 profit, because the profit benchmarks are available for 2006 only. Sources: Internal Revenue Service, Statistics of Income, Sole Proprietorship Returns, Tax Year 2006, Table 1, for the single-store benchmark, and U.S. Census Bureau, Annual Retail Trade Survey: 2022 and County Business Patterns: 2006, for the all-establishment benchmark.

Most funds set one deductible for every facility they cover. The levels run from $500 in Kentucky, Maine, and Virginia to $55,000 in Ohio. The Texas private contract carries a $5,000 deductible on one million dollars of coverage per occurrence, the federal reference

terms every carrier files against.

The fee ignores a gap the schedule prices. A fund charges an old single-walled facility and a young facility with no single-walled tank the same $50 per tank, and the two sit $4,125 a year apart in expected cleanup cost. The uniform fee ignores that difference. The Texas schedule prices it, and every year an old tank stays in the ground under that schedule its owner is charged for the risk the fund had absorbed. What a facility does when that charge arrives is the question the rest of the paper answers.

2.5 Texas and control facilities at the reform

Texas and control facilities operate at the same scale. The median facility runs 3 tanks in Texas against 3 in the controls and holds 24 thousand gallons against 20. Both groups carry the same stock profile at the reform: a mean tank age of 15.4 years in Texas against 16.0 in the controls, 0.74 of tanks installed before 1989, and 0.90 single-walled. Table 2 reports the comparison in full for the 127,987 facilities in the estimation sample, 21,440 in Texas and 107,044 in the controls, across 2,776,307 facility-years.

Table 2: Baseline characteristics, estimation sample

	Texas	Control	Norm. diff.
Stations	21,440	107,044	
Tanks per station, mean (standard deviation)	3.42 (2.87)	3.55 (2.75)	−0.05
Tanks per station, median [interquartile range]	3 [2–4]	3 [2–4]	−0.05
Total capacity, median [interquartile range] (thousand gallons)	24 [14–35]	20 [6–31]	−0.01
Mean tank age at the reform (years)	15.4	16.0	−0.05
Single-walled share	0.90	0.90	+0.01
Share installed before 1989	0.78	0.73	+0.13

Notes: Estimation sample: facilities holding at least one tank active at the December 23, 1998 reform, in Texas and the 21 control states. Facility characteristics are measured at the reform date. Cells show means with standard deviations in parentheses, medians with interquartile ranges in brackets, and shares for indicator variables. Total capacity is in thousands of gallons. Mean tank age is the mean across the 406,343 incumbent tanks with a recorded install year, and its normalized difference is computed across tanks. Every other row is computed across facilities. The last column is the Imbens and Rubin normalized difference, the Texas minus control gap over the pooled standard deviation, and 0.25 in absolute value is the conventional balance threshold. Pre-reform outcome rates, over 1990 to 1998, are reported in the text.

The two groups match on scale and composition. Texas tanks run slightly younger at the reform, 15.4 years against 16.0, a normalized difference of -0.05, and cell-by-year effects absorb that gap.

Pre-reform outcome rates give the denominators for every estimate that follows. A control facility closes at least one tank in 3.3 percent of its pre-1998 years, cuts capacity in 2.3

percent, and replaces a tank in 1.5 percent.

Exit does not appear in that list, because the estimation sample records no pre-reform exit at all. That zero is mechanical. The incumbent restriction admits only facilities holding an active tank on the reform date and therefore excludes by construction every facility that had already left. The unrestricted registry carries that behavior. This sample cannot speak to it. Section 3 states what the mechanical zero does to the event study.

2.6 The Texas repricing

The fund failed on its own finances. Texas met the federal mandate through a fund that failed for reasons unrelated to the tanks it covered. The legislature created the Petroleum Storage Tank Remediation Fund in 1989 under the Texas Water Code. The fund reimbursed cleanups, charged a uniform $50 per tank per year, and drew the balance from gasoline delivery fees. Its obligations outgrew that revenue. Actuarial review found the fund structurally unsustainable by 1995, and companion legislation that session authorized transferring up to $120 million from general revenue to keep it paying claims (House Research Organization 1995).

The legislature closed it with three and a half years of notice. The legislature closed the fund rather than repair it. House Bill 2587 barred reimbursement of any release reported after December 22, 1998, making December 23 the first day under private coverage (Texas Legislature 1995). The sunset passed in 1995, which gave operators three and a half years of notice. 21,658 facilities moved from a uniform fee to risk-rated premiums on that date, in a single step.

The federal upgrade deadline and the Texas repricing fall on the same date, because the legislature set the fund's closing date to match it. That deadline applied in all fifty states, and facilities in the comparison states met it while keeping their uniform fee. Its burden also fell unevenly across tanks. Release detection phased in by install cohort between 1989 and 1993, and tanks installed after 1988 owed nothing at all. Marcus (2021) uses the deadline itself as a treatment, to measure the health effects of leak exposure. This paper uses the repricing. Section 3 states how the empirical design controls for the federal deadline.

Two features of that sequence carry the identification. The fund closed because of its own finances. Neither the decision nor its timing responded to the condition of any particular facility's tanks. The fund also never repriced. It stopped writing entirely, and a private market already rating the same product in never-fund states absorbed its policyholders. The facilities in this study met a change in the price of risk that originated outside them. Section 3 builds the comparison that measures their response.

3 Research Design

After December 1998, a Texas tank and a control-state tank of identical wall construction, fuel, capacity, and install year faced different prices for remaining in service. Texas closed the fund that had charged a uniform fee of $50 per tank, and private carriers priced the characteristics recorded in the registry. The same December date coincided with a federal

upgrade deadline that applied in every state and fell hardest on the oldest tanks. The difference-in-differences design therefore compares Texas tanks only with control-state tanks of the same construction and vintage, which separates the insurance price change from the response to the federal deadline.

Constructing the comparison group

The ideal experiment would compare the retirement decision for the same tank under a uniform fee and a risk-rated contract in the same year. Such a comparison is not feasible because no tank is observed under both pricing regimes. I therefore develop an empirical design that approximates this comparison using the detailed tank- and facility-level data. I compare a Texas tank only to control-state tanks that match it on every trait that carriers price: wall construction, fuel grade, capacity class, and install-cohort year. These traits define the cell c. A facility's cell is the cell of its oldest tank crossed with the facility's tank count, in four bins of one, two to three, four to six, and seven or more tanks. A Texas facility therefore enters the comparison against control facilities holding the same oldest asset at the same scale, and a difference in station size cannot be read as a treatment effect. The rate manuals recovered in Section 2.2 determine the cell dimensions because they read wall construction and age from the state registry and set the price from them. Section 2.4 reports the resulting price gradient. The estimation sample contains 1,086 such cells.

Comparison cells defined this narrowly are ordinarily vulnerable to thin overlap. The overlap in this sample is substantial. 983 of the 1,086 cells contain both Texas and control tanks, and those cells hold 99.7 percent of tank-years. Only 8 cells contain Texas tanks with no control counterpart. The facility design retains 93.3 percent of facilities. Section A.5 reports the matched age distributions at the reform date.

The cell-by-year interaction is necessary to control for federal requirements that vary by install cohort and tank construction. Release detection phased in between 1989 and 1993 according to install year, and tanks installed after 1988 owed nothing at all. Every state faced the same requirements, but the resulting response depended on the composition of its tank stock. A facility-and-year design would therefore fold cohort-specific regulatory responses into the treatment term because Texas and the control states hold different distributions of vintages. The cell-by-year effect instead absorbs any trend common to tanks of the same construction and vintage anywhere in the sample, including the response to the federal deadline. This absorption is exact because each federal mandate indicator is a deterministic function of install cohort and year, and the cell-by-year effect partitions the sample more finely than those indicators do.

Estimating equations

One equation carries the design at both levels of observation.

$$Y_{it} \;=\; \beta\,\mathrm{TX}_i \times \mathrm{Post}_t \;+\; \alpha_f \;+\; \delta_{ct} \;+\; \varepsilon_{it} \tag{3}$$

Y_{it} denotes the outcome for unit i in year t, TX_i indicates a Texas unit, Post_t takes the value one from 1999 onward, the first full year following the December 23, 1998 reform, α_f

is a facility fixed effect, and δ_{ct} is the cell-by-year fixed effect defined above. β is the average effect of risk-based pricing on the Texas units repriced by the reform.

The unit of observation varies with the outcome, and no other element of the specification varies with it. Tank-year observations carry tank closure. Facility-year observations carry the portfolio margins, which are any closure, exit, downsizing, consolidation, upward reconfiguration, replacement, and capacity reduction. The two regressions differ only in what indexes c, because a facility's cell is constructed from its oldest tank rather than from the tank being closed.

The event-study specification tests the assumption on which the static equation rests and traces the response over time.

$$Y_{it} \;=\; \sum_{k \neq -1} \beta_k \, \mathbf{1}[k_{it} = k] \times \mathrm{TX}_i \;+\; \alpha_f \;+\; \delta_{ct} \;+\; \varepsilon_{it} \tag{4}$$

I omit $k = -1$, and every coefficient therefore reads against the year preceding the reform. The pre-reform coefficients test whether Texas and its matched controls followed the same closure path. The post-reform coefficients show how long the response lasts. A risk-rated schedule prices age and a uniform fee does not, and the price gap between a Texas tank and its control therefore widens each year as the repriced stock ages. The pricing mechanism predicts a response that persists.

The heterogeneity specification splits the treatment term by the age of the tank at the reform, the trait on which the filed schedules raise the price most steeply.

$$\begin{aligned} Y_{it} \;=\; & \beta \, \mathrm{TX}_i \times \mathrm{Post}_t \;+\; \sum_{q=2}^{4} \gamma_q \, \mathrm{TX}_i \times \mathrm{Post}_t \times \mathbf{1}[q_i = q] \\ & +\; \sum_{q=2}^{4} \lambda_q \, \mathrm{TX}_i \times \mathbf{1}[q_i = q] \;+\; \alpha_f \;+\; \delta_{ct} \;+\; \varepsilon_{it} \end{aligned} \tag{5}$$

q_i is the quartile of the tank's age at the reform date, 1998 less its install year, with cut points of 8, 13, and 20 years taken from the tanks active at the end of 1998 in Texas and the control states. The youngest quartile is the reference, and the reform effect for quartile q is therefore $\beta + \gamma_q$. The terms λ_q allow Texas tanks of each age quartile to close at a different rate throughout the sample, a difference the facility fixed effect does not absorb because one facility holds tanks of several ages. The fixed effects and sample are those of Equation 3 at the tank level. Age at the reform is fixed on the reform date and cannot respond to it. Section 4.2 reports the four effects.

Treatment is assigned at the state level, and one state receives it. The regressions carry 22 clusters and 1 treated cluster, a configuration in which cluster-robust t-tests over-reject substantially. I therefore report a placebo distribution in Section B.9, following Conley and Taber (2011). I re-estimate Equation 3 once for each control state with that state assigned pseudo-treatment and Texas kept in the sample and coded as untreated. Control states

are matched to Texas rather than to one another, and the sample remains fixed across the placebo runs. A shock common to 1998 would move the control-state estimates as well, and the position of the Texas estimate within that distribution is the test. With 21 control states the smallest attainable placebo p-value is 0.045, just below the conventional five percent, and a rejection at that level requires the Texas estimate to rank first among the 21 control-state estimates. The test therefore reports the rank of the Texas estimate rather than a rejection at conventional levels. I report the placebo p-value in Section B.9 alongside the cluster-robust standard error rather than in place of it.

Identifying assumption

Identification requires that, absent the reform, a Texas tank and its same-cell control tanks would have followed the same closure path. Under that assumption, β in Equation 3 recovers the effect of the reform. Without it, β combines that effect with whatever else differentiates the two groups after 1998.

The variation originates in a fund that exhausted its revenue. The Texas legislature set the closing date to the fund's financial position, and neither the decision nor its timing responded to the condition of any particular facility's tanks. The fund did not reprice. It ceased writing coverage entirely, and a private market already rating the identical product in never-fund states absorbed its policyholders. The price of risk facing a Texas facility therefore changed for reasons originating outside that facility.

Four other forces move a tank's closure date, and each meets a different element of the design. The federal deadline moves it, and the cell-by-year effect absorbs any response common to tanks of the same construction and vintage. Fuel demand and station margins move it, and the facility effect absorbs a facility's own level while the cell-by-year effect absorbs national movement specific to a tank type. Local competition moves it, and no fixed effect in Equation 3 absorbs a change in it after the reform. Section B.4 reports the estimate split by market density. Access to capital moves it, and I hold no measure of it. Reverse causation does not arise because carriers file rates statewide with the Texas Department of Insurance and those rates apply to a class rather than to an individual facility.

The reform altered two features of the price of holding a tank, and the design distinguishes them. A level change raises the cost of holding every tank by approximately the same amount and should raise closure evenly across the stock. A slope change raises the cost most for the tanks most likely to leak and should concentrate closure at the upper end of the risk distribution. Texas experienced both because the fund's fee was flat and set below cost. A level change alone should raise closure by about the same amount at every age at the reform. Section 4.2 tests whether the response instead rises with age.

Threats to identification

Cell-by-year effects absorb national policy and absorb nothing that an individual state enacts. States legislate above the federal floor, and most of what they add is a technical standard governing new systems. A smaller number order existing tanks out of the ground on criteria that bind within the sample window. Maine required the permanent closure of tanks installed before 1986 that had been retrofitted to meet leak detection and spill requirements, effective

in 2008. Kentucky required the permanent closure of steel tanks and piping lacking corrosion protection in 2012, and required interior-lined steel tanks either to add cathodic protection or to close in 2013. Both states are controls, and all three rules bind the old unprotected steel this design studies. Treatment varies at the state-by-year level, and no fixed effect in Equation 3 absorbs a mandate of that form. The direction of the resulting bias is nevertheless determinate. A forced removal concentrates closures within a control state in a particular year and raises that state's closure rate, which narrows the Texas-minus-control difference. These mandates therefore bias the estimates reported below toward zero.

A second kind of state rule reaches the sample through a different channel. Every state in the sample now requires double-wall construction on new installations, on a date it set for itself, and Texas adopted that requirement on January 1, 2009. A construction standard binding new tanks cannot force an old one to close, and grandfathering delays retirement rather than hastening it. The standard does raise the cost of replacing a tank because the replacement must meet the current standard at roughly seventy percent more per tank. Replacement is a margin this paper measures separately, and the Texas date falls a decade after the reform and inside the event-study window. Section 4.3 reports the replacement response.

Enforcement intensity is the form of this concern that regulatory logic cannot dispose of, because a state that inspects more often finds more occasions to close a tank. Nothing in Equation 3 absorbs a change in inspection effort specific to one state in one year, and the direction of the bias follows the state that changes. A rise in Texas inspections after 1998 would raise Texas closures for a reason other than price and bias the estimate upward. A rise in a control state would bias it toward zero.

Two measures bound the concern without disposing of it. Since the Energy Policy Act of 2005, federal law has required every state to inspect each regulated tank at least once every three years, and the EPA reports for every state and fiscal year how often inspectors looked and what they found. Texas looked less often than every control state, 102.4 inspections per thousand active tanks a year against a control mean of 168.5 over fiscal 2008 to 2024, and it found fewer violations when it looked, 21.1 percent of inspected facilities out of compliance against 32.8 percent under the standard reported through fiscal 2021 and 15.2 against 42.7 percent under its successor, ranked 3 of 22 and 2 of 22 from the lowest rate upward. The closures a state's enforcement forces are the product of how much of its stock it inspects, how often an inspection finds a violation, and how often a violation ends in a closure. The first two factors are measured, and both run against the concern. For enforcement to have raised Texas closures relative to the controls in the years the measures cover, Texas would have had to close tanks per violation found at least 2.5 times as often as the control states under the first standard and 4.6 times under the second, and at least 1.5 and 2.8 times had it inspected as large a share of its stock as they did. What the measures cannot do is sign the bias. Both series begin after the reform, and the bias in Equation 3 is the change in the Texas-minus-control enforcement gap at the reform rather than its later level. Section B.10 reports the measures, the bound, and the assumptions it rests on.

Closure is an absorbing outcome, which governs what a long-horizon coefficient can represent. The tanks still contributing to a coefficient seventeen years after the reform are the tanks that have not yet closed. If the reform advanced the closures of the tanks most exposed to

it, the surviving stock becomes progressively less sensitive, and the late coefficients therefore understate the response rather than overstate it. The late coefficients are also noisier because the sample thins as tanks close.

Anticipation operates in the same direction. The sunset passed in 1995 and gave operators three and a half years of notice, and a closure that an operator advanced falls in the pre-period and reduces the estimated increase. That lead time also places $k = -1$ within the announcement window rather than before it, which makes the event study's base year an imperfect untreated benchmark.

The design cannot address exit before the reform. The estimation sample admits only facilities holding an active tank on December 23, 1998, and a facility that closed in 1994 is therefore absent, which makes the pre-reform exit rate zero as a matter of construction rather than behavior. The unrestricted registry carries those exits. An annual exit series for 1990 through 1998, constructed without the incumbent restriction, is the evidence bearing on whether Texas and the control states followed the same pre-reform path.

4 The Firm's Response

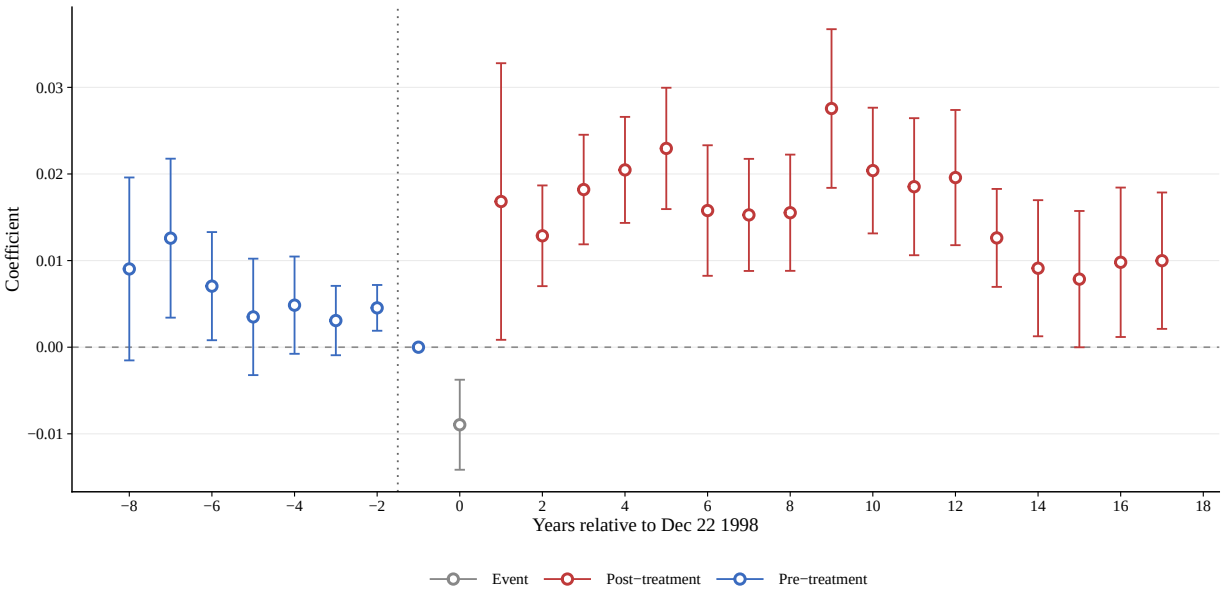
4.1 The closure response

Figure 3 plots the coefficients from Equation 4 on the annual probability that a tank is permanently removed, the Texas-minus-control closure gap by year relative to the reform. The figure shows four facts. First, the pre-reform coefficients are small. 57 percent fall below half a point, the largest reaches 1.26 points, and the few that differ statistically from zero reflect the precision of 7,862,172 tank-years rather than an economically meaningful pre-trend. The coefficient for 1998, event time zero, is negative because Texas facilities removed fewer tanks at the December 1998 federal upgrade deadline than facilities in the control states did.

Second, the gap opens at the reform and stays elevated in every post-reform year. Five years out a Texas tank is 2.3 points more likely to close than its matched control peer (s.e. 0.4), and ten years out the gap is 2.0 points (s.e. 0.4). Third, the path is not monotone. The gap is largest around year nine and smaller in years fourteen to seventeen, and the effect persists through all seventeen post-reform years. Fourth, the later coefficients are noisier because the sample thins as tanks close. They also understate the response. Closure is absorbing, and Section 3 shows survivorship makes the surviving stock progressively less sensitive. I read the late gap as a floor on the effect of the risk-rated regime.

Table 9 in Section B.3 reports estimates of Equation 3 on the annual probability that a tank is permanently removed. Following the transition to the private risk-rated market, a Texas tank closes 1.3 points more often than control-state tanks in the same cell of construction, fuel, capacity, and vintage (s.e. 0.4), an increase of about 47 percent on the pre-reform control mean of 2.71 percent. Column (7), with facility and cell-by-year fixed effects, is the preferred specification. The estimate is positive in all seven columns and lies between 1.1 and 1.6 points in six of them. Column (4), with tank and year effects but no cell-by-year

Figure 3: Event study of tank closure, Texas relative to control states



Notes: Coefficients on the interaction of a Texas indicator with an indicator for each year relative to 1998, from Equation 4 estimated on the tank-year panel. The outcome is an indicator that the tank is permanently closed in the year. Coefficients are in probability units, and 0.01 is one percentage point. The year before the reform is the omitted category. Event time zero is 1998. The fund stopped covering new releases after December 22 of that year, and private coverage began on December 23. The sample is 7,862,172 tank-years from 1990 to 2015, on tanks active at the reform, in Texas and 21 control states. The regression includes facility fixed effects and cell-by-year fixed effects, where a cell is a tank's wall construction, fuel, capacity class, and install year. Vertical bars are 95 percent confidence intervals from standard errors clustered by state.

effect, gives 0.4 points (s.e. 0.4).

The same comparison at the facility level, where the operator makes the decision, gives 1.8 points (s.e. 0.3) on a control mean of 3.31 percent, and is column (1) of Table 4. Figure 19 in Section B.7 replicates the event study at the facility level, and its path has the same shape as the tank-level one.

4.2 Who responded

Table 3 reports the heterogeneity specification Equation 5, and Figure 4 plots the reform effect for each quartile of age at the reform. The response rises with age. A tank in the youngest quartile, 8 years old or younger at the reform, closes 0.5 points more often after the reform (s.e. 0.1). The effect is 0.8 points (s.e. 0.2) for tanks aged 9 to 13, 1.6 points (s.e. 0.6) for tanks aged 14 to 20, and 4.1 points (s.e. 0.8) for tanks 21 years or older, about 8.8 times the point estimate for the youngest quartile. The oldest quartile's effect exceeds the

youngest quartile's by 3.6 points (s.e. 0.8).

Table 3: Closure response by tank age at the reform

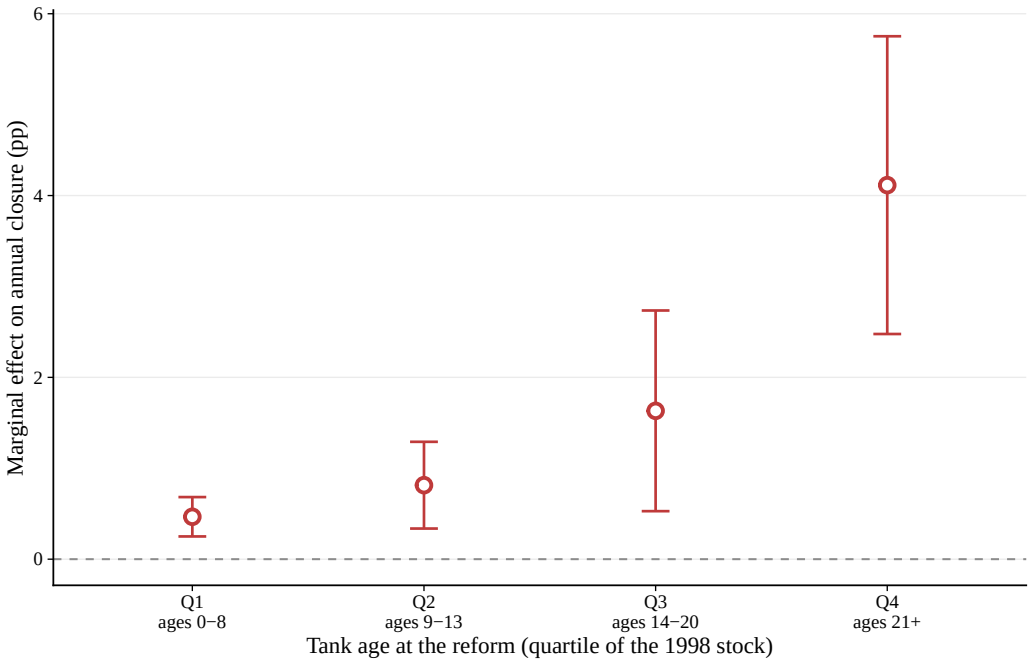
	Annual tank closure (pp)
TX × Post	0.47
	(0.11)
TX × Q2 (ages 9–13)	4.20
	(0.75)
TX × Q3 (ages 14–20)	3.32
	(0.88)
TX × Q4 (ages 21+)	4.83
	(1.05)
TX × Post × Q2 (ages 9–13)	0.35
	(0.21)
TX × Post × Q3 (ages 14–20)	1.17
	(0.54)
TX × Post × Q4 (ages 21+)	3.65
	(0.80)
Facility fixed effects	Yes
Cell-by-year fixed effects	Yes
Observations	7,862,172
States (clusters)	22

Notes: Difference-in-differences estimates of the reform effect on annual tank closure, split by the quartile of the age of the tank at the December 23, 1998 reform, in percentage points. Quartile cut points are 8, 13, and 20 years, from the tanks active at the end of 1998. The youngest quartile is the reference. The TX x Post row is therefore its effect, and each TX x Post x quartile row is the difference from it. The TX x quartile rows are level differences for Texas tanks of each age that the facility fixed effect does not absorb. The sample is the tank-year panel from 1990 to 2015, on tanks active at the reform in Texas and the control states, unweighted, with facility and cell-by-year fixed effects, as in column 7 of Table 9. Standard errors clustered by state with 22 clusters are in parentheses. With one treated state these analytic errors understate the uncertainty, and no randomization-inference or bootstrap p-values are computed for this split.

The oldest tanks are both the riskiest part of the stock and the part the reform repriced most. Panel A of Figure 1 shows the release gap of a single-walled tank widening with age, the filed schedule of Section 2.4 charges more as a tank ages, and the closure response rises in the same direction. The reform raised the price of risk most where the risk is largest, and the response followed the price.

The split also separates the two features of the price change named in Section 3. A level change alone would raise closure by about the same amount at every age, and the youngest

Figure 4: Reform effect on annual tank closure by tank age at the reform



Notes: Each point is the estimated effect of the reform on the annual probability of tank closure for the tanks in one quartile of age at the reform, $\beta+\gamma_q$ from Equation 5 and β for the youngest quartile, in percentage points. Age at the reform is 1998 less the install year. The quartile cut points are 8, 13, and 20 years, from the tanks active at the end of 1998 in Texas and the control states. The sample is 7,862,172 tank-years from 1990 to 2015, on tanks active at the December 23, 1998 reform, in Texas and 21 control states. The regression includes facility fixed effects and cell-by-year fixed effects. Vertical bars are 95 percent confidence intervals from the covariance matrix of the coefficients, clustered by state with 22 clusters. The dashed line marks zero. Table 3 reports the coefficients.

quartile's response of 0.5 points bounds from above what a level change produced. The oldest quartile's response of 4.1 points is that level plus the slope the schedule adds for age.

4.3 How they responded

Table 4 reports Equation 3 estimated separately on each portfolio margin at the facility level, where the portfolio decision is made. Three margins move. The probability that a facility closes at least one matched tank rises 1.8 points (s.e. 0.3), facility exit rises 1.2 points (s.e. 0.4), and capacity reduction rises 1.6 points (s.e. 0.3). Exit carries no pre-reform base rate because the sample holds only facilities operating at the reform.

The downsizing and replacement averages are 0.3 points (s.e. 0.2) and 0.4 points (s.e. 0.2), and both conceal responses that change sign with facility size. In the smallest capacity bin the downsizing effect is -0.6 points (s.e. 0.2) and the replacement effect is -0.3 points (s.e.

Table 4: Facility-level estimates by portfolio margin

	Any closure (1)	Facility exit (2)	Downsize (3)	Consolidate (4)	Reconfigure-up (5)	Any replacement (6)	Capacity cut (7)
Reform × Post	0.0183***	0.0123***	0.0033**	-0.0005**	5.2×10^{-5}	0.0038**	0.0158***
	(0.0031)	(0.0040)	(0.0016)	(0.0002)	(0.0003)	(0.0015)	(0.0031)
Observations	2,776,307	2,776,307	2,776,307	2,776,307	2,776,307	2,776,307	2,776,307
Facility fixed effects	✓	✓	✓	✓	✓	✓	✓
Composition × year fixed effects	✓	✓	✓	✓	✓	✓	✓

Notes: Facility-level estimates of the design equation of Section 3, one regression per column, in probability units on the annual facility-year indicator named in each column head. A coefficient of 0.01 is one percentage point. The unit of observation is the facility-year over 1990 to 2015, on Texas and the 21 control states, restricted to facilities holding at least one tank active at the December 23, 1998 reform. Any closure is a facility-year in which the facility closes at least one matched tank. Facility exit is the facility-year in which the facility closes its last tank. Downsizing is a facility-year in which the facility does not exit, ends the year with fewer tanks than it held at the end of the previous year, and cuts its total capacity by more than five percent. Consolidation is a facility-year in which the facility does not exit, ends the year with fewer tanks than it held at the end of the previous year, and changes its total capacity by no more than five percent in either direction. Reconfiguring up is a facility-year in which the facility does not exit, ends the year with fewer tanks than it held at the end of the previous year, and raises its total capacity by more than five percent. Replacement is a facility-year in which the facility closes at least one tank and installs a new tank on a later date. A capacity cut is a facility-year in which the facility ends the year with less total capacity than it held at the end of the previous year. Section A.3 gives the exact rules. All columns carry facility and portfolio-composition-by-year fixed effects and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines. Standard errors clustered by state with 22 clusters are in parentheses, and stars mark 0.1, 0.05, and 0.01. With one treated cluster those errors over-reject. Section 2 reports the pre-reform control base rates.

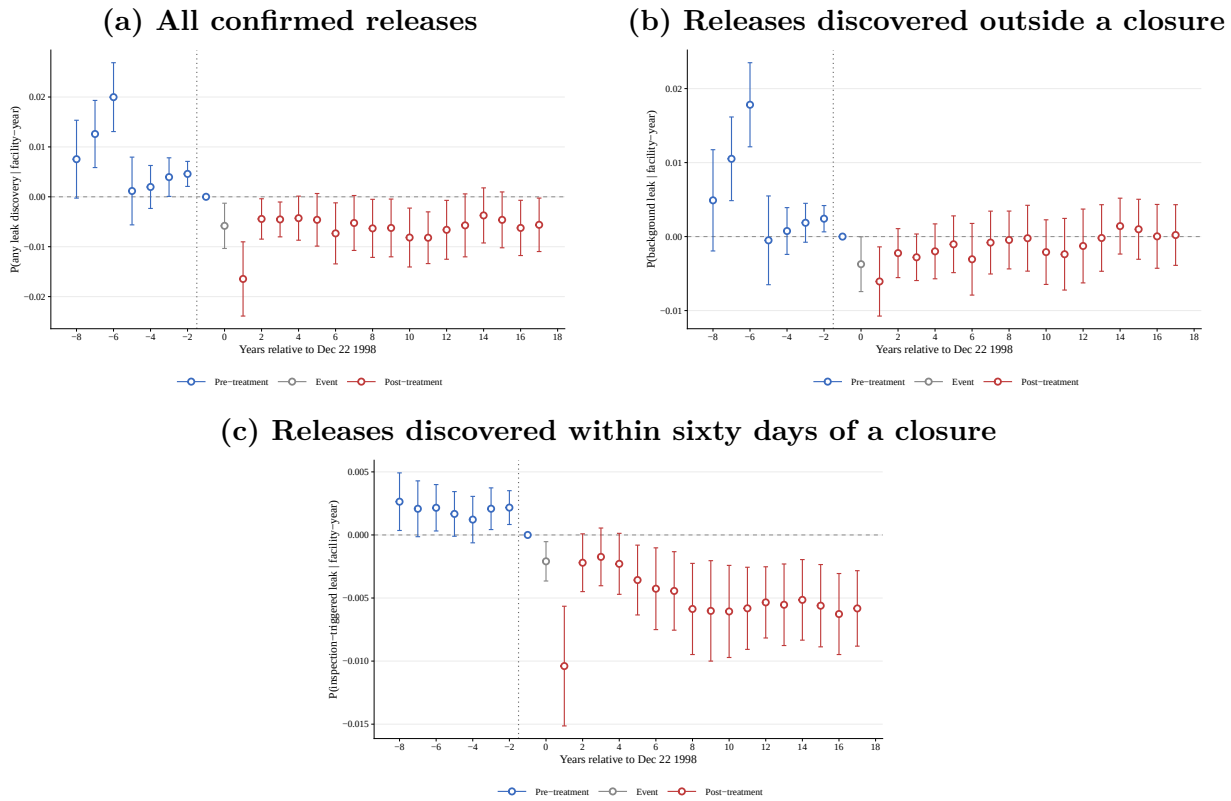
0.2), while the same facilities exit at 5.5 points (s.e. 0.7). In the largest bin the downsizing effect is 1.7 points (s.e. 0.2) and the replacement effect is 1.3 points (s.e. 0.2). The average of two groups moving in opposite directions is close to zero and describes neither.

Small facilities respond on the extensive margin, and facilities with more capacity respond by re-sizing what they hold. A price on tank risk therefore does not shrink the stock uniformly. It removes the operators for whom the priced risk was worth least to carry, and reorganizes the ones for whom it was worth more. Wall construction cannot be changed on an installed tank, and a firm seeking a lower risk class removes the tank rather than modifying it. The response therefore appears on the closure and exit margins rather than as retrofitting. Entry of new stations shows no perceptible response to the reform (Section B.8).

4.4 Confirmed releases

Confirmed release discoveries fall. Figure 5 reports the release response: the pooled series in panel (a), releases discovered on their own in panel (b), and releases discovered during a closure inspection in panel (c). After the reform, the annual probability of a confirmed release discovery at a Texas facility falls by 1.2 points relative to control-state facilities in the same cell (s.e. 0.3). Releases discovered on their own fall by 0.5 points (s.e. 0.2), and releases discovered during a closure inspection fall by 0.6 points (s.e. 0.2). Table 12 in Section B.6 reports the static estimates.

Figure 5: Event study of confirmed release discovery, Texas relative to control states



Notes: Coefficients on the interaction of a Texas indicator with an indicator for each year relative to 1998, from Equation 4 estimated on the facility-year panel. The outcome is an indicator that a confirmed release is discovered at the facility in the year. Panel (a) pools every confirmed release, panel (b) keeps releases discovered outside a closure inspection, and panel (c) keeps releases reported within the sixty days following a closure. Coefficients are in probability units, and 0.01 is one percentage point. The year before the reform is the omitted category. Sample, fixed effects, and controls follow Table 4. Vertical bars are 95 percent confidence intervals from standard errors clustered by state.

Release discovery differs from release incidence because closing a tank triggers an inspection that can find a release already in the ground, and Section A.3 separates releases reported within the sixty days following a closure from the rest. The structural model prices avoided

releases rather than counting discovered ones.

5 Dynamic Structural Model

I develop a dynamic model of facility tank-portfolio management to quantify what pricing tank risk is worth relative to a flat fee, a replacement subsidy, and a retirement mandate, and to value the response Section 4 measured. I use the model to estimate (i) the value facilities place on operating revenue, (ii) the weight they place on a dollar of coverage cost, the premium plus the expected out-of-pocket cost of a leak, and (iii) the costs of removing tanks, installing them, and exiting. A separate step estimates the cost of opening a new station. Together with the measured premium schedules and hazard, these estimates deliver facility behavior and welfare under the observed contracts and the counterfactual designs of Section 6.

Facilities holding portfolios of tanks that differ in age and wall construction earn revenue on their storage capacity and face a premium and an expected out-of-pocket leak cost that depend on the tanks they hold. Facilities also differ in private payoff shocks that carry the operating value no schedule observes. My empirical strategy first measures every price and risk input outside the model, the premium from the carriers' filed rate schedules, the hazard from the leak model of Section 2.3, and revenue from fuel margins and storage capacity, and then estimates the payoff parameters by nested pseudo-likelihood from how choices respond as those inputs vary across the flat-fee control states and Texas carrier-year pricing environments in the sample.

5.1 The model

The agent is the facility rather than the tank. Closure in the data is a portfolio decision, several tanks removed in one act, and a model of one representative tank cannot price the partial-closure margin Section 4.3 found. A facility enters each year holding a portfolio of tanks that differ in age and wall construction. It earns revenue on the fuel its tanks store, pays the premium its insurer rates on the portfolio, and bears the deductible if a leak arrives. While its tanks are young, the premium and the hazard are both small, and the facility maintains the portfolio and operates. Each year its tanks grow older. The chance of a leak grows with them, and the expected out-of-pocket cost therefore climbs under any contract. The premium is the part the contract decides: under a risk-rated schedule it climbs with the tanks, and under a flat fee it does not move at all.

As the charges rise, the facility weighs four options. It can pay the charges for another year. It can pay once to remove its oldest tanks and operate with fewer. It can pay more to replace them with new double-walled tanks that are cheap to insure. Or it can decide the revenue no longer justifies any of it, collect its scrap value, and close for good. Whatever it chooses, the surviving tanks age, and the facility begins the next year with what its choice has left.

No facility in the model responds to another. The reform repriced each facility's own risk rather than the structure of local competition, and the responses the model must reproduce are choices over a facility's own capital: removing, replacing, and exiting. The incumbent's

problem is therefore a single-agent dynamic discrete-choice problem in the tradition of Rust (1987), and new stations enter by a rule that involves no strategic interaction either. The cost of that scope is that the model is silent on how pump prices and fuel volumes adjust as stations exit and enter. The return is exactness: the state reduces to one facility's own portfolio, and the 298,448-state problem solves with no approximation to the value function.

The state is $s = (c, B, g)$. The portfolio c counts a facility's tanks over 16 cells, two wall types crossed with eight five-year age bands. The oldest band is absorbing, and fleet size runs between one and six tanks. Enumerating the count vectors gives 74,612 portfolios. The capacity bin B sorts facilities into the four storage classes of Section 2.2, and the environment g names the pricing regime the facility faces, a flat-fee state or a Texas carrier-year, and with it the premium schedule and the deductible. For a Texas facility the environment includes the carrier writing its coverage. Facilities in the data switch carriers, and the model takes each year's carrier as given rather than pricing the choice of carrier. Capacity sits apart from the portfolio because the two drive different payoffs. Capacity drives revenue, while tank count, age, and wall drive the premium and the hazard. Crossing portfolios with capacity bins gives the 298,448 states.

The portfolio matters because the premium and the hazard are steep in exactly the cells a removal can empty. Consider two facilities under the same filed schedule. The first holds three tanks, two single-walled tanks in the oldest band and one new double-walled tank. The schedule charges the old single-walled pair the highest per-tank rate on the card and the new tank the lowest, and the facility's hazard reads off the top of the risk gradient in Section 2.3. Removing those two tanks is one action that eliminates most of the facility's premium and moves its rating from the top of the gradient to the bottom. The second facility holds four double-walled tanks of middle age. Its total premium is larger because it insures more tanks, but every tank carries the same low rate, its hazard sits near the bottom of the gradient, and no single removal moves either charge by much. A tank count calls the second facility the larger risk, and an average age calls the two similar. The portfolio state is what lets the model see that the first facility can buy a large reduction in both charges with one removal and the second cannot.

Actions. Each year the facility chooses (k, m), remove k tanks and install m, or exits. Maintaining is $(0, 0)$, downsizing removes without installing, replacement does both, and exit is absorbing. At most four tanks move in either direction in a year, a work action cannot remove the last tank, and the post-action fleet respects the six-tank cap.

Three assumptions turn the counts into a law of motion. **A1**, removal takes the oldest single-walled tanks first, then the oldest double-walled. The facility chooses how many tanks to shed and the rule chooses which. The data show the margin is nearly deterministic: the rule names the removed category correctly in 84.6 percent of removal events (Appendix D). **A2**, every installed tank is new and double-walled, mirroring the secondary-containment requirements that ended single-walled installation over the sample. **A3**, each surviving tank either advances one age band or stays, with band-specific probabilities estimated from the registry's observed transitions. The facility cannot slow its tanks' aging. Its only lever on the fleet's risk is removal and installation, and that is the margin the premium prices.

Revenue. A facility's operating revenue is the fuel margin in its state and era, the retail price net of wholesale and taxes, scaled by the capacity of its bin. The result is a lookup table $R_{B,g,e}$, in units of \$10,000. Revenue is measured outside the model, following Sweeting (2013), and enters the payoff through one estimated weight, ψ: the data pin what a facility of a given size earns, and the model estimates how much a dollar of it moves choices. The construction makes revenue exogenous to the dynamic problem. A facility changes its revenue only by changing its capacity bin, through the removal and installation choices the model already prices. Appendix D details the construction and its cost.

Payoffs. An operating action earns the facility its revenue and charges it for its risk and its adjustments,

$$u_{(k,m)}(s) = \psi R + \alpha_g - \gamma\,[P(n') + H(n')\,D] - k\,c^{\text{rem}} - m\,c^{\text{inst}}, \qquad u_X(s) = \kappa_1\,N(n),$$

where n' is the fleet the action leaves in the ground. The premium $P(n')$ comes from the carriers' filed rate schedules applied to that fleet (Section A.6), and the expected out-of-pocket cost is the chance some tank leaks, $H(n')$ from Section 2.3, times the deductible D the contract sets. Both are dollars the facility pays for carrying its risk, and I call their sum the coverage cost. One weight, γ, applies to it. The level term α_g absorbs what is persistently different about operating in an environment. Exit pays a scrap value that scales with the fleet, with the constant part normalized to zero.

Each year the facility also draws one private payoff shock per action, Type-I extreme value with scale normalized to one, and every weight in the payoff therefore reads in units of the shock. The shocks carry the facility-specific operating value the revenue lookup cannot, and they are why two facilities in identical states choose differently. All else equal, a facility whose premium and hazard are concentrated in a few old single-walled tanks gains more from removal than one whose charges are spread evenly, and a steeper schedule widens that difference.

The value function. Within a year, events run in a fixed order. The facility observes its state, draws one private shock per action on its menu, and acts. The action reshapes the fleet at once, the insurer re-rates the post-action fleet, the year's payoff is realized on it, and the surviving tanks then age into next year's state. The facility knows the current premium schedule, its hazard, the deductible, and its own shocks. It treats the policy environment as permanent and forecasts its tanks' aging and leaking with the same processes the model estimates. It does not observe future shocks. The facility takes the action with the highest choice-specific value, its flow payoff plus the discounted value of the state it leads to,

$$v_d(s) = u_d(s) + \beta \sum_{s'} F_d(s' \mid s)\,V(s'), \qquad V(s) = \log \textstyle\sum_d \exp v_d(s).$$

The discount factor $\beta = 0.95$ is calibrated rather than estimated. F_d is the deterministic reshaping of the fleet followed by stochastic aging, and exit carries no continuation value.

The dynamic tradeoff is the model's core economics. Keeping an old tank means paying its premium and bearing its hazard again next year, while removing it means paying the adjustment cost once, and the choice turns on how long the facility expects to keep operating.

The same repricing therefore moves a facility near the end of its horizon differently than one with decades of revenue ahead, and that difference is what a static model of the same choice cannot represent. With logit shocks the value function solves in one exact linear system, and Section D.2 gives the solver.

Entry. New facilities open each year from a pool of potential entrants in each state. A potential entrant opens when the value of a new facility exceeds what it costs to open one, and the probability that it does is

$$\Pr(\text{enter}_g) = \Lambda(\bar{V}_g - c^{\text{inst}}\,\bar{n}_g - ec), \qquad \Lambda(x) = \frac{1}{1+e^{-x}},$$

where $\bar{V}_g$ is the incumbent's value function in environment g averaged over the sizes at which facilities open there, $\bar{n}_g$ is the mean number of tanks an entrant installs, and ec is the entry cost. An entrant opens with new double-walled tanks, pays the install cost on each of them and the entry cost once, and from then on solves the incumbent's problem above. The logistic form carries the same unit shock scale as the incumbent's choice. Entrants are not strategic. Each compares its own value of opening with its own cost and does not condition on how many others open. A contract changes entry through $\bar{V}_g$ alone, because a contract that lowers the cost of holding tanks raises the value of a new facility.

The pool of potential entrants is not observed, and I set it by rule. The pool in a state and year is the number of facilities the state's population could support less the facilities operating at the start of the year, $\lambda\,\text{POP}_{g,y} - n_{g,y-1}$. The rate λ is the largest number of facilities per resident, counting the year's entrants, observed in any state and year of the sample. The rule treats the most saturated market in the data as full. Facilities open at the tank counts and capacities observed among entrants in each state, and the model does not price the choice of size. Section D.6 gives the construction.

The contract enters this model in exactly one place, the schedule $P(\cdot)$. A flat fee makes P flat in the portfolio, and the risk terms a facility can manage shrink to the deductible channel. A risk-rated schedule makes P steep in age and wall, and removing the right tank pays twice, once in premium and once in expected out-of-pocket cost. The counterfactuals of Section 6 are this substitution performed on the estimated model, one $P(\cdot)$ swapped for another with everything else held fixed. Entry responds to the same substitution through the value of a new facility.

5.2 Estimation

Estimation of the incumbent's problem runs in two stages, and the first is fixed before the second begins. The first stage builds every payoff input from data outside the dynamic model: the premium schedules from the carrier rate engines of Section A.6, the hazard from the leak model of Section 2.3, the deductible from the contracts, the revenue table from the fuel-margin construction above, and the aging kernel from the registry's observed transitions. None is estimated jointly with the dynamic parameters, and the evidence on prices and risk therefore stands on its own before the model reads behavior against it. The second stage estimates the five payoff parameters and the state effects from the observed choices.

I estimate the parameters by nested pseudo-likelihood (Aguirregabiria and Mira 2007). The estimator starts from the choice probabilities observed in the data, recovers the value function they imply through the linear system above, and chooses the parameters that maximize the pseudo-likelihood of the observed actions. It then updates the probabilities toward the model's implied ones, damped to stabilize the iteration, and repeats until successive parameter changes fall below 10^{-5}. Dynamic models with larger or continuous state spaces approximate the value function to make this loop feasible. The portfolio state is finite, and every inner solve here is exact. Section D.2 gives the solver and the runtimes.

The estimating sample is the incumbent panel at annual frequency, 1,654,626 facility-years. Texas enters from 2006 through 2020, with the start set by when its carrier premium data begin, and the control states enter from 1999 through 2020. Control states number 21 in the reduced form and 14 enter estimation. Kansas and Maryland are excluded because no per-tank fund fee could be verified for either, and an unverifiable fee is missing data rather than a price: several control states charge no per-tank fee at all, and those verified zeros enter the model as true prices. Missouri is excluded for holding too few events to identify a state level. One parameter vector applies across every environment, with Texas the reference level for the state effects. Standard errors come from the outer product of the score at the estimates.

The entry cost is estimated in a separate step, after the incumbent's problem is solved. Given the value of a new facility, the install cost, and the pool, the number of entrants in a state and year is binomial with the entry probability of the model section, and I choose ec to maximize that likelihood. The sample is 259 state-years in 14 states, the control states from 1999 through 2017 and Texas from 2006 through 2017, in which 12,554 facilities open. The window ends in 2017 because the population series does. A state-year whose pool is smaller than its own count of entrants is excluded. The standard error on ec is clustered by state and takes the value of a new facility as given. Section D.6 reports the pool, the fit by state, and the estimates under larger pools.

5.3 Identification

If every facility of a given risk faced one price, no choice could reveal the weight on a dollar of coverage cost, because the cost would move only when the tanks moved, and its effect could not be separated from the effect of the risk itself. The sample prices risk in enough different ways to separate the two. Within Texas, carriers file their own rate schedules with the state's insurance regulator, and two facilities holding identical tanks in the same year pay different premiums because different carriers write them. In the control states the insurer is the state fund, and each fund sets its own per-tank fee and its own deductible. A fund's fee moves a facility's premium with its tank count and never with its tanks' age or wall construction, and both the fee and the deductible differ from state to state. Together these institutions supply four kinds of identifying variation: premiums that rise with a tank's age in Texas and stay flat in the fund states, premiums that differ across carriers at fixed risk, deductibles that differ across states at fixed premium, and hazards that vary with age and wall construction under fee schedules that never price either. Each parameter below is pinned by a named piece of this variation.

The revenue weight ψ is identified from how operating decisions move with measured revenue. Measured revenue differs across facilities for two reasons the model takes as given: a larger capacity bin stores more fuel, and a richer state fuel margin earns more on each gallon sold. Facilities with higher measured revenue exit less often and downsize less often, and ψ is the weight that reproduces how strongly their choices move with it. The concern is that the direction could run backward: a facility intending to stay might acquire capacity, rather than capacity keeping it in business. The model's timing rules that out within the year. This year's revenue is set by the capacity bin the facility held entering the year, and a removal or installation this year changes revenue only from next year onward, through the continuation value. A data limitation bounds what ψ can mean. I observe no facility's own gallons sold and no facility's own prices, and the revenue measure is built from what is observable instead, the capacity bin and the state fuel margin. Two consequences follow. The measure does not observe how many times a facility turns its storage over in a year, and ψ absorbs that turnover rate. And the measure assigns every facility in a bin, state, and era the same revenue, and true within-cell differences enter the estimate as measurement error, which attenuates ψ toward zero. Both forces push the same way, and the estimate in Section 6 should therefore be read as a lower bound on how revenue moves choices, a weight on the constructed measure rather than the value of a literal dollar.

The coverage-cost weight γ is identified from variation in what a facility pays for its risk that the risk itself does not explain. Two sources supply it, the premium and the expected out-of-pocket cost, and both move the same term, $P(n') + H(n')\,D$. The premium supplies two comparisons. The first is across contracts. In Texas the filed premium rises with a tank's age, and in the fund states the fee does not move with age at all. Facilities holding the same portfolio therefore face a premium that climbs with age in one group and stays flat in the other, and the difference in how removal and exit respond to age across the two groups pins γ. The second is within Texas. Two facilities with identical portfolios pay different premiums when different carriers write them, and that difference moves the bill without moving the tank. Every premium is priced at the same reference contract, \$1,000,000 in limits with a \$5,000 deductible, and the dispersion across carriers is therefore rate-schedule dispersion alone rather than differences in chosen coverage. The dispersion comes from the carriers with recoverable filings, and facilities carrying an imputed market-mean premium contribute none of it. The model takes each year's carrier as given, and if facilities switch toward carriers that price their portfolio cheaper, γ understates the response to a premium a facility could not shop away. Figure 35, in Section D.4, plots the filed premium by fleet age for Texas against the flat-fee control states, where the control schedules are flat in age by construction.

The expected out-of-pocket cost $H(n')\,D$ supplies variation the premium does not share, and a data limitation shapes how. The financial-responsibility contract records carry neither the deductible nor the coverage form a facility chose, and the rate engines therefore price every Texas contract at the carriers' filing base, \$1,000,000 in coverage with a \$5,000 deductible. I impose that \$5,000 deductible on every Texas facility. The assumption buys a clean within-Texas lever: the out-of-pocket cost then moves only through the hazard, and a fleet's wall construction shifts the hazard at fixed age in a way a pure age effect cannot mimic. Its cost is measurement error in the out-of-pocket level, because actual deductibles that vary across

Texas facilities differ from the imposed one, and if the choice of deductible tracks a facility's risk the error is not random. The cross-state variation carries the identification of the level for that reason. Each control-state fund sets its deductible by rule, in most states at one level for every facility it covers. The level differs from state to state, and no facility chooses it. One further limit comes from the leak data: releases are recorded at the facility, and the hazard is therefore a facility-level rate rather than a rate per tank. Figure 36, in Section D.4, plots this expected cost by fleet age, one line per state, with the deductible setting the level of a line and the hazard its slope.

The adjustment costs are identified from how often facilities act and how many tanks move when they do. The frequency of downsizing and the number of tanks shed at once pin c^{rem}, and the frequency and size of installations pin c^{inst}, with the lumpiness separating the per-tank cost from a fixed cost of acting. The scrap slope κ_1 is identified from how the exit rate falls with tank count at a fixed state. Timing driven by forces outside the model, the end of a lease for one, adds noise to these estimates rather than bias, provided it is uncorrelated with the fleet's risk exposure.

The environment effects α_g are identified from the operate-versus-exit margin. A state's α_g is read off how often its facilities keep operating, relative to Texas facilities holding the same portfolio at the same prices. Both α_g and κ_1 touch exit, and they separate by shape, a per-state level against a slope in tank count. Texas is the reference, $\alpha_{\text{TX}} = 0$, and no state's level pins at a bound in the estimation.

The entry cost is identified from the rate at which new facilities open. Given the value of a new facility and the pool, ec is the cost at which the predicted number of entrants equals the observed number. The pool is set by rule rather than observed, and the level of ec depends on it. A larger pool implies a lower entry probability for the same count of entrants and therefore a higher entry cost. Section D.6 reports the estimate with pools two and five times as large and with no limit on the pool.

What is imposed rather than estimated. The remaining assumptions are forced by the data or by the model class, and each carries a stated cost. The discount factor is calibrated at $\beta = 0.95$, because choice data cannot separate patience from the flow payoffs. The payoff shocks are Type-I extreme value with scale one, and the scale is a normalization: every parameter reads in shock units, and ratios of parameters carry the economic content. Fixing the shock scale and estimating ψ is the same normalization Sweeting (2013) makes from the opposite side, holding measured revenue at par and estimating the shock scale instead. The constant part of exit value is normalized to zero, which pins the level of the value function, and the data therefore identify how exit value scales with tank count and not its level. The environment is stationary, a facility treats current rules as permanent, and a rule change a facility in fact foresaw would be read by the model as a response to current prices. Owners forecast their tanks' aging and leaking with the same processes the first stage estimates, and the facility and the model share one forecast. The removal order of A1 is imposed rather than chosen, and where the rule misnames the removed tank the premium and the hazard carry small assignment errors (Appendix D). Revenue is a bin-level proxy rather than an observed account, and the idiosyncratic part of profitability is carried by the payoff shocks rather than measured. Entry adds three assumptions. Entrants do not respond to one another. The

pool of potential entrants is set from the highest observed number of facilities per resident rather than estimated. And facilities open at the sizes observed among entrants, with no choice of size. The estimates hold conditional on these assumptions, and Section 6 inherits them.

Identification is cross-sectional throughout. The parameters are pinned by dispersion across facilities and environments within years, not by the before-and-after of the Texas reform, whose pre-period never enters the estimation window.

6 Estimates, Counterfactuals, and Welfare

The estimated model prices the contracts. This section reports the parameter estimates and what they imply in dollars, checks the model's fit, defines three alternative contracts and a first-best benchmark, and scores every contract on one welfare identity.

6.1 Estimates and model fit

Table 5 reports the five payoff parameters and the entry cost with standard errors, in payoff units, and the dollar figures below are payoff units divided by γ, the weight on a dollar of coverage cost, stated in premium-equivalent dollars. Three magnitudes carry the results. First, facilities respond to the coverage cost. The weight on a dollar of premium or expected out-of-pocket cost is $\gamma = 0.5504$ (s.e. 0.0067). The weight on measured revenue is $\psi = 0.0725$ (s.e. 0.0018), and it is low as the construction of the revenue measure anticipates: ψ absorbs the turnover normalization and the attenuation Section 5 describes, and I read it as a lower bound on how revenue moves choices.

Second, the adjustment costs are large. Removing a tank costs $c^{\text{rem}} = 4.9064$ (s.e. 0.0093) in payoff units, about \$89,142 in premium-equivalent dollars, and installing one costs $c^{\text{inst}} = 0.9308$ (s.e. 0.0106), about \$16,911. These are full project costs rather than the price of a tank. They cover the labor, the inspections, the environmental review, and the replacement capital, along with disruption, downtime, financing, and the closure-triggered site assessment, and their size is what rationalizes how rarely facilities move.

Third, leaving is worth little and opening is expensive. The exit value is small, $\kappa_1 = 0.0473$ (s.e. 0.0095), about \$859 of scrap per tank, and what disciplines exit is therefore forgone operating value rather than salvage. The entry cost is $ec = 12.76$ (s.e. 0.88), about \$231,875 for a new facility before the install cost of its tanks. The state fixed effects are estimated alongside and not reported. The iteration converges in the parameters at iteration 13, and Section D.2 documents the convergence diagnostics.

The adjustment costs are recovered from choices, and it is worth asking whether they land where engineering estimates do. On the total they do. Removing a tank and installing its replacement costs about \$106,054 in the model. A county project under Indiana's trust-fund reimbursement program put the cost of removing and replacing three tanks at \$133,000 to \$167,000 per tank (Indiana Capital Chronicle 2026). Turnkey contractor quotes approach \$250,000 once dispenser work is included (UST Contractors 2026). The EPA's own 1988

Table 5: Structural parameter estimates

Parameter	Description	Estimate	Standard error	Dollar equivalent
ψ	Weight on measured revenue	0.072	(0.002)	
γ	Weight on coverage cost	0.550	(0.007)	
c^{rem}	Removal cost, per tank	4.906	(0.009)	\$89,142
c^{inst}	Install cost, per tank	0.931	(0.011)	\$16,911
κ_1	Exit value, per tank	0.047	(0.009)	\$859
ec	Entry cost, per facility	12.762	(0.884)	\$231,875

Notes: Estimates of the payoff parameters of the dynamic model of Section 5 and of the entry cost, with standard errors in parentheses. One model unit is 10,000 dollars per facility-year. The dollar equivalent of a cost or a value is its estimate divided by γ, the weight on a dollar of coverage cost, and multiplied by 10,000. The weights ψ and γ have no dollar equivalent. The removal and install costs are full project costs per tank, covering labor, inspections, environmental review, and replacement capital. The five payoff parameters are estimated by nested pseudo-likelihood on 1,654,626 facility-years, Texas from 2006 to 2020 and 14 control states from 1999 to 2020, with standard errors from the outer product of the score. The control-state fixed effects are estimated and not reported, and Texas is the omitted state. The entry cost is estimated in a separate binomial step on 259 state-years, at the baseline pool of potential entrants, with a standard error clustered by state that takes the value of a new facility as given. The entry cost excludes the install cost of the entrant's tanks.

estimate implies per-tank installed costs of \$65,000 to \$86,000 in 2023 dollars (EPA 1988), and an industry survey from 1987 puts an installed double-walled tank at \$98,000 to \$105,000 (U.S. General Accounting Office 1992).

The split runs opposite to the engineering menu. Clean removal costs roughly \$10,000 to \$25,000 per tank in state program records and reimbursement schedules (Association of State and Territorial Solid Waste Management Officials 2017; Kentucky Energy and Environment Cabinet 2024), while the model puts removal at about 5 times the install. The reason is what each parameter absorbs. The benchmarks price the tank work, and the model's costs are the cost of the whole project. Every action that opens the ground pays c^{rem}, and the parameter therefore carries the common cost of acting: labor, inspections, environmental review, downtime, demolition, financing, and above all the closure-triggered site assessment. Contractor guidance for an old single-walled tank is to budget two to three times the clean-closure cost against contamination discovered on removal (UST Contractors 2026). The claims records that feed the model's severity put the mean cleanup at \$348,000 once a release is found, above the EPA's national figure of about \$154,000 (EPA 2024a), and an expected-discovery term of a fifth to two-fifths of that cleanup closes most of the distance between the excavation bill and the estimate. Read against that, c^{inst} is the incremental cost of lowering a new tank into a hole that is already open. Benchmarks span 1987 to 2026. The 1987 and 1988 figures and the removal range are in 2023 dollars, and the Indiana and contractor figures are in the dollars of their own years. They are a plausibility check on

magnitudes rather than an independent estimate of either parameter.

Table 21 in Section D.3 sets the model's predicted action shares against the observed shares, overall and split by regime. The aggregate fit is close, maintaining at 97.40 percent predicted against 97.80 observed. The regime split shows where one parameter vector strains: within Texas the model over-predicts downsizing and under-predicts exit, attributing part of the Texas response to the partial margin the premium prices most directly rather than to full exit. The shares are moments the estimation touches only through the likelihood, not targets matched by construction.

6.2 Counterfactuals

A state that wants old tanks out of the ground has instruments other than the price of coverage, and two of the three experiments below are those instruments. A subsidy for replacement prices an action rather than the risk it removes. It pays every firm that replaces, including the firms that would have replaced anyway, and by lowering the cost of putting a new tank in the ground it raises the value of continuing to operate the site, working weakly against retirement even as it modernizes the stock. A mandate that retires every tank at a fixed age removes tanks on their observable condition alone and ignores the rent the site earns, the very information a price uses. It removes hazardous capital reliably and profitable capital alongside it, with the loss governed by how much rent varies among tanks of the same age.

A legislature could levy the risk charge directly as a tax on old tanks rather than route it through an insurance premium, and in the model the two are the same instrument, since both enter the payoff as $P(x)$. What separates them is who sets the schedule and on what information: a tax is legislated, changes at the pace of statute, and is graded on the categories the statute names, while a premium is set by carriers who underwrite the risk, is revised as claims accumulate, and is graded as finely as their rating plans allow. Nor do I run a no-coverage benchmark. Every state in the sample mandates coverage throughout, no variation in the data identifies behavior without it, and the exposure an uninsured, judgment-proof operator faces is not an object this model's payoff can represent.

Each counterfactual fixes the estimated parameters, changes one policy, and re-solves the facility's problem and the entry rule. The state fixed effects stay in, both in the facility's re-optimization and in the welfare level, because they stand for persistent features of an environment rather than the policy under study. Welfare starts from the observed stock, with no incumbent added, removed, or moved at the start, and the baseline therefore reproduces observed behavior. Entry then adds facilities at the rate the entry rule gives under each contract.

The uniform flat fee. The first experiment asks what Texas would look like under uniform pricing, and it is the paper's headline experiment: the reduced-form reform run forward as a policy. I replace the Texas risk-rated contract, premium schedule and deductible together, with a representative uniform-pricing contract, and re-solve. A maintaining Texas facility

earns, at baseline,

$$u_{(0,0)}(s) = \psi R - \gamma\,[P(n') + H(n')\,D], \qquad \alpha_{\mathrm{TX}} = 0,$$

and under the uniform contract,

$$u_{(0,0)}(s) = \psi R - \gamma\,[\underbrace{\tau\,N(n')}_{\text{flat fee replaces } P(n')} + H(n')\,\underbrace{D^{\mathrm{unif}}}_{\text{fund deductible}}].$$

The revenue, the hazard, and every cost hold fixed. Only the price of risk changes. One caveat holds throughout: risk-based pricing means Texas in this sample, and the contrast identifies the pricing mechanism suggestively rather than cleanly.

The replacement subsidy. The second experiment lowers the install cost for new double-walled tanks by 50 percent. A replacing action pays

$$u_{(k,m)}(s) = \psi R + \alpha_g - \gamma\,[P(n') + H(n')\,D] - k\,c^{\mathrm{rem}} - m\,\underbrace{(1-s)\,c^{\mathrm{inst}}}_{\text{subsidized install}},$$

and the government pays the remaining $s\,c^{\mathrm{inst}}$ on each installed tank, the tanks of new facilities included. That payment is the one direct outlay in the accounting below.

The removal mandate. The third experiment imposes a rule: a single-walled tank past twenty years must be removed. Twenty is not arbitrary. It is where the single-walled release hazard peaks (Section 2.3), and filed premiums imply that replacement beats deferral for single-walled systems near that age (Yin, Kunreuther, and White 2011). A regulator writing an age rule on the available evidence would land close to it, and the cost reported below is driven by what a fixed age cutoff does rather than by where it is set, because a mandate at any age retires tanks on age alone. The mandate changes no payoff and shrinks the choice set: a facility holding a single-walled tank past the threshold cannot maintain, and its feasible set narrows to the actions that clear every such tank,

$$D^{\mathrm{mandate}}(s) = \{\, d \in D(s) : n'(d) \text{ holds no single-walled tank past age twenty} \,\},$$

met by downsizing or replacing at the usual adjustment costs.

The first-best premium. The fourth experiment is a benchmark rather than a policy a state has run. Every facility faces a risk-rated premium equal to the marginal social cost of its own leak risk,

$$P^{\mathrm{fb}}(a,s) \;=\; h_a(s)\,[\,S_{aw}(a,s) + \bar{S}_{>L} + E_{\mathrm{ext}}\,],$$

the post-action hazard times the expected cleanup the insurer pays, the expected loss above the cap, and the damage a leak does beyond the cleanup, E_{ext}. The premium is indexed by the composition the action leaves behind, and replacing a tank therefore lowers next period's premium while exiting drives it to zero. The premium works as an incentive rather than a lump sum. The deductible is unchanged, because the facility already bears it in the payoff and the fitted severity is net of it. The one term the model does not estimate is E_{ext}. I set it to the infant-health damage Marcus (2021) attributes to a leak, \$25,600 per leak in 2023

dollars, or 2.56 model units.[2] Marcus counts infant health only and names what she omits, adult health, cancer, property values, and unreported leaks. The benchmark therefore prices a lower bound on the damage.

6.3 The welfare accounting

A contract changes welfare only through the choices it changes. Premiums and fund fees move money between a facility and its insurer, and they cancel in a social total. The resources that do not cancel are the operating value facilities earn, the costs of removing and installing tanks, the cleanups, and the damage from leaks. I measure net welfare as the present value of those resources per facility in the observed stock, for the facilities in the ground at the start and for the facilities that open afterward,

$$
\begin{aligned}
W &= W^I + W^E, \\
W^I &= PS + \text{Insurer} - \text{Gov} - \text{FirmAboveCap} - \text{External}, \\
W^E &= PS^E - \text{EntryCost} + \text{Insurer}^E - \text{Gov}^E - \text{FirmAboveCap}^E - \text{External}^E.
\end{aligned}
$$

Producer surplus PS is the facility's value function stated in premium-equivalent dollars. Insurer is premium revenue less expected claims. A claim is the fitted cleanup cost of a leak, up to the one-million-dollar coverage limit. FirmAboveCap is the expected cleanup cost above that limit, borne by the facility, and Gov is government outlay, nonzero only under the subsidy. External is the damage a leak does beyond the cleanup, charged at \$25,600 per leak under every contract, the baseline included. The entrant block W^E applies the same flows to the facilities that open, with PS^E the value of a new facility and EntryCost the entry cost plus the install cost the entrant pays on its tanks. Section D.5 gives the formula for each term. Section 5 notes that γ understates the response to a premium if facilities shop across carriers. In that case the dollar magnitudes below are overstated, and the rankings are more reliable than the levels.

The entrant block leaves two things out. The ledger has no consumer side, and the value of a new facility to the drivers it serves is not counted. An entrant is booked at the mean value of a new facility less the cost of opening it, and the private shock that makes entry worthwhile for the facilities that do open is not counted either. Both omissions lower the measured value of entry. At the estimated entry cost the mean value of a new facility is below the cost of opening one in every environment, and each entrant therefore lowers measured welfare under every contract.

6.4 Welfare results

Table 6 ranks the risk-rated baseline and the three policy experiments against the first-best benchmark, each in present value per facility, counting incumbents and entrants, and each

[2] Marcus (2021) reports total damages over the leaks in her sample without stating a dollar base year. Mortality is 75.5 percent of the total and is valued at the EPA's value of a statistical life on its 2006-dollar convention. I therefore deflate the per-leak figure from 2006 to 2023 by a factor of 1.5114. Every other dollar in the premium is already in 2023 dollars.

charged the same infant-health damage per leak. The gap to the first best is first-best net welfare less the row's net welfare, and the share below the first best is that gap as a percent of first-best net welfare. Among the contracts a state has run the risk-rated baseline ranks highest, 57.8 percent below the first best, followed by the uniform flat fee at 62.2 percent, the replacement subsidy at 66.7 percent, and the removal mandate at 70.5 percent. Table 23 in Section D.6 splits each row into its incumbent and entrant blocks and reports the totals with larger pools of potential entrants. The risk-rated contract ranks first at every pool size.

Table 6: Counterfactual welfare ranked against the first best

Scenario	Net welfare	Gap to first best	Below first best (percent)	Rank
First best	56,868	—	—	—
Risk-rated pricing (baseline)	23,981	32,887	57.8	1
Uniform flat fee	21,472	35,397	62.2	2
Replacement subsidy	18,953	37,916	66.7	3
Age-20 removal mandate	16,759	40,110	70.5	4

Notes: Present value of net welfare in 2023 dollars per facility in the observed stock, counting incumbents and entrants, with the pool of potential entrants at its baseline size. Each row holds the estimated parameters fixed, changes one policy, and re-solves the facility's problem and the entry rule, at an annual discount factor of 0.95. Net welfare is the identity of Section 6: producer surplus plus the insurer's position, less government outlay, the cleanup cost above the coverage limit, and the external damage, for incumbents and for entrants, with the entrants' producer surplus net of the cost of opening. Every row, the first best included, charges the same infant-health damage of 25,600 dollars per leak. The gap to the first best is first-best net welfare less the row's net welfare, and the next column states that gap as a percent of first-best net welfare. Rank orders the baseline and the three policy experiments by net welfare, with 1 the closest to the first best.

The uniform flat fee ranks below the risk-rated contract, $2,510 per facility lower in net welfare. The replacement subsidy and the removal mandate both rank below the flat fee. A subsidy pays for replacements that would have happened without it. A mandate retires sound tanks along with risky ones, and it ranks last, $7,223 per facility below the risk-rated contract. A price lets each facility choose the margin it adjusts, and the mandate chooses one margin for every facility.

The first-best premium raises net welfare from $23,981 under the risk-rated contract to $56,868 per facility. For incumbents the insurer's position turns positive under the first best, since the premium collects the above-cap loss and the external damage that the baseline leaves with facilities and third parties, and their net welfare then equals producer surplus (Section D.5).

The annual facility exit rate rises from 1.6 percent at baseline to 12.9 percent under the first best, and the annual tank closure rate rises by a factor of 6.7. Both are one-year rates on the observed stock, before the stock adjusts to the premium. The reform Section 4 measures raised tank closures by 1.3 points a year on a control base of 2.71 percent. Set against

the first best, the Texas repricing moved a small fraction of what full internalization would move, because the filed schedule prices the cleanup the insurer expects to pay and not the loss above the cap or the damage beyond it.

7 Conclusion

A compulsory price of coverage that carries no information about risk holds the riskiest capital in service. This paper measured that retention, and the effect of ending it, using Texas's 1998 fund closure, which moved 21,658 facilities from a flat per-tank fee to risk-rated private insurance at once. Closures of observably identical tanks rose 1.3 points a year (s.e. 0.4) on a 2.71 percent control base, and the rise was steepest among the oldest tanks, which the filed schedules charge most, at 4.1 points a year for tanks 21 years or older at the reform. Small and old facilities exited, and large facilities downsized and replaced instead.

The structural model, estimated on the filed prices and on the choices facilities make under the two contracts, ranks the risk-rated contract first at every size of the entrant pool. Facilities pay about $89,142 in full project cost to remove a tank, and that cost is why facilities move rarely. Counting incumbents and entrants, the risk-rated contract comes closest to a first-best premium that prices both the cleanup cost and the health damage of a leak, 57.8 percent below it. The uniform flat fee is 62.2 percent below the first best, a replacement subsidy of 50 percent of the install cost is 66.7 percent below, and an age-twenty removal mandate is 70.5 percent below. A subsidy pays for replacements that would have happened without it. A mandate retires sound tanks along with risky ones.

Five boundaries limit how far these results carry. Texas is one treated state, and the inference rests on a single treated cluster and its 21 control states rather than on repeated treatment. The setting operates on the extensive margin, because an installed tank's wall cannot be changed, and the elasticities here need not transfer to hazards where care is continuous. The model prices the reform Texas ran, holding filed prices fixed and letting facilities enter without responding to one another, rather than the equilibrium a national repricing would produce. The welfare ledger counts no benefit of a new facility to the drivers it serves, and the ranking of the risk-rated contract over the flat fee rests on how entry is valued. And the dollar levels lean on measured inputs: a fitted mean severity transferred from other states' claims, and a coverage-cost weight that understates the response to a premium when facilities shop across carriers, whose division makes the levels high-side even where the ranking is not.

The choice Texas made in 1998 is the choice the remaining fund states face now. Of the 36 state funds the EPA counts as financial-responsibility mechanisms today, six carry a statutory sunset, a claim cutoff, or a fee repeal dated between 2028 and 2030 (EPA 2026b).[3] A fund that closes repeats the Texas decision with better information than Texas had. For those states the operative finding is the ordering. Pricing risk retires the riskiest capital at a lower cost than commanding its retirement, and a fund that keeps its uniform fee keeps the retention that fee finances. The result extends beyond tanks. Wherever compulsory

[3]The six are Nebraska, Utah, Vermont, Illinois, Washington, and Missouri. Legislatures have extended such dates before, and each of these can be extended as well.

coverage is met with uniform pricing, the same retention operates, and the price of coverage is a policy instrument whether or not a state uses it as one.

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A Data Appendix

A.1 Sources

Tank registries. The panel begins from the EPA's UST Finder, the compendium that harmonizes every state's tank registry into one national schema. That compendium leaves some states missing fields the design needs. I rebuild the registry for seven states directly from the state agency's own database rather than the compendium extract: Texas, Arkansas, Colorado, Louisiana, Maine, Oklahoma, and Tennessee. Every rebuilt registry still reports the five fields the design requires: install date, closure date, wall type, fuel, and capacity. Texas records one field none of the other six carry, the carrier writing each facility's financial-responsibility coverage in each year. That field maps a facility's tanks to the rate filing that priced them.

Rate filings. Section A.6 documents the six carrier rate engines built for Texas, the licensing rule that bounds their coverage, and the imputation applied where no filing survives.

Premium coverage and imputation. I reconstruct each Texas facility's premium from six filed rate manuals covering the largest carriers in the market. Those manuals price 70.9 percent of insured Texas facility-years from 2008 through 2020. Two limits bound that coverage. A carrier writing without a Texas license files no rate manual with the state. That carrier leaves no schedule for me to recover. A tail of small carriers writes the remainder, each holding well under one percent of facility-years. I impute a premium for every unpriced facility-year and flag the imputation rather than drop the facility.

Selecting the largest carriers determines what the premium variable measures. These carriers write most of the market. The schedules I recover therefore capture what a typical Texas facility paid to hold a tank one more year. The unpriced facilities do not form a random subset. They concentrate in the later years of the panel, and reconstructed coverage thins after 2014. Section A.6 documents the licensing rule, the carrier counts, and the imputation.

Trust fund claims. I collect facility-level cleanup claims from state remediation trust funds, covering 6 states. Appendix C states which cost field each fund reports and the rule that harmonizes them across states.

Release records. Every release record carries a report date, a facility identifier, and a release identifier. Section A.3 states the rule that classifies each release as standalone, revealed by a closure, or already known at the time of a closure.

Fund fees. For the control states I use the ASTSWMO annual State Fund Survey. The survey reports each state's uniform per-tank fee as raw text rather than a parsed number. Texas's pre-reform entry states a combined figure for both tank types, one dollar amount for above-ground tanks and another for underground tanks, and I take the amount tied to the underground-tank label rather than the first dollar figure the string reports.

EPA performance measures. Every state reports to the EPA twice a year the percentage

of its recently inspected tank facilities found in compliance with the federal technical standards, under one standard through fiscal 2021 and a differently defined one from fiscal 2019. Section B.10 builds the enforcement comparison from the complement of that percentage.

A.2 Sample construction

Three fields decide whether a state can enter the sample at all: a tank's install date, its closure date, and its wall type. The two dates place the tank in event time. Wall type, together with age, defines the risk class the market prices. State registries do not report these three fields consistently. I therefore begin with the 42 states that operated a uniform per-tank fund and keep the 22 whose registries report all three. That screen leaves Texas and 21 control states. I then restrict those registries to incumbents, the facilities holding at least one active tank on December 23, 1998. The reform repriced this stock. Tanks installed later entered a market that was already risk-rated and carry no before-and-after. Table 8 reports the sample remaining after each restriction.

Table 7 reports the registry audit behind that restriction, state by state. Each row gives the state, the provenance group its registry entered the panel through, and the share of tank records missing three of the fields the tank-level design needs: the closure date, the install date, and the wall type. A state's registry enters through one of two documented routes, a bespoke cleaner reading the state's own database or the EPA compendium, or it sits outside both routes for the reasons Section A.1 gives. Heavy missingness on any one of the three fields removes a state from the tank-level design regardless of which route its registry took.

Table 7: Registry field completeness by state

		Percent missing					Percent missing		
State	Group	Closure	Install	Wall	State	Group	Closure	Install	Wall
TX	Target (TX)	0.0	0.0	9.9	FL	Excluded (Treated)	100.0	28.8	100.0
AL	Control Tier 1 (Custom)	9.4	24.1	85.0	IA	Excluded (Treated)	0.0	10.4	1.9
AR	Control Tier 1 (Custom)	7.7	18.5	21.7	MI	Excluded (Treated)	0.2	15.9	10.8
CO	Control Tier 1 (Custom)	6.9	21.6	24.7	WI	Excluded (Treated)	100.0	79.7	59.3
LA	Control Tier 1 (Custom)	0.0	0.1	33.7	OR	Excluded (Other)	2.4	18.6	100.0
ME	Control Tier 1 (Custom)	0.6	0.0	0.2	WA	Excluded (Other)	93.7	7.1	100.0
MO	Control Tier 1 (Custom)	26.0	35.3	0.3	AK	Remainder	100.0	15.0	100.0
ND	Control Tier 1 (Custom)	1.2	6.7	0.0	CA	Remainder		2.6	0.2
NJ	Control Tier 1 (Custom)	4.3	0.1	0.5	DE	Remainder	8.6	18.1	100.0
NM	Control Tier 1 (Custom)	2.5	15.9	100.0	GA	Remainder	100.0	3.3	0.0
OK	Control Tier 1 (Custom)	11.1	13.2	0.0	HI	Remainder	100.0	19.4	100.0
PA	Control Tier 1 (Custom)	20.0	17.4	16.1	IL	Remainder	6.4	48.4	100.0
TN	Control Tier 1 (Custom)	10.8	0.6	9.6	IN	Remainder	100.0	53.3	100.0
UT	Control Tier 1 (Custom)	0.1	11.8	14.9	MS	Remainder	100.0	100.0	100.0
ID	Control Tier 2 (EPA Std)	0.8	2.2	0.0	MT	Remainder	4.7	9.9	12.2
KS	Control Tier 2 (EPA Std)	16.5	9.4	0.0	NE	Remainder	100.0	3.8	99.7
KY	Control Tier 2 (EPA Std)	5.2	18.0	0.0	NH	Remainder	100.0	1.7	100.0
MA	Control Tier 2 (EPA Std)	5.7	8.2	45.7	NV	Remainder	100.0	1.8	0.0
MD	Control Tier 2 (EPA Std)	15.5	18.1	0.0	NY	Remainder	100.0	30.0	100.0
MN	Control Tier 2 (EPA Std)	6.8	22.3	0.0	RI	Remainder	100.0	1.4	78.0
NC	Control Tier 2 (EPA Std)	0.3	1.4	8.2	SC	Remainder	100.0	100.0	100.0
OH	Control Tier 2 (EPA Std)	3.0	28.4	0.0	SD	Remainder	80.0	9.1	0.0
VA	Control Tier 2 (EPA Std)	26.7	14.1	0.0	VT	Remainder	0.5	100.0	100.0
AZ	Excluded (Treated)	0.3	15.4	37.9	WV	Remainder	100.0	16.4	0.0
CT	Excluded (Treated)	26.2	4.7	81.1	WY	Remainder	100.0	13.0	100.0

Notes: The three numeric columns are in percent. Closure is the percent of tanks recorded as closed that have no closure date, and it is blank for a registry that records no closed tank. Install is the percent of tank records with no install date, and Wall is the percent with a missing or unknown wall type. These are the three fields the tank-level design requires. All three are computed from the state's whole harmonized registry rather than from the study sample. Group is the study-group label from the registry build. Control Tier 1 (Custom) marks a state whose registry is read from the state's own database, and Control Tier 2 (EPA Std) marks a state whose registry comes from the standard extract of the Environmental Protection Agency (EPA) compendium. Target (TX) is Texas, Excluded (Treated) and Excluded (Other) mark states the build sets aside from the control tiers, and Remainder is every other state. Group is a provenance label and not a statement of whether a state enters the estimation sample, and Table 8 gives the sample restrictions. States are sorted by group and then by postal code, and the list continues in the right-hand block. Source: the registry completeness audit, Code/Verification/state_design_coverage.R.

Table 8: Sample construction

Restriction	Facilities	Tanks	Facility-years	Dropped
Harmonized state registries, all tanks	761,106	2,326,539	—	—
States in the study sample (Texas and 21 control states)	361,817	1,081,742	—	1,244,797 (53.5%)
Tanks with a recorded installation date	318,123	956,274	—	125,468 (11.6%)
Tanks with a recorded wall type, fuel type and installation cohort	296,475	884,190	—	72,084 (7.5%)
Tank-years in the annual panel, 1985–2020	289,600	856,900	5,312,231	27,290 (3.1%)
Drop tank-years that install and close in the same year	286,728	846,853	5,309,027	10,047 (1.2%)
Tanks installed before the 1998 reform	266,937	768,979	4,931,303	77,874 (9.2%)
Facilities holding at least one tank active at the reform	131,063	464,474	3,784,674	304,505 (39.6%)
Birth-cohort coarsened-exact match on build, vintage and state	128,484	453,718	3,704,963	10,756 (2.3%)
Panel years 1990–2015 (North Carolina to 2012)	128,484	434,604	2,788,928	19,114 (4.2%)
Estimation sample: facility-year difference-in-differences	127,987	434,604	2,776,307	0 (roll-up)

Notes: Each row applies one restriction to the row above. Dropped reports the tanks removed by that row and the share of the tanks in the row above. Facility-years are undefined before the registry is expanded into an annual panel and are shown as dashes in the first four rows. A blank closure date is treated as an open tank and right-censored at 2020. No tank is dropped for a missing closure date. States whose registries never record closures contribute no closure events. The wall-type row also removes tanks with an unrecorded fuel type or installation cohort. Wall type, fuel type, and installation cohort enter the sample definition together. The final row removes no tanks: it aggregates the matched tank roster to facility-years, the unit of the regression. The panel-years row keeps the years every estimate uses. A tank drops there when none of its panel years falls inside them. North Carolina's registry stops recording closures around 2012. Its years end there.

A.3 Variable definitions

The closure decision and the portfolio decision happen at different levels, and the analysis carries a panel for each. The tank-year panel records every tank in every year it sits in the ground and measures closure at the finest resolution the registries support. The facility-year panel aggregates those tanks into a portfolio and measures the decisions an operator makes for the site as a whole, including replacement, consolidation, and exit.

Closure. I code the tank-level outcome as an annual closure indicator. It takes the value one in the calendar year matching the tank's recorded permanent-removal date and zero in every other year the tank appears in the panel. For tank i, let t_i^{close} be that year. The indicator is

$$\text{closure}_{it} = \mathbf{1}\{t = t_i^{\text{close}}\}.$$

A tank with no recorded closure date stays in the panel through 2020 with the indicator at zero throughout. I treat a tank still in service at the end of the panel as right-censored rather than as one that never closed. The facility-level indicator takes the value one in any year when at least one of the facility's tanks records a closure. Let c_{it} be the number of tanks facility i removes in year t, and let K_{it} be its total capacity. The facility indicators are

$$\text{any_closure}_{it} = \mathbf{1}\{c_{it} \geq 1\}, \qquad \text{cap_cut}_{it} = \mathbf{1}\{K_{it} < K_{i,t-1}\}.$$

That distinction matters because the average facility operates 3.53 tanks and can retire one while continuing to trade on the others.

Replacement. I then separate closures by what the facility does next. I classify a tank closure as a replacement when the facility installs at least one new tank on a date later than that closure. The rule compares each closure date against the latest install date recorded anywhere in the facility's history. A tank closed in 2003 and followed by an installation in 2011 therefore counts as a replacement. Tanks closed in the same calendar year they were installed are excluded before the classification runs. A closure that fails the test counts as permanent.

Contraction margins. The remaining margins compare the facility's tank count and total capacity against the previous year. A facility contracts in a given year when it does not exit that year, ends the year holding strictly fewer tanks than it held the year before, and held positive capacity in the previous year. Within contracting facility-years I separate three cases by the proportional change in capacity,

$$g_{it} = \frac{\Delta C_{it}}{C_{i,t-1}} = \frac{C_{it} - C_{i,t-1}}{C_{i,t-1}}.$$

A facility consolidates when capacity stays within five percent in absolute value, the case of three tanks giving way to one tank of comparable total volume. It downsizes when capacity falls by more than five percent. It reconfigures upward when capacity rises by more than five percent, the case of several small tanks giving way to one substantially larger one. The three cases are

$$\text{downsize} : g_{it} < -0.05, \qquad \text{consolidate} : |g_{it}| \leq 0.05, \qquad \text{reconfigure up} : g_{it} > 0.05.$$

Exit and censoring. Consolidation earns its own margin because carriers price per tank. Cutting the number of tanks cuts the premium even when the facility keeps every gallon of capacity. A facility exits in the last year it appears in the panel, provided that year falls strictly before 2020. Let t_i^{last} be the last year facility i operates any tank. Exit is

$$\text{exit}_{it} = \mathbf{1}\{t = t_i^{\text{last}} \text{ and } t_i^{\text{last}} < 2020\}.$$

That proviso is the censoring rule. I code the last row of a facility still operating in the panel's final year as zero rather than as an exit.

Releases. I attach the release records to the panel to track pollution outcomes. Each release enters with a report date and a facility identifier. A tank closure requires an excavation. The excavation doubles as an inspection and can surface contamination already in the ground. I therefore classify each release by the number of days between its report date and the nearest tank closure at the same facility. I treat a release reported on the day of a closure, or within the sixty days following it, as revealed by that closure. I treat a release reported in a facility-year with no closure as standalone. I treat a release reported more than one hundred eighty days before a closure as already known at the time of that closure rather than revealed by it. I construct the same three-way split at thirty, forty-five, and ninety days as a check on the window.

A.4 Pre-reform covariates

Every covariate in the heterogeneity analysis is measured at the reform and held there, because the reform itself changes local market structure. A competitor count measured after 1998 would register an outcome of the treatment, and conditioning on it would absorb part of the effect the design estimates. Fixing each covariate before the treatment date keeps it a pre-treatment trait.

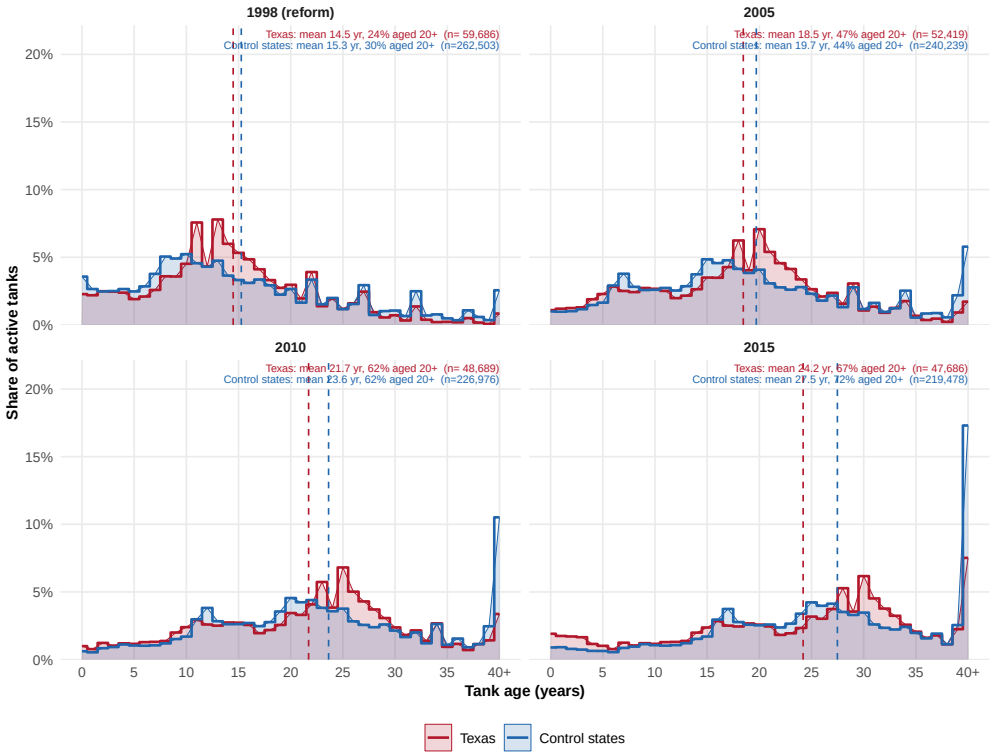
I place each facility in its 2000 Census tract by matching its coordinates to the tract polygon containing it, and every tract measure comes from that tract. Four indicators describe the neighborhood. A facility is rural when more than half its tract's population is rural, under the Census 2000 urban-rural classification. It is low-population when its tract's population density, in people per square mile, sits in the bottom third of the sample. It is low-income when its tract's median household income, in 1999 dollars, is below the sample median. It is high-poverty when its tract's poverty rate is above the sample median. One indicator describes the local market. For each facility I count the other gasoline-retailing facilities whose coordinates fall within one mile, or 1,609 meters, restricting the count to those holding at least one active tank on December 23, 1998. A facility is in a thin market when that count is one or fewer, and a retailer facing at most one nearby competitor can pass a cost increase through without losing customers.

Every covariate here enters only as a split of the treatment effect, never as a control in the headline regression.

A.5 Age distributions at the reform

Figure 6 plots the age distribution of the active tank stock in Texas and in the control states at the reform date and at three later snapshots, for the matched estimation sample. The two groups hold a comparable stock at the reform, and the post-reform divergence appears as old tanks leaving the Texas distribution rather than as new tanks entering it.

Figure 6: Age distribution of the active tank stock, Texas and control states, at four snapshots



Notes: Share of active tanks at each single year of age, with ages of 40 and above grouped in the last bin. Age is the calendar year less the install year. The panels show the stock in 1998, the year of the reform, and in 2005, 2010, and 2015. Shares sum to one within each group and panel. Texas is red and the control states are blue, and dashed vertical lines mark each group's mean age. The text in each panel gives each group's mean age in years, the percent of its tanks aged 20 or more, and the number of active tanks. The sample is the matched sample of Table 8, in Texas and the control states. A tank is not counted in the year it is removed.

A.6 Carrier rate engines

For Texas I price each facility from the carriers' filed rate manuals, collected from the Texas and Iowa electronic rate-filing systems and linked to facilities through the Texas Commission on Environmental Quality contract records that name each facility's carrier. Six carriers have recoverable filings: Mid-Continent, Tank Owners Members (TOMICS), Great American, Zurich American, ACE, and Commerce and Industry (AIG). Together they price 70.9 percent of insured Texas facility-years. The main gap is two excess-and-surplus-lines carriers, Ironshore Specialty and Colony Insurance. Neither makes an admitted-market rate filing, and no schedule can be reconstructed for either. A facility insured by one of these keeps its contract record but not a reconstructed premium, and its premium is imputed as the priceable-market mean for its age, wall, and era cell and flagged as imputed.

Every manual levies the premium per tank and loads it on the tank's age and its wall construction, and beyond that the six differ more than one description can carry. Three of them set the base rate on age directly. TOMICS sets its base on install vintage, keyed to the 2006 Energy Act cutoff, and carries age as an additive load on top. Mid-Continent and Great American hold the base flat and move age through a load or a factor. The double-wall credit is the one feature common to all six, though the mechanism varies: Zurich keeps separate single-walled and double-walled base tables, ACE bundles double-wall with clad steel under a single construction code, and Mid-Continent's credit requires double-walled steel. A double-walled fiberglass tank earns nothing under that credit. Only two of the six price the gallons a tank holds. That scarcity is why cutting capacity within a tank is the one adjustment margin the schedules largely ignore.

Rebuilding a premium means fixing three contract terms the records do not report. The first is the coverage limit. I hold every facility at the federal reference of one million dollars per occurrence and in aggregate, and the premium is therefore the price of the mandated floor rather than of any higher limit a firm bought. The second is the deductible, and I hold every facility at the filing's reference deductible of five thousand dollars. The third is the underwriter's schedule credit or debit, the discretionary adjustment for firm-specific risk. I cannot observe it. I price each facility at the schedule-neutral rate and report that as the standard premium, and I report a minimum and a maximum by moving the schedule to the edges of its filed band. Those bands are not uniform. Five of the six carriers apply a band of forty percent in each direction. Great American files no schedule rating at all, and its minimum, standard, and maximum coincide. Every carrier applies a minimum policy premium as a floor, and those floors run from two hundred and twenty-five to five hundred and twenty-five dollars across carriers. Weighted across carriers and deflated to 2023 dollars the floor is $563 per policy-year, or $219 per tank-year split across a facility's tanks, and it binds for 15.0 percent of Texas facilities.

The manuals are filed as countrywide rate pages adjusted to each state, and I scale a national age-and-wall schedule to each state's level by Harwayne's method (Boor 1996). I assign the rate era from the policy effective date. I read each rate load from the facility's records: tank age, wall construction, corrosion protection, overfill protection, stored product, and prior release. I set any load the records do not report to the filing's reference tier. Zurich files its manual in Iowa, and I apply it to Texas facilities as an Iowa-rate proxy with a neutral territory factor. Each engine is coded from the filed factor tables, with every table read twice and reconciled before it is encoded, and each carries a self-test that reproduces hand-computed premiums from those tables. One filing required a departure from its printed text. Great American prints its composition rule as a sum, but the same filing states that its rates derive from ACE's program, ACE's manual states that the adjustments are multiplicative and ACE's own worked sheets confirm it, and the additive reading would leave Great American's age and construction factors inert. I price Great American multiplicatively. The resulting cards are stated on the same five-year age grid as the model's state space, and the carrier-by-year variation they preserve is the dispersion that identifies γ in Section 5.

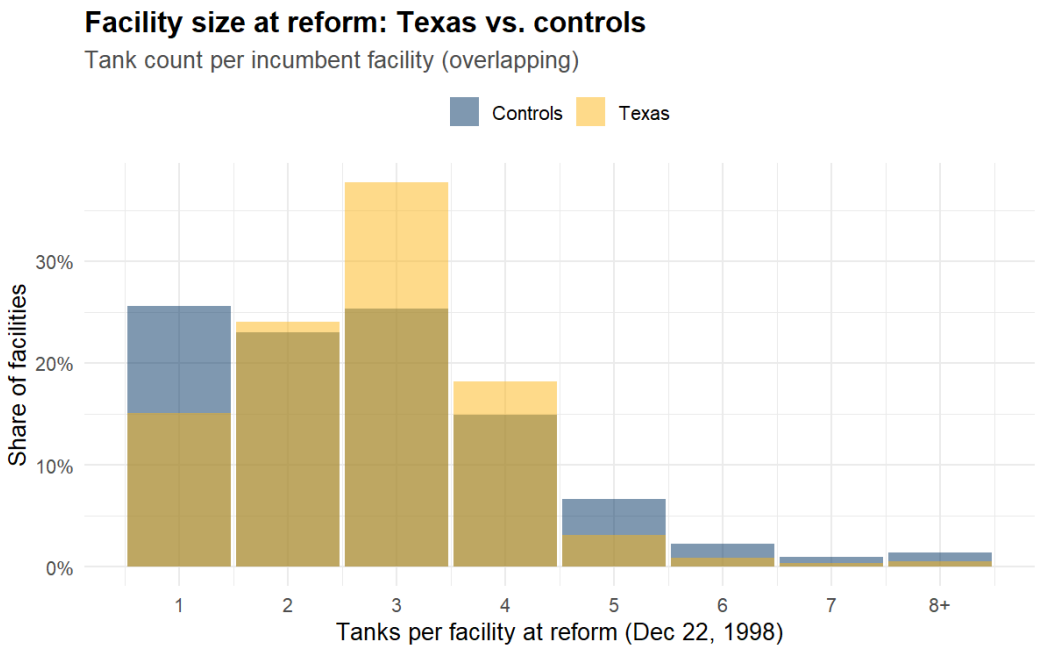
A.7 The stock over time

The exhibits below describe the estimation sample rather than the reform's effect. The first pair fixes the stock at the reform date. Figure 7 gives the distribution of active tanks per incumbent facility on the day the Texas rule took effect, and Figure 8 gives the distribution of total storage capacity per facility on the same date. Both are drawn for Texas and the control states together. The two groups enter the panel holding comparable portfolios, the condition the matched design requires.

The next four follow the same quantities through the panel. Figure 9 and Figure 10 plot tanks per facility and capacity per facility by year, and the Texas and control series separate after the reform rather than before it. Figure 11 plots the mean age of tanks at removal. That age falls in Texas after the reform, as facilities retire tanks earlier in their lives than the control states do. Figure 12 plots the raw annual closure probability, the unadjusted counterpart to the event study of Section B.7.

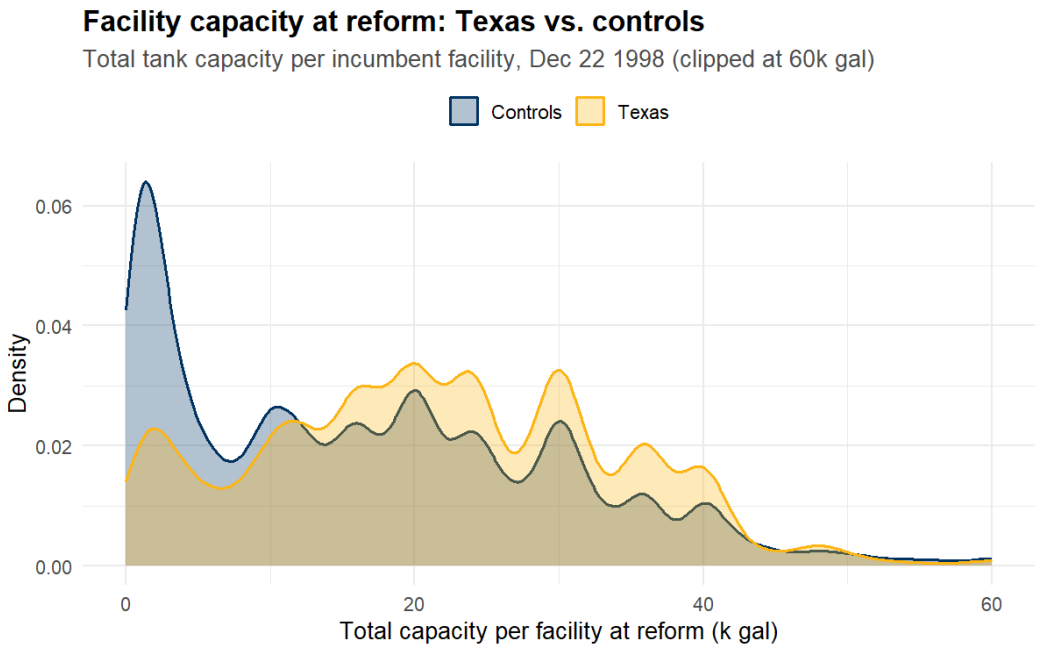
Four further exhibits belonging to this appendix are drawn elsewhere because the sections that argue from them own them. The cleanup-cost distribution is Figure 17 in Section B.2, and the premium exhibits Figure 31, Figure 30, and Figure 32 are drawn in Appendix C.

Figure 7: Tank count per incumbent facility at the reform date



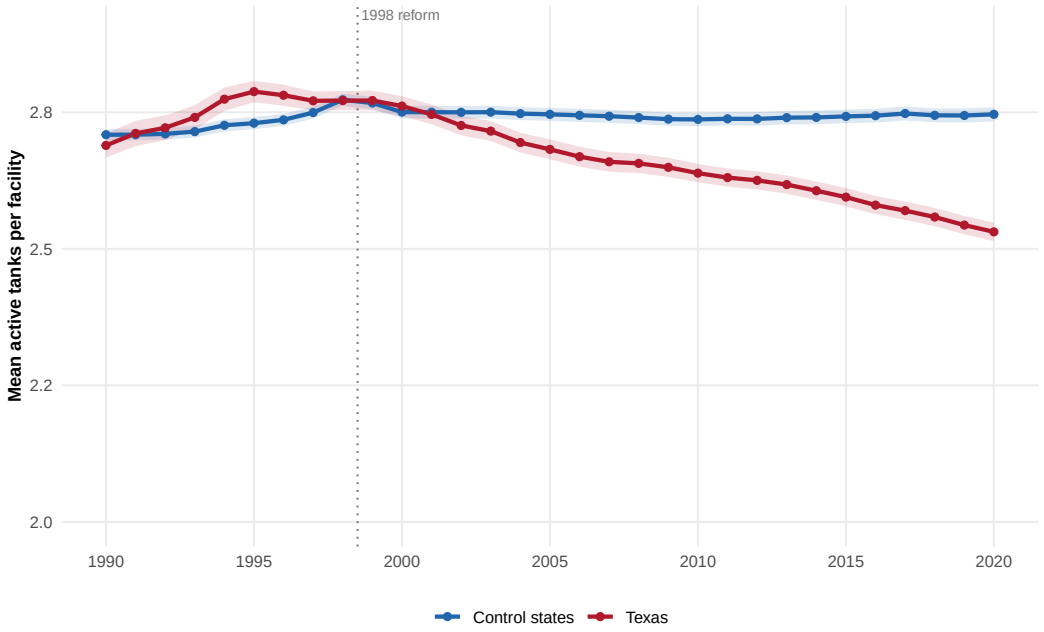
Notes: Distribution of the number of active tanks per facility on December 23, 1998, for incumbent facilities in Texas and the control states. Incumbent facilities hold at least one active tank at the reform, and the sample is every incumbent facility before matching. Each bar is the share of a group's facilities with that tank count, and the shares sum to one within each group. Counts of eight or more are grouped in the last bar. Texas is gold and the control states are blue, and the bars of the two groups overlap.

Figure 8: Total tank capacity per incumbent facility at the reform date



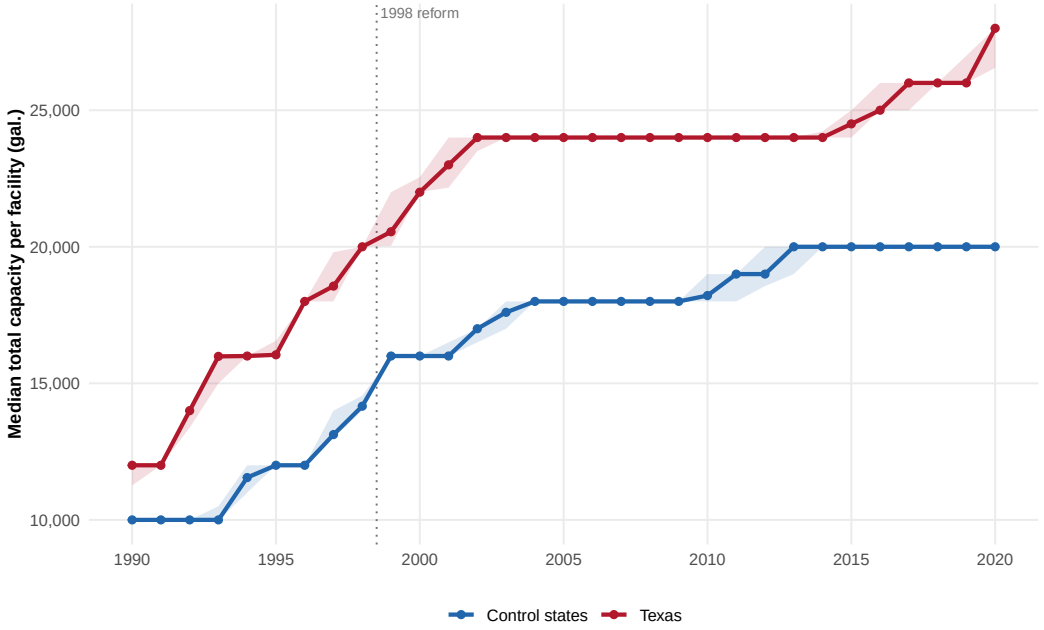
Notes: Distribution of total storage capacity per facility on December 23, 1998, in thousands of gallons. Curves are kernel density estimates and the vertical axis is density. Texas is gold and the control states are blue. The sample is the incumbent facilities of Figure 7, before matching, omitting facilities with zero capacity or more than 60,000 gallons.

Figure 9: Tanks per facility by year, Texas versus control states



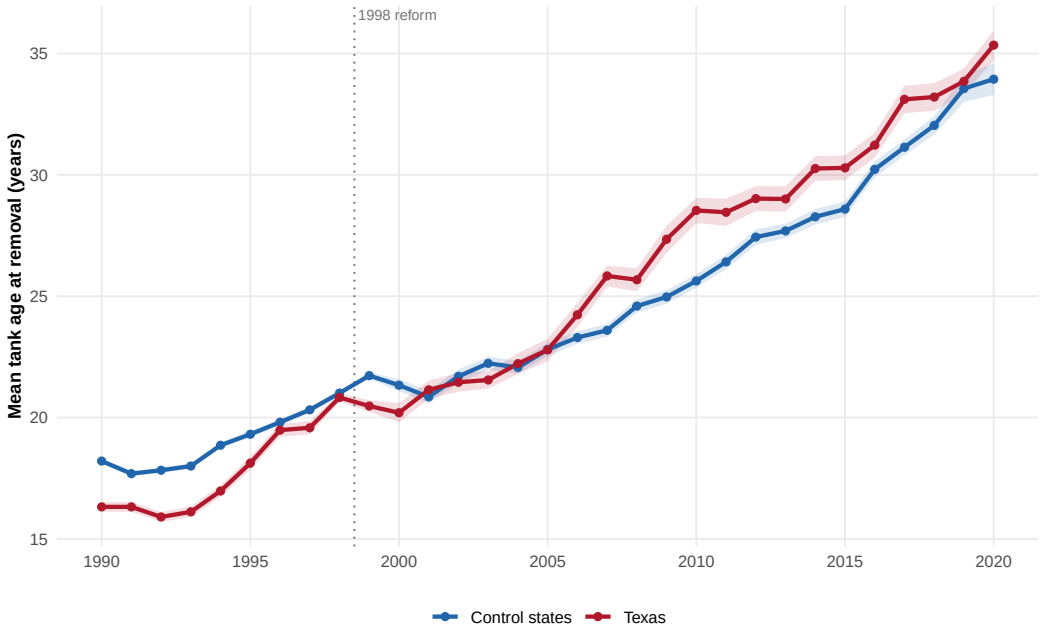
Notes: Mean number of active tanks per facility by year, 1990 to 2020, for Texas (red) and the 21 control states (blue). The unit of observation is the facility-year, and the sample is the unmatched annual facility-year panel of these states, entrants included. A tank is active in a year if it is in the ground at any point in the year, and tanks installed and removed in the same year are excluded. In 2019 and 2020, 10 control states record no closures, and North Carolina's registry stops recording closures around 2012. The dotted vertical line marks the 1998 reform. Shaded bands are 95 percent confidence intervals for the mean, computed with facility-years treated as independent. The vertical axis does not start at zero.

Figure 10: Median facility capacity by year, Texas versus control states



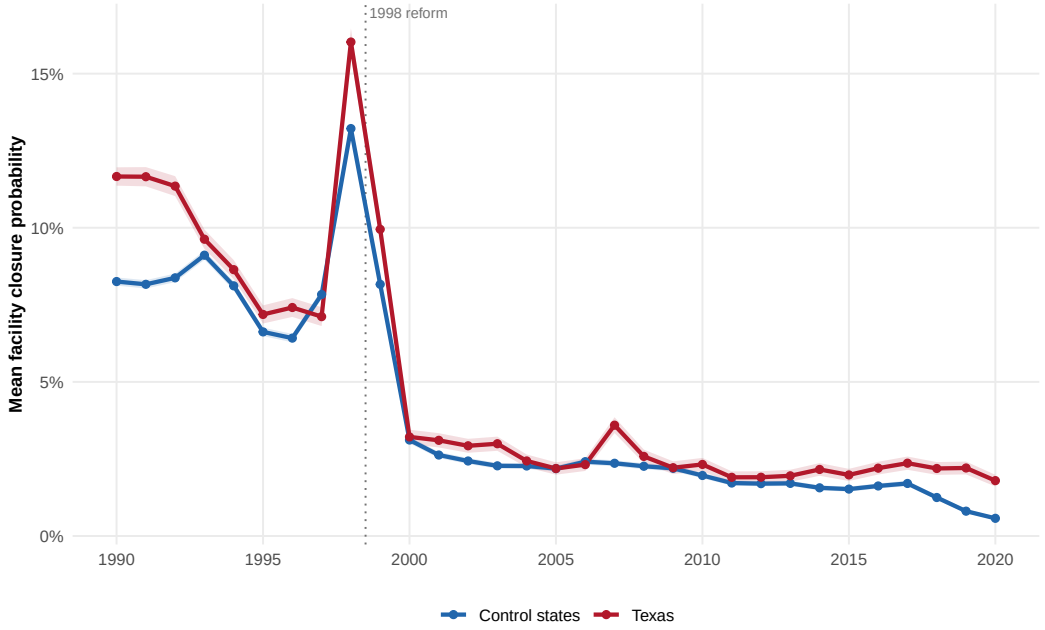
Notes: Median total tank capacity per facility, in gallons, by year, 1990 to 2020, for Texas (red) and the 21 control states (blue). The unit of observation is the facility-year, and the sample is the unmatched annual facility-year panel of these states, entrants included. A facility's capacity is the sum of the capacities of its tanks in the ground at any point in the year, excluding tanks installed and removed in the same year. The median is taken across the facilities operating in each year. In 2019 and 2020, 10 control states record no closures, and North Carolina's registry stops recording closures around 2012. The dotted vertical line marks the 1998 reform. Shaded bands are distribution-free 95 percent confidence intervals for the median, computed with facility-years treated as independent.

Figure 11: Mean age of tanks at removal by year, Texas versus control states



Notes: Mean age, in years, of the tanks removed in each year, by removal year, 1990 to 2020, for Texas (red) and the 21 control states (blue). Age is the removal year less the install year. The unit of observation is a tank removal, and the sample is the matched sample of Table 8. In 2019 and 2020, 10 control states record no closures, and North Carolina's registry stops recording closures around 2012. The dotted vertical line marks the 1998 reform. Shaded bands are 95 percent confidence intervals for the mean, computed with removals treated as independent.

Figure 12: Share of facilities closing a tank by year, Texas versus control states



Notes: Percent of facilities that close at least one tank in the year, by year, 1990 to 2020, for Texas (red) and the 21 control states (blue). The unit of observation is the facility-year, the sample is the unmatched annual facility-year panel of these states, entrants included, and tanks installed and removed in the same year are not counted as closures. In 2019 and 2020, 10 control states record no closures, and North Carolina's registry stops recording closures around 2012. The dotted vertical line marks the 1998 reform, the year of the federal upgrade deadline in every state. Shaded bands are 95 percent confidence intervals for the proportion, from the normal approximation with facility-years treated as independent.

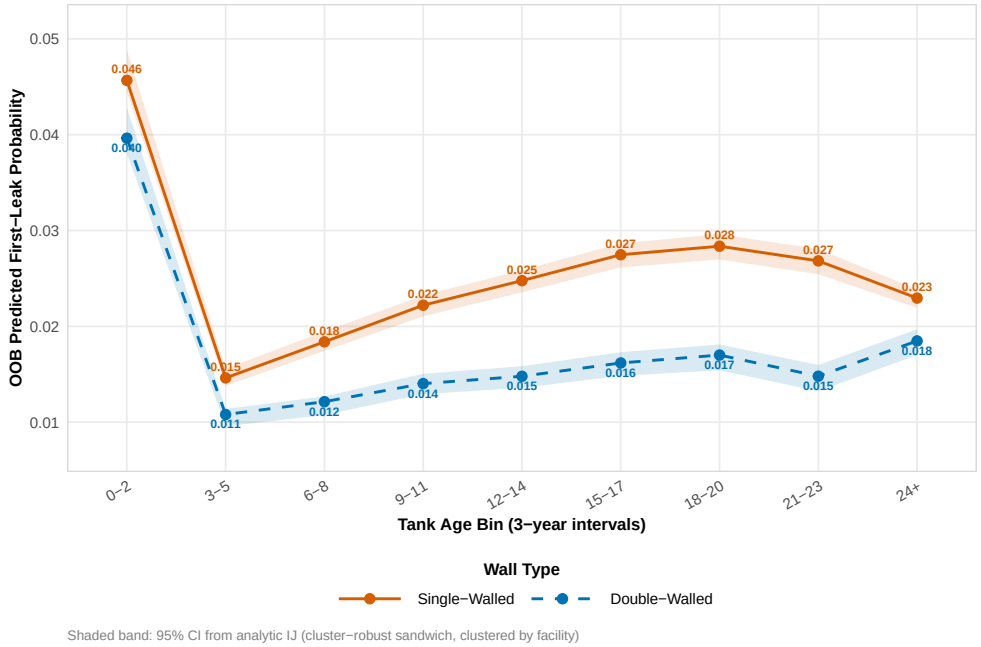
B Additional Results

This appendix holds the results the main text states but does not display. Section B.1 reports the full battery of checks behind the risk model's fit, cell by cell and facility by facility. Section B.2 draws the cleanup-cost distribution behind the pure premium, and Section B.3 reports the tank-level difference-in-differences estimates behind the static closure result. Section B.4 reports which facilities responded to the reform and where, and Section B.5 splits the closure response by county population growth. Section B.6 carries the static and event-study estimates behind the release response, and Section B.7 collects the remaining event studies by margin. Section B.8 reports the county entry analysis. Section B.9 gathers the placebo distribution and the formal sensitivity analysis behind the headline estimate, and Section B.10 reports the enforcement-intensity comparison that the design section raises as a threat to identification.

B.1 Risk prediction model

Figure 13 plots the out-of-fold predicted release probability from the risk equation of Section 2.3 by average tank age and wall construction, averaged within cells of three-year average-tank-age bins by wall construction, the cells of the observed rates quoted in Section 2.3. The predictions recover the observed profile cell by cell: the double-walled line rises with age, the single-walled line sits above it at every age, and the installation spike in the youngest bin appears in both. The fit statistics quoted in the main text, the AUC and the share of releases in the highest-risk decile, are computed from the same out-of-fold predictions.

Figure 13: Predicted release probability by tank age and wall construction

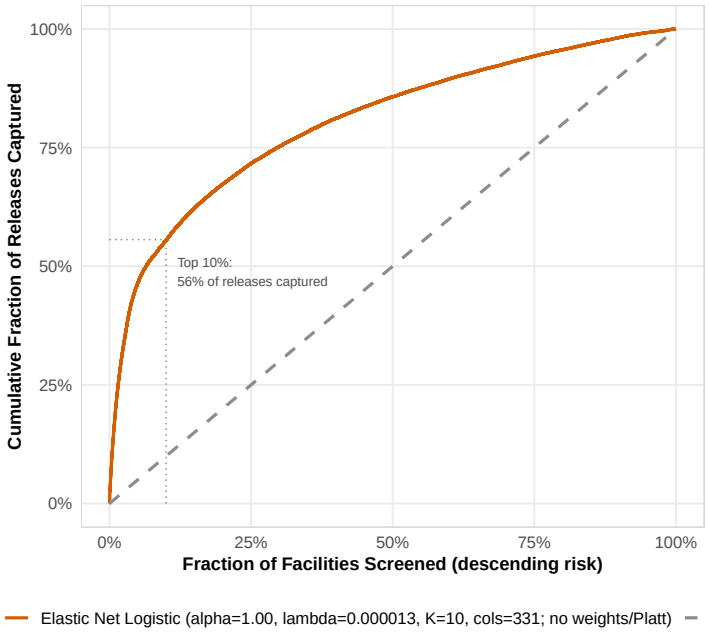


Notes: Each point is the mean out-of-fold predicted probability that a facility records its first confirmed release in the year, from the lasso logit of the risk equation of Section 2.3, within a cell of the facility's average tank age in three-year bins and wall construction. Values are in probability units, and 0.01 is one percentage point. The sample is facility-years before the facility's first confirmed release in 23 states, Texas from 1990 to 1998 and the control states from 1990 onward. The solid line is facilities holding at least one single-walled tank, and the dashed line is facilities holding none. Facilities in New Mexico, whose registry records no wall construction, are not drawn. No facility is scored by a model fit that saw it. Shaded bands are pointwise 95 percent confidence intervals, from the delta method applied to a logistic refit without the penalty, with standard errors clustered by facility.

This appendix reports the three checks behind that fit in full. Every predicted probability is cross-fitted, as in the main text, and no facility is scored by a model that saw it.

The first check is concentration. If the model carries signal, the facilities it flags as riskiest should hold a large share of the releases that actually occur. Figure 14 ranks every facility by predicted risk, from highest to lowest, and plots the share of next year's releases the screened facilities contain against the share of facilities screened. A model with no signal traces the diagonal, since screening any share of facilities at random catches that same share of releases. My curve rises well above it. The highest-risk tenth of facilities holds 55.6 percent of the next year's releases, and the highest-risk fifth holds 67.4 percent.

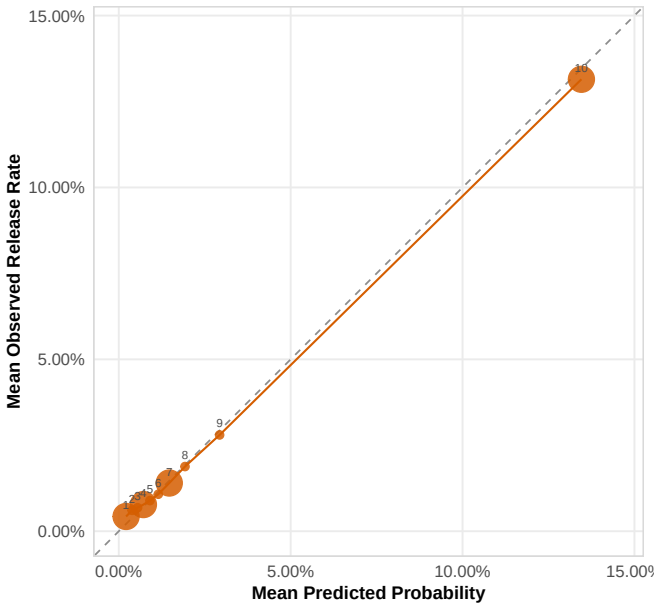
Figure 14: Risk concentration in the highest-ranked facilities



Notes: Out-of-sample concentration of risk. Facility-years are ranked by out-of-fold predicted probability of a first confirmed release from the cross-validated risk model, from the highest to the lowest, and the curve plots the cumulative share of first confirmed releases captured against the cumulative share of facility-years screened, both in percent. The 45-degree line is the no-information benchmark, and the dotted lines mark the highest-risk tenth of facility-years. The sample is 2,444,454 facility-years before the facility's first confirmed release in 23 states, Texas from 1990 to 1998 and the control states from 1990 onward. No facility is scored by a model fit that saw it.

A model can rank facilities correctly and still misstate how risky they are. The second check, calibration, asks whether the predicted probabilities match the rates facilities actually realize. Figure 15 sorts facility-years into ten bins by predicted score and plots each bin's average predicted probability against its observed release rate. A perfectly calibrated model places every point on the forty-five degree line, where predicted risk equals realized risk. My points sit close to it. In the riskiest bin the model predicts a 13.5 percent chance of a release, and those facilities leak 13.1 percent of the time.

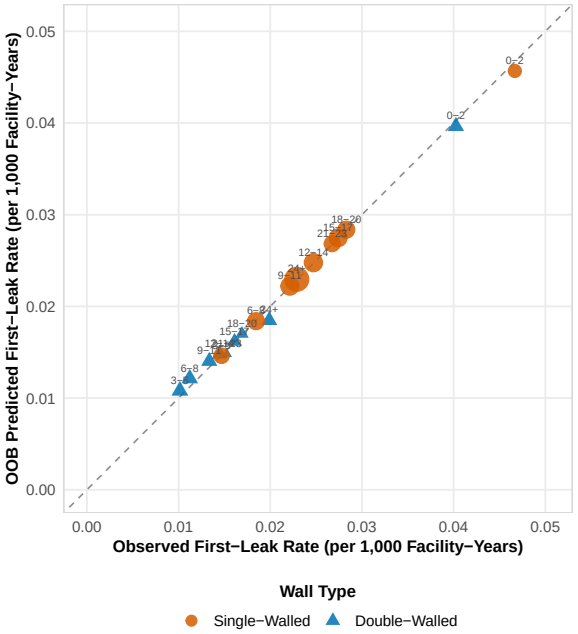
Figure 15: Calibration of predicted first-release probability



Notes: Out-of-sample calibration of the risk model. Facility-years are grouped into ten bins at the deciles of the out-of-fold predicted probability of a first confirmed release, and each point plots the bin mean of predicted first-release probability against the observed release rate in that bin, both in percent. The numbers label the bins from the lowest to the highest predicted probability, and the dashed line is the 45-degree line. The sample is 2,444,454 facility-years before the facility's first confirmed release in 23 states, Texas from 1990 to 1998 and the control states from 1990 onward. No facility is scored by a model fit that saw it.

The third check is the risk map itself. The model should reproduce the release rate in each age-and-wall cell, not merely rank facilities along it. Figure 16 plots, for each cell, the rate the model predicts against the rate actually observed. My cells fall close to the forty-five degree line. As a single summary of how well the model separates future leakers from non-leakers out of sample, the area under the ROC curve is 0.81, with a 95 percent interval of 0.811 to 0.815, where 0.5 is chance and 1.0 is perfect. Against a base rate of 2.37 percent the area under the precision-recall curve is 0.237, with an interval of 0.233 to 0.241. The model recovers the level of risk in each cell. A risk-based premium prices exactly that.

Figure 16: Predicted versus observed release probability, by age-and-wall cell



Notes: Out-of-sample fit of the risk model at the cell level. Each point is a cell of the facility's average tank age in three-year bins and wall construction, plotting the mean out-of-fold predicted probability of a first confirmed release in the year against the raw release rate observed in that cell. Both axes are in probability units, and 0.01 is one percentage point. Single-walled marks facilities holding at least one single-walled tank, and double-walled marks facilities holding none. Each label gives the age bin, and larger points have more facility-years. The dashed line is the 45-degree line. The sample is facility-years before the facility's first confirmed release, Texas from 1990 to 1998 and the control states from 1990 onward, excluding New Mexico, whose registry records no wall construction.

The youngest tanks carry a spike of their own. Tanks age zero to two hold 6.2 percent of the facility-years behind Figure 13 and 11.9 percent of its releases. I keep this band in the figure and in the fitted model, because a release found at installation is a real claim against the fund and belongs in the premium. It does not carry the age gradient, and no result in this appendix depends on it.

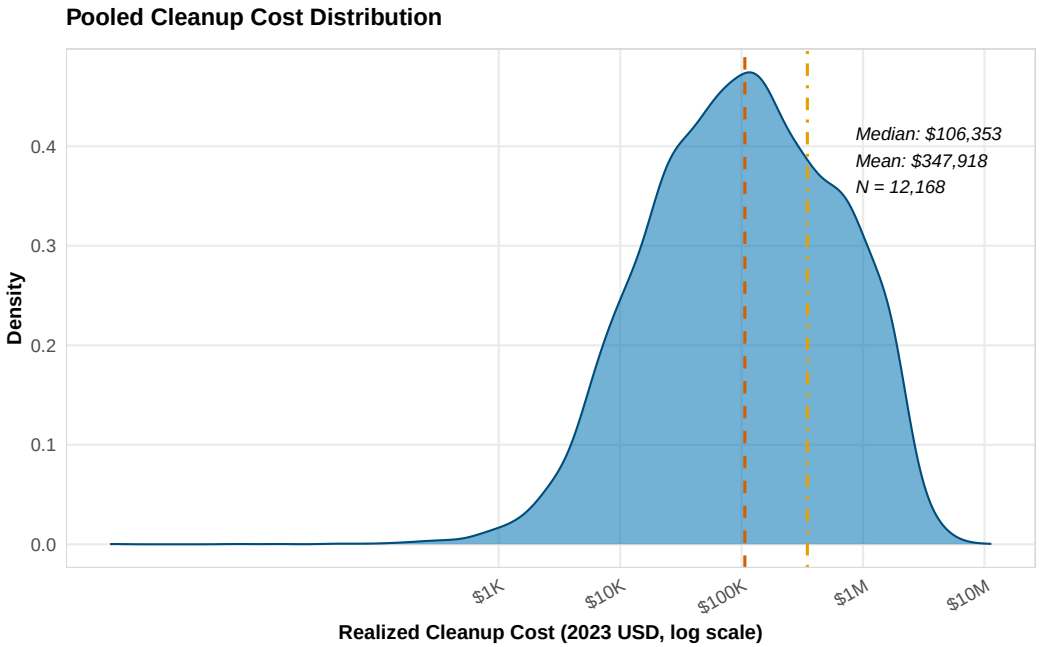
The three checks above, and the cell map of Figure 13 and Figure 16, require a known wall type. New Mexico's registry does not record one. Its facility-years enter the discrimination and calibration checks and drop out only of the cell-level figures. The excluded facility-years carry a release rate close to the sample average, and the restriction does not select on risk.

B.2 The distribution of cleanup cost

Figure 17 plots the distribution of cleanup cost across the 12,168 claims paid by 6 state trust funds, in 2023 dollars. The median claim runs $106,000 against a mean of $348,000. The

distribution is right-skewed, a minority of releases accounts for most of the spending, and the mean is the object a pure premium is built on. Appendix C states how each state's cost field is harmonized.

Figure 17: The distribution of cleanup cost across trust-fund claims



Notes: Kernel density of the cleanup cost of each claim paid by a state trust fund, pooled across states, in 2023 dollars on a logarithmic horizontal axis. The sample is 12,168 claims with a positive cost from the trust funds of 6 states (Colorado, Louisiana, New Mexico, Pennsylvania, Tennessee, and Utah). Costs are deflated to 2023 dollars by the month the claim starts. The dashed vertical line marks the median and the dot-dash vertical line marks the mean. Sources: state trust-fund claim records, and the consumer price index for all urban consumers from the Federal Reserve Bank of St. Louis (series CPIAUCSL).

B.3 Tank-level difference-in-differences

Table 9 estimates Equation 3 as a linear probability model of the annual tank-closure indicator under seven sets of fixed effects.

Columns (1) to (4) carry year effects and the three federal mandate indicators, with state effects in columns (1) and (2), facility effects in column (3), and tank effects in column (4). Column (2) adds cell effects. Columns (5) to (7) carry cell-by-year effects in place of year effects, and these absorb the mandate indicators exactly. State effects enter column (5), tank effects column (6), and facility effects column (7).

The estimate in column (7), the specification of Equation 3, is 1.3 points (s.e. 0.4), the static estimate of Section 4.1. It is positive in all seven columns and lies between 1.1 and 1.6 points

in six of them. Column (4), with tank and year effects but no cell-by-year effect, gives 0.4 points (s.e. 0.4).

Table 9: Difference-in-differences estimates of the tank-closure response

	(1) *(state+yr)*	(2) *(state+cell+yr)*	(3) *(fac+yr)*	(4) *(tank+yr)*	(5) *(state+cell×yr)*	(6) *(tank+cell×yr)*	(7) *(fac+cell×yr)*
Texas reform ($\hat{\beta}$)	0.0120	0.0105	0.0157	0.0037	0.0108	0.0107	0.0127
	(0.0026)	(0.0027)	(0.0035)	(0.0043)	(0.0023)	(0.0052)	(0.0038)
	[0.117]	[0.306]	[0.171]	[0.452]	[0.174]	[0.520]	[0.379]
State FE (explicit)	Y	Y	—	—	Y	—	—
Year FE	Y	Y	Y	Y	—	—	—
Cell FE	—	Y	—	—	—	—	—
Facility FE	—	—	Y	—	—	—	Y
Tank FE	—	—	—	Y	—	Y	—
Cell × year FE	—	—	—	—	Y	Y	Y
Mandate controls	Y	Y	Y	Y	—	—	—
Observations	7,862,897	7,862,896	7,862,897	7,852,369	7,862,174	7,851,644	7,862,172
Adj. R^2	0.0086	0.0174	0.0404	0.0875	0.0296	0.1045	0.0891
Wild cluster B	9,999	9,999	9,999	9,999	9,999	9,999	9,999

Notes: Linear probability model of the annual tank-closure indicator on the Texas reform interacted with the post-1998 period. The unit of observation is the tank-year over 1990 to 2015, on tanks active at the December 23, 1998 reform, unweighted. Coefficients are in probability units. A coefficient of 0.01 is one percentage point. Analytic cluster-robust standard errors, clustered by state with 22 clusters, are in parentheses. Restricted wild cluster bootstrap p-values from 9,999 draws are in brackets. The bottom rows record which fixed effects enter each column. A cell is the wall construction, fuel, capacity class, and install year of a tank. Mandate controls are three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines. Each is a deterministic function of install cohort and calendar year, and the cell-by-year effect absorbs all three exactly. Columns 5 through 7 therefore do not carry them. Column 7 is the estimating equation of Section 3.

The table reports one average over the post-reform years. The event study of Figure 3 reports the timing. With one treated state the analytic standard errors in parentheses over-reject. The bracketed row reports restricted wild cluster bootstrap p-values from 9,999 draws, and with one treated cluster this bootstrap is known to under-reject (MacKinnon, Nielsen, and Webb 2023). The smallest of the seven bracketed p-values is 0.117. Section B.9 reports the placebo distribution.

B.4 Which facilities respond

The average response hides wide variation, and its shape is a test of the mechanism. If a risk-rated premium drove the closures, the response should track the price of risk and the room a facility has to absorb it. It should fall hardest on the facilities whose premium rose most and lightest on those that can pass the cost to a captive market. I test this by letting the treatment effect vary with traits fixed at the reform, sorted into two questions. Who responded, by firm type, and where, by local market and neighborhood.

I use three firm traits and five market-and-place traits, each set at or before the reform so no split can pick up the reform's own effect. The firm traits are whether a facility is a motor-fuel retailer, defined as ever having sold gasoline, its total tank capacity in four bins, and the vintage of its oldest tank. The market-and-place traits, built in Section A.4 from 1998 geography and the 2000 Census, are whether a facility sits in a thin market of one or fewer competing gasoline retailers within a mile, and whether its Census tract is rural, low-population, low-income, or high-poverty. Each captures a distinct reason a facility might respond more or less. Capacity and vintage stand for how much the premium rose. Motor-fuel status and the thin-market count stand for how captive a facility's demand is, and so how easily it can pass the premium through instead of exiting.

I estimate the heterogeneity with the facility regression of Section 4.3, interacting the treatment with one trait at a time,

$$Y_{ft} = \beta\,\mathrm{TX}_f \times \mathrm{Post}_t + \eta'\left(\mathrm{TX}_f \times \mathrm{Post}_t\right) \times Z_f + \gamma' M_{ft} + \alpha_f + \delta_{c(f)t} + \varepsilon_{ft},$$

where Z_f is the trait, a single indicator or a set of category dummies fixed at the reform, whose level the facility fixed effect absorbs. The reform effect for the omitted group is β, and the effect for a group marked by Z_f is $\beta + \eta$. Rather than reporting β and η separately, Table 10 reports each group's own post-reform effect, one row per group and one column per margin. The entry for a given group and margin is therefore how much the reform moved that outcome for that kind of facility, and the difference between two rows in a column is the heterogeneity.

The first firm trait is size, and Table 10 shows it separates leaving from reinvesting. Reading down the capacity rows of the exit column, the smallest facilities leave. A facility under 9,000 gallons raised its exit rate 5.46 points (s.e. 0.72) and its closure rate 5.02 points (s.e. 0.62), the largest response on either margin in the table and several times that of any larger bin. Exit falls steeply with size, to 1.02 points at 9,000 to 20,000 gallons (s.e. 0.43) and to under a point in the two largest bins. The reinvestment columns run the other way. A facility over 30,000 gallons raised its downsizing 1.67 points (s.e. 0.17) and its tank replacement 1.33 points (s.e. 0.17), while the smallest bin cut both. Across the four size rows, exit falls as reinvestment rises.

The vintage rows of Table 10 sort the market the same way, consistent with the age gradient of Section 4.2. The oldest facilities, whose tanks predate 1975, raised closures 2.80 points (s.e. 0.41) and exits 2.14 points (s.e. 0.48) and did not reinvest on either margin. The youngest, with tanks installed from 1989 to 1998, barely closed and instead raised replacement 0.68 points (s.e. 0.24) and downsizing 0.39 points (s.e. 0.12).

The fuel rows separate the two kinds of response rather than separating response from silence. A facility that ever sold gasoline raised its closure 3.28 points (s.e. 0.33) and its exit 1.96 points (s.e. 0.30), and it raised downsizing and replacement as well. A facility that never sold gasoline did not raise its closure rate at all, and its point estimate is negative and indistinguishable from zero. It moved instead by cutting the reinvestment margins, lowering downsizing 0.42 points and replacement 0.75 points. Read together, the firm-trait rows

show the marginal capital, small, old, and fuel-retailing, leaving the market, while the viable capital, large and modern, shrank, replaced, and stayed.

Where a facility sits decided how far it moved, and the dividing line in the market rows of Table 10 is local market power. In a competitive market, a facility with more than one gasoline retailer within a mile raised its closures 2.13 points (s.e. 0.35) and its exits 1.85 points (s.e. 0.38). A facility holding captive demand, with one or fewer competitors within a mile, moved about half as far on closures, 0.89 points (s.e. 0.38), and its exit response of 0.52 points (s.e. 0.42) is not different from zero. Captive demand let these stations pass part of the premium through and keep their tanks. Rural facilities fall between the two, closing 1.62 points (s.e. 0.43) and exiting 1.08 points (s.e. 0.46), a real response but a smaller one than in the denser markets where a facility cannot pass the premium to customers who have nowhere else to go.

Poorer tracts respond nearly as much as richer ones, and almost entirely by leaving. In the neighborhood rows of Table 10, low-income facilities raised their exit 1.57 points (s.e. 0.42) against 1.58 points for higher-income ones, the same response within rounding, and high-poverty facilities exited 1.59 points against 1.56. Their closure response is modestly smaller, 1.79 points against 2.09 (s.e. 0.38), and their reinvestment margins are flat. Because these tracts overlap with rural and captive markets, I read the small closure gap as another expression of the competition channel rather than a poverty effect of its own. Where the split is informative, it shows poor and rich areas retiring capital at the same rate.

Together, the facility evidence tells one story. The reform sorted the market along the price of risk and the room to absorb it. The facilities whose premium rose most, the small, the old, and the gasoline retailers, retired tanks and left. The largest and most modern shrank and replaced instead of leaving, and stayed open. The response fell hardest where gasoline retailers compete and lightest where a station holds captive demand. Poorer areas exited at the same rate as richer ones. Consolidation, the one move that changes nothing physical, stays near zero in every group, as it did on average. The response concentrated where the premium rose most and where competition left no room to pass it on, not evenly across Texas.

The facilities that exited once risk was priced are the small, old, high-risk facilities that a risk-rated premium charges most. The pattern matches the signature the bonding evidence reads as efficient reallocation, exit by small operators with poor environmental records while productive assets transfer rather than shut (Boomhower 2019; Yin, Kunreuther, and White 2011). The pattern is necessary for an efficiency reading, not sufficient. Whether the retired capital was worth retiring depends on weighing each exiter's forgone profit against the cleanup costs it would have imposed, and that weighing is the welfare accounting the structural model supplies.

Table 10: Heterogeneous reform effects, by facility group

Panel A. Firm traits

	Any closure	Facility exit	Downsize	Consolidate	Any replacement
Size (total capacity)					
Under 9k gal (reference group)	+5.02***	+5.46***	-0.65***	-0.01	-0.29*
	(0.62)	(0.72)	(0.19)	(0.03)	(0.15)
9–20k gal	+0.85***	+1.02**	-0.48**	-0.06*	-0.16
	(0.30)	(0.43)	(0.21)	(0.03)	(0.19)
20–30k gal	+0.71***	-0.14	+0.66***	-0.17***	+0.57***
	(0.23)	(0.31)	(0.15)	(0.03)	(0.14)
Over 30k gal	+2.46***	+0.40**	+1.67***	+0.08	+1.33***
	(0.31)	(0.19)	(0.17)	(0.06)	(0.17)
Vintage (oldest tank)					
1989–1998 (reference group)	-0.05	-0.09	+0.39***	-0.01	+0.68***
	(0.17)	(0.15)	(0.12)	(0.02)	(0.24)
1985–1988	+1.22***	+1.14***	+0.11	-0.07**	+0.05
	(0.28)	(0.33)	(0.10)	(0.03)	(0.09)
1975–1984	+1.96***	+0.98*	+0.45**	-0.05	+0.63***
	(0.37)	(0.48)	(0.18)	(0.03)	(0.17)
Pre-1975	+2.80***	+2.14***	+0.28	-0.05	+0.13
	(0.41)	(0.48)	(0.25)	(0.05)	(0.29)
Fuel					
Non-gasoline (reference group)	-1.27*	-0.33	-0.42	-0.20***	-0.75
	(0.64)	(0.56)	(0.56)	(0.06)	(0.56)
Gasoline	+3.28***	+1.96***	+0.68***	+0.02	+0.91***
	(0.33)	(0.30)	(0.09)	(0.03)	(0.11)

Table 10: Heterogeneous reform effects, by facility group (continued)

Panel B. Market and place traits

	Any closure	Facility exit	Downsize	Consolidate	Any replacement
Local competition					
Competitive (reference group)	+2.13***	+1.85***	+0.18*	-0.09***	+0.25**
	(0.35)	(0.38)	(0.09)	(0.02)	(0.10)
Thin (≤1 gas competitor)	+0.89**	+0.52	+0.16*	+0.09***	+0.19*
	(0.38)	(0.42)	(0.09)	(0.02)	(0.09)
Rural					
Urban (reference group)	+2.01***	+1.71***	+0.18**	-0.08***	+0.28**
	(0.36)	(0.40)	(0.08)	(0.02)	(0.09)
Rural	+1.62***	+1.08**	+0.28**	+0.04	+0.25**
	(0.43)	(0.46)	(0.10)	(0.02)	(0.09)
Population density					
Higher density (reference group)	+1.96***	+1.63***	+0.18**	-0.08***	+0.30***
	(0.36)	(0.40)	(0.08)	(0.02)	(0.09)
Low density	+1.85***	+1.41***	+0.27**	+0.03	+0.20**
	(0.42)	(0.46)	(0.10)	(0.03)	(0.09)
Income					
Higher income (reference group)	+2.09***	+1.58***	+0.31***	-0.02	+0.44***
	(0.38)	(0.40)	(0.07)	(0.03)	(0.09)
Low income	+1.79***	+1.57***	+0.11	-0.08***	+0.13
	(0.38)	(0.42)	(0.10)	(0.03)	(0.10)
Poverty					
Lower poverty (reference group)	+2.06***	+1.56***	+0.29***	+0.02	+0.41***
	(0.38)	(0.40)	(0.08)	(0.03)	(0.09)
High poverty	+1.86***	+1.59***	+0.15	-0.10***	+0.19*
	(0.38)	(0.42)	(0.09)	(0.02)	(0.10)

Notes: Panel A splits the reform effect by firm traits and Panel B by market and place traits. Each cell is that group's own reform-by-post effect on the facility-year outcome named in the column head, in percentage points, not a differential against the reference group. Within each block the first row is the reference level and its cell is the effect at that level. Every other cell is the reference effect plus the interaction, with the standard error taken from the fit covariance rather than from the interaction alone. Columns are any closure, facility exit, downsizing, consolidation, and any replacement. Section A.3 defines each outcome and Section A.4 defines each group. The unit of observation is the facility-year over 1990 to 2015, with facility fixed effects, portfolio-composition-by-year fixed effects, and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines. The size block rests on 2,775,559 facility-years, and the vintage and fuel blocks on 2,776,307, in Texas and the 21 control states. The local competition block rests on 2,429,811 facility-years in Texas and 17 control states, and the rural, density, income, and poverty blocks on 2,309,424 to 2,314,145 in Texas and 16 control states. The geographic files omit Montana, North Dakota, Pennsylvania, and Utah, and Missouri's facilities carry identifiers in the tract file that do not merge to the panel. Standard errors clustered by state are in parentheses, with 22 clusters in the size, vintage, and fuel blocks, 18 in the local competition block, and 17 in the tract blocks. Stars mark 0.1, 0.05, and 0.01. With one treated cluster those errors over-reject and are provisional. Thin market marks a facility with at most one gasoline competitor within one mile. Source: 2000 Census tract data, matched to facility coordinates.

B.5 Population growth

The closure response is larger in counties whose population grew more. Counties grew at very different rates after 1998, and a growing county can raise the value of keeping a station open.

Growth is measured over 1998 to 2015, as the percent change in county population from the Bureau of Economic Analysis county series. I give each tank the growth of its county and sort tanks into quartiles of growth, with cut points of 1, 11, and 24 percent set on the tanks open at the end of 1998 in Texas and the control states. I estimate the specification of Equation 5 with growth quartile in place of age quartile, on 7,808,442 tank-years. The slowest-growing quartile is the reference. A facility sits in one county, and the facility effect therefore absorbs the Texas-by-quartile terms λ_q. The specification adds growth-quartile-by-year effects. These absorb any post-1998 closure trend that counties of the same growth share in every state. A continuous version replaces the quartiles with county growth in units of ten percentage points, centered at the mean of the 1998 stock, 15.5 percent.

Table 11 reports the estimates, and Figure 18 plots the reform effect for each quartile. In the slowest-growing quartile a Texas tank closes 0.38 points more often after the reform than a control-state tank (s.e. 0.47). That effect is not distinguishable from zero. The effect is 1.01 points (s.e. 0.43) in the second quartile, 1.57 points (s.e. 0.39) in the third, and 1.78 points (s.e. 0.37) in the fastest-growing quartile. The effect in the fastest-growing quartile exceeds the effect in the slowest-growing quartile by 1.40 points (s.e. 0.33). In the continuous version the effect rises by 0.15 points (s.e. 0.05) for each ten points of county growth.

Table 11: Closure response by county population growth

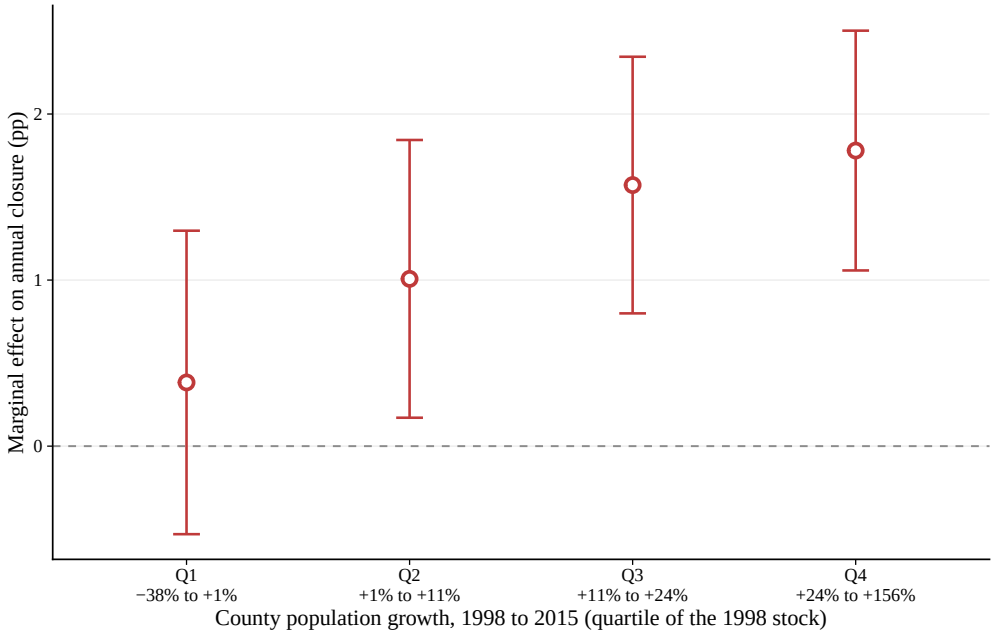
	Quartiles (1)	Continuous (2)
Panel A. Coefficients (pp of annual closure)		
TX × Post	0.38	1.26
	(0.47)	(0.38)
TX × Post × Q2 (+1% to +11%)	0.62	
	(0.17)	
TX × Post × Q3 (+11% to +24%)	1.19	
	(0.20)	
TX × Post × Q4 (+24% to +156%)	1.40	
	(0.33)	
TX × Post × growth (per 10 points)		0.15
		(0.05)
Growth quartile × year FE / growth × year slopes	Yes	Yes
Facility FE and cell × year FE	Yes	Yes
Observations	7,808,442	7,808,442

Panel B. Reform effect and closure rates, by growth quartile

	Reform effect	Control closure rate (%)		Texas closure rate (%)	
Growth quartile	(pp)	pre	post	pre	post
Q1 (-38% to +1%)	0.38 (0.47)	2.55	2.75	2.03	2.78
Q2 (+1% to +11%)	1.01 (0.43)	2.60	2.66	1.92	3.00
Q3 (+11% to +24%)	1.57 (0.39)	3.04	2.81	1.96	3.32
Q4 (+24% to +156%)	1.78 (0.37)	2.71	2.61	1.89	3.19

Notes: Linear probability model of the annual tank-closure indicator on the Texas reform interacted with the post-1998 period, in percentage points, estimated on 7,808,442 tank-years over 1990 to 2015 for tanks active at the December 23, 1998 reform, in Texas and the 21 control states. County growth is the percent change in county population over 1998 to 2015. Quartile cut points are 1, 11, and 24 percent, from the tanks open at the end of 1998. Every column carries facility and cell-by-year fixed effects. Column (1) of Panel A also carries growth-quartile-by-year effects and splits the reform effect by growth quartile, with the slowest-growing quartile as the reference. The TX × Post row is its effect, and each TX × Post × quartile row is the difference from it. Column (2) carries growth-by-year slopes and replaces the quartiles with county growth in units of ten percentage points, centered at the mean of the 1998 stock, 15.5 percent. The TX × Post row is the effect at mean growth. Panel B reports the reform effect for each quartile, the TX × Post coefficient plus the interaction for that quartile, and the closure rate in percent for control-state and Texas tanks before and after the reform. Standard errors clustered by state with 22 clusters are in parentheses. With one treated state these errors understate the uncertainty. Source: Bureau of Economic Analysis, county population series (table CAINC1).

Figure 18: Reform effect on annual tank closure by county population growth



Notes: Each point is the reform effect for one quartile of county population growth over 1998 to 2015, the TX × Post coefficient plus the interaction for that quartile from Table 11, in percentage points. The quartile labels give the range of county growth in percent, and the cut points are 1, 11, and 24 percent, from the tanks open at the end of 1998. The sample is 7,808,442 tank-years from 1990 to 2015, on tanks active at the December 23, 1998 reform, in Texas and 21 control states. The regression includes facility fixed effects, cell-by-year fixed effects, and growth-quartile-by-year fixed effects. Vertical bars are 95 percent confidence intervals from the covariance matrix of the coefficients, clustered by state with 22 clusters. The dashed line marks zero. Source: Bureau of Economic Analysis, county population series (table CAINC1).

Growth is measured after the reform, and reading it as a moderator assumes that the reform did not change county population. The fastest-growing quartile holds 54 percent of Texas tanks and 19 percent of control-state tanks. The headline estimate of Section 4.1 therefore weights high-growth counties most. With one treated state the clustered standard errors understate the uncertainty, and no placebo distribution is computed for this split.

B.6 Release discovery estimates

Section 4.4 reads the release response from the three-panel event study. This section reports the estimates behind that reading and nothing beyond them. Table 12 gives the static reform-times-post effect for each discovery channel under the two fixed-effect versions the paper uses. Pooled across channels, the decline is 1.15 points (s.e. 0.30) under the portfolio-composition effects and 1.93 points (s.e. 0.33) under the full-portfolio version. The static estimate sets the

whole post-period against a pre-period in which Texas discovery ran above the control states and was already falling toward them, and it therefore overstates the reform's contribution.

The event-study coefficients carry the timing. Texas discovery ran 0.75 points above the controls eight years before the reform, 0.46 points below them five years after it (s.e. 0.27), and 0.82 points below ten years after (s.e. 0.30), or 1.40 points under the full-portfolio version. By channel, releases found in routine monitoring with no closure nearby are 0.21 points lower at ten years with a standard error of 0.22, and releases surfaced by the inspection a closure triggers are 0.61 points lower (s.e. 0.19). Why the second channel is mechanical is stated in Section 4.4.

Table 12: Leak-discovery response, by margin

	Any leak (1)	Standalone (2)	Induced (3)
Reform × Post	-0.0115***	-0.0052**	-0.0063***
	(0.0030)	(0.0020)	(0.0016)
Observations	2,776,307	2,776,307	2,776,307
Facility fixed effects	✓	✓	✓
Composition × year fixed effects	✓	✓	✓

Notes: Estimates are in probability units on the annual facility-year leak-discovery indicator. A coefficient of 0.01 is one percentage point. Reform × Post is the interaction of a Texas indicator with an indicator for the years after 1998. Any leak pools the two channels. Standalone releases are found in routine monitoring. Induced releases are found at the inspection a closure triggers, within sixty days of it. The unit of observation is the facility-year over 1990 to 2015, on facilities active at the December 23, 1998 reform in Texas and the 21 control states, unweighted. The regression includes facility fixed effects, portfolio-composition-by-year fixed effects, and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines. Standard errors clustered by state with 22 clusters are in parentheses, and stars mark 0.1, 0.05 and 0.01. With one treated cluster those errors over-reject, and the event study of Figure 5 shows the timing.

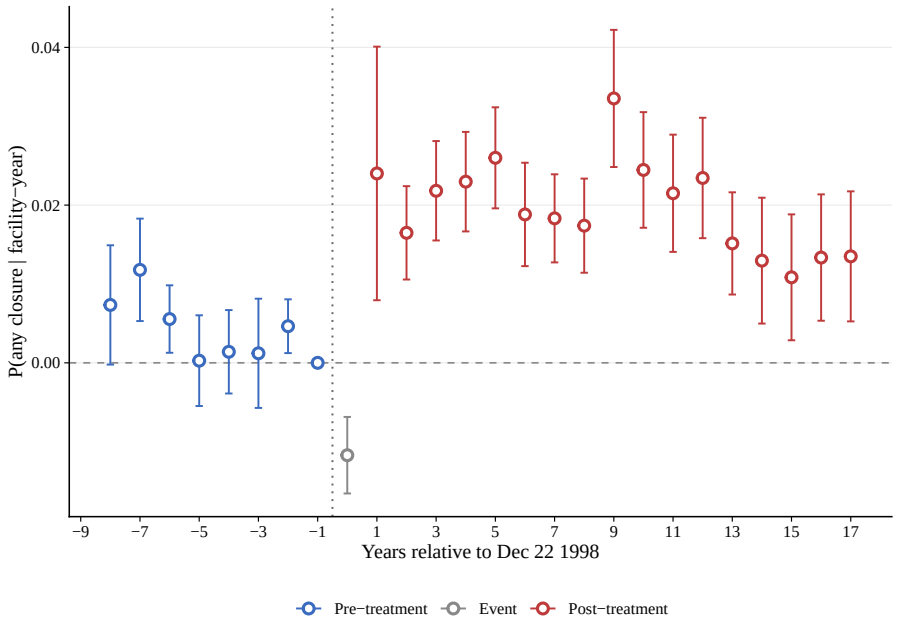
B.7 Additional reduced-form results

This section collects the event studies and the alternative specification the main text summarizes but does not display. Figure 19 replicates the tank-closure event study of Figure 3 at the facility level, where the operator decides. The outcome equals one in any year the facility closes at least one tank. The path has the same shape. The pre-reform coefficients are small, the largest reaching 1.18 points, and the gap opens at the reform, at 2.6 points five years out (s.e. 0.3) and 2.4 points ten years out (s.e. 0.4). Whether the paper counts tanks or facilities, the reform raises closures by a similar proportion of the base rate.

The remaining margins are reported on the same design. Figure 20 plots facility downsizing, Figure 21 plots tank replacement, and Figure 22 plots consolidation. The first two open after the reform and stay open, which is the reinvestment response the heterogeneity of Section B.4 attributes to the larger and more modern facilities. The pre-reform path of downsizing is not flat, and I report that margin as descriptive rather than as an effect. Several pre-reform years of the replacement path run elevated, and I read that margin as suggestive. Consolidation, the one adjustment that moves no capital in or out, stays near zero throughout, and its flatness is a useful negative control on the design.

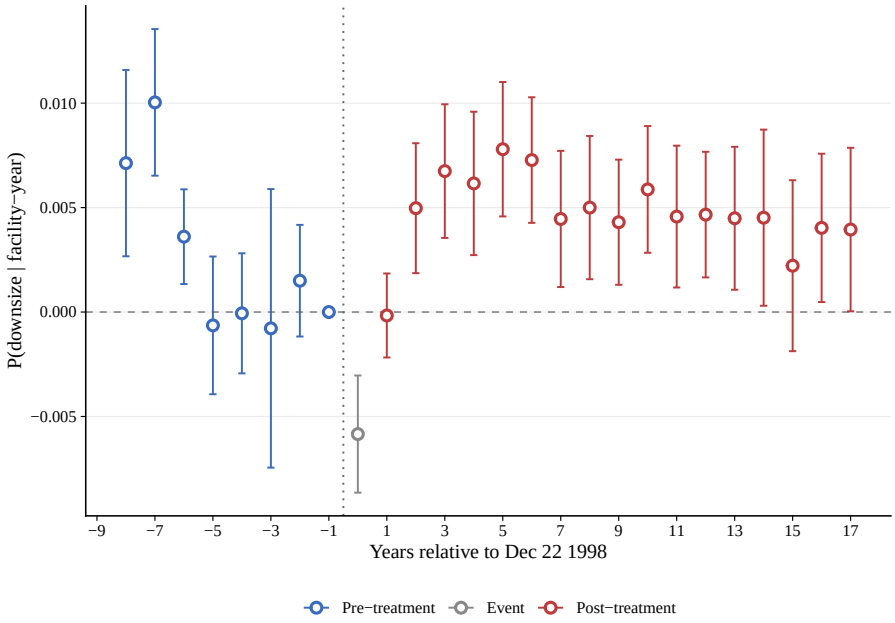
Table 13 repeats the facility regression under full-portfolio fixed effects, which absorb a facility's entire tank composition interacted with year rather than the coarser cell used in the main text. The specification is stricter and the sample smaller, because a composition cell observed in only one state contributes nothing once the interaction is absorbed. The margins keep their sign and their ordering under it.

Figure 19: Event study of tank closure at the facility level, Texas relative to control states



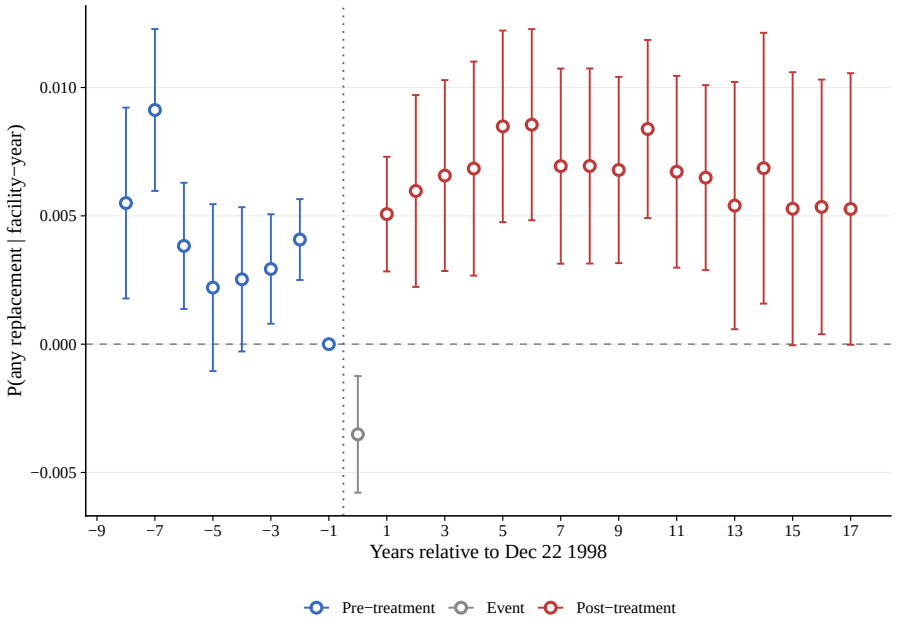
Notes: Coefficients on the interaction of a Texas indicator with an indicator for each year relative to 1998, from Equation 4 estimated on the facility-year panel. The outcome is an indicator that the facility closes at least one tank in the year. Coefficients are in probability units, and 0.01 is one percentage point. The year before the reform is the omitted category. The sample is 2,776,307 facility-years from 1990 to 2015, on facilities holding at least one tank active at the December 23, 1998 reform, in Texas and 21 control states. The regression includes facility fixed effects, portfolio-composition-by-year fixed effects, and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines, as in column 1 of Table 4. Vertical bars are 95 percent confidence intervals from standard errors clustered by state.

Figure 20: Event study of facility downsizing, Texas relative to control states



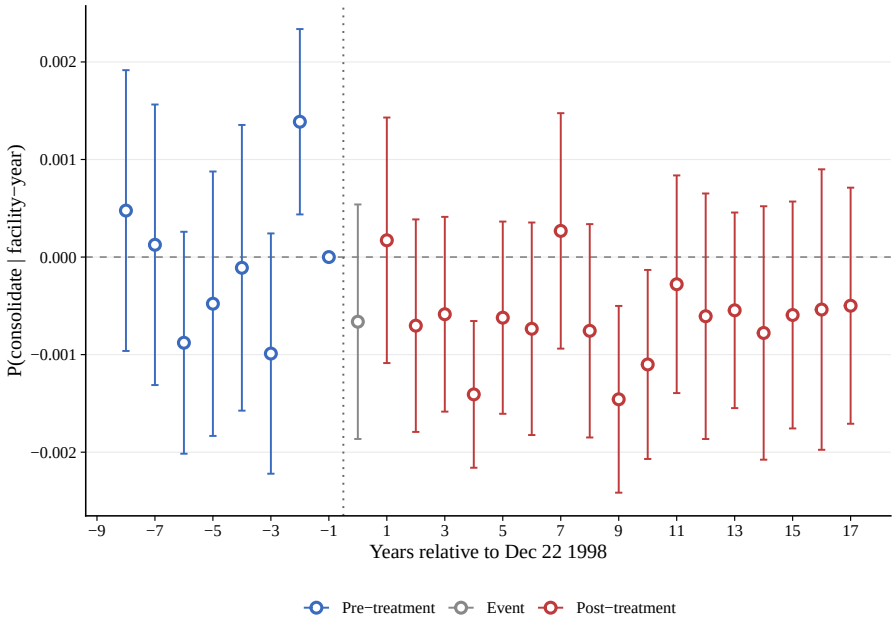
Notes: Coefficients on the interaction of a Texas indicator with an indicator for each year relative to 1998, from Equation 4 estimated on the facility-year panel. The outcome is an indicator that the facility downsizes in the year. Downsizing is a facility-year in which the facility does not exit, ends the year with fewer tanks than it held at the end of the previous year, and cuts its total capacity by more than five percent. Coefficients are in probability units, and 0.01 is one percentage point. The year before the reform is the omitted category. The sample is 2,776,307 facility-years from 1990 to 2015, on facilities holding at least one tank active at the December 23, 1998 reform, in Texas and 21 control states. The regression includes facility fixed effects, portfolio-composition-by-year fixed effects, and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines, as in Table 4. Vertical bars are 95 percent confidence intervals from standard errors clustered by state.

Figure 21: Event study of facility replacement, Texas relative to control states



Notes: Coefficients on the interaction of a Texas indicator with an indicator for each year relative to 1998, from Equation 4 estimated on the facility-year panel. The outcome is an indicator that the facility replaces a tank in the year. Replacement is a facility-year in which the facility closes at least one tank and installs a new tank on a later date. Coefficients are in probability units, and 0.01 is one percentage point. The year before the reform is the omitted category. The sample is 2,776,307 facility-years from 1990 to 2015, on facilities holding at least one tank active at the December 23, 1998 reform, in Texas and 21 control states. The regression includes facility fixed effects, portfolio-composition-by-year fixed effects, and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines, as in Table 4. Vertical bars are 95 percent confidence intervals from standard errors clustered by state.

Figure 22: Event study of facility consolidation, Texas relative to control states



Notes: Coefficients on the interaction of a Texas indicator with an indicator for each year relative to 1998, from Equation 4 estimated on the facility-year panel. The outcome is an indicator that the facility consolidates in the year. Consolidation is a facility-year in which the facility does not exit, ends the year with fewer tanks than it held at the end of the previous year, and changes its total capacity by no more than five percent in either direction. Coefficients are in probability units, and 0.01 is one percentage point. The year before the reform is the omitted category. The sample is 2,776,307 facility-years from 1990 to 2015, on facilities holding at least one tank active at the December 23, 1998 reform, in Texas and 21 control states. The regression includes facility fixed effects, portfolio-composition-by-year fixed effects, and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines, as in Table 4. Vertical bars are 95 percent confidence intervals from standard errors clustered by state.

Table 13: Facility portfolio response, by margin, full-portfolio fixed effects

	Any closure (1)	Facility exit (2)	Downsize (3)	Consolidate (4)	Reconfigure-up (5)	Any replacement (6)	Capacity cut (7)
Reform × Post	0.0239***	0.0204***	0.0030***	-0.0007***	0.0018***	0.0035***	0.0219***
	(0.0062)	(0.0060)	(0.0005)	(0.0002)	(0.0002)	(0.0007)	(0.0061)
Observations	1,612,261	1,612,261	1,612,261	1,612,261	1,612,261	1,612,261	1,612,261
Facility fixed effects	✓	✓	✓	✓	✓	✓	✓
Full portfolio × year fixed effects	✓	✓	✓	✓	✓	✓	✓

Notes: Robustness version of Table 4, identical in outcome, sample definition, and clustering. The unit of observation is the facility-year over 1990 to 2015, and the outcome in each column is the indicator named in the column head. The cell-by-year fixed effect here interacts a facility's full tank multiset, the exact set of type-and-vintage cells it operates, with the year, rather than the coarser composition cell of the headline design. A Texas facility is therefore compared only against control facilities holding an identical fleet in the same year. The comparison is stricter and the sample smaller, because a multiset observed in only one state contributes nothing once the interaction is absorbed. Every column also carries facility fixed effects and the three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines. Estimates are in probability units, and a coefficient of 0.01 is one percentage point. Standard errors clustered by state are in parentheses, and stars mark 0.1, 0.05 and 0.01.

B.8 County entry, synthetic control

The reform had no perceptible effect on entry. The closure results describe incumbent facilities, and this section asks whether the rate at which new facilities open changed with the reform.

A facility enters in the year of its earliest tank installation. I sum entrants from the county panel to state-year counts, for facilities with no undated tank, and compare Texas with the 21 control states on four outcomes. These are entrants per 100,000 residents, the log of entrants, entrants as a percent of the previous year's stock of facilities, and the raw count. I estimate the effect by synthetic difference-in-differences and by synthetic control. Both choose weights on the control states to match Texas over 1990 to 1998. As a comparison I also use difference-in-differences with equal weights. I report the average effect over 1999 to 2006 and over 1999 to 2017. Inference is by placebo. Each control state is assigned the treatment in turn, with Texas kept as a donor, and the placebo p-value is one plus the number of control states whose estimate is at least as large in absolute value as the Texas estimate, divided by 22.

Table 14 reports the estimates, and Figure 23 plots Texas against its synthetic counterparts. On entrants per 100,000 residents the synthetic difference-in-differences, synthetic control, and difference-in-differences estimates for 1999 to 2006 are 12.3, 7.8, and 35.2 percent, with placebo p-values of 0.82, 0.68, and 0.41.

Ten of the twelve estimates have placebo p-values of 0.09 or higher in both windows. They

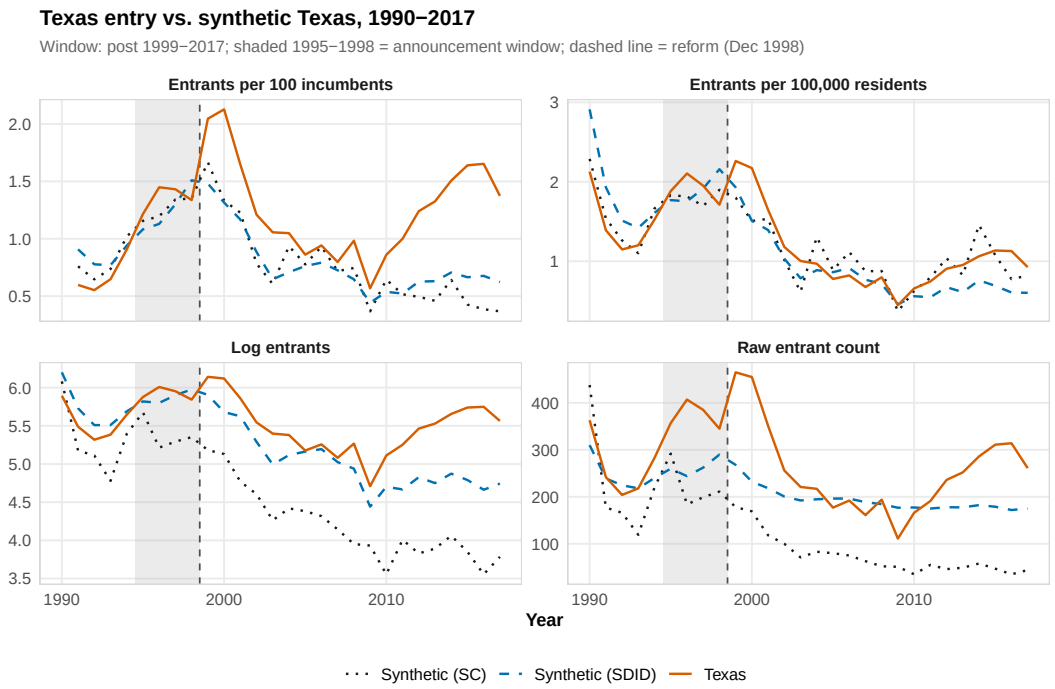
are the three estimators on entrants per 100,000 residents and on the entry rate, and the synthetic difference-in-differences and difference-in-differences estimators on log entrants and raw entrants. The exceptions are the synthetic-control estimates for log entrants and raw entrants, with all of their weight on Pennsylvania and placebo p-values of 0.045 in both windows.

Table 14: Effect of the reform on entry of new facilities, by outcome and estimator

Outcome	Estimator	Effect 1999–2006	p 1999–2006	Effect 1999–2017	p 1999–2017	Pre-period RMSPE
Entrants per 100k	SDID	+12.3%	0.818	+12.5%	0.864	0.35
Entrants per 100k	SC	+7.8%	0.682	+3.0%	0.864	0.18
Entrants per 100k	DiD	+35.2%	0.409	+37.0%	0.455	0.65
Log entrants	SDID	+30.4%	0.409	+57.4%	0.273	1.83
Log entrants	SC	+165.5%	0.045	+260.8%	0.045	0.47
Log entrants	DiD	+47.6%	0.182	+85.4%	0.136	1.71
Entry rate (% of stock)	SDID	+38.9%	0.227	+48.3%	0.182	0.19
Entry rate (% of stock)	SC	+32.9%	0.318	+51.0%	0.091	0.13
Entry rate (% of stock)	DiD	+52.6%	0.136	+63.5%	0.091	0.32
Raw entrants	SDID	+20.6%	0.136	+19.5%	0.136	204.28
Raw entrants	SC	+58.6%	0.045	+57.6%	0.045	120.91
Raw entrants	DiD	+3.8%	0.727	-4.1%	0.727	242.38

Notes: Estimated effect of the reform on the entry of new facilities, Texas against the 21 control states, by synthetic difference-in-differences (SDID), synthetic control (SC), and difference-in-differences with equal weights (DiD). A facility enters in the year of its earliest tank installation. Entrants are summed from the county panel to state-year counts, for facilities with no undated tank, over 1990 to 2017. Entrants per 100k is entrants per 100,000 residents, log entrants is the natural log of the count, the entry rate is entrants as a percent of the previous year's stock of facilities, and raw entrants is the count. Effects are averages over 1999 to 2006 and over 1999 to 2017, in percent of the Texas mean over 1990 to 1998 (1991 to 1998 for the entry rate), and for log entrants as 100 times the exponential of the estimate minus one. The p-value is a placebo p-value. Each control state is assigned the treatment in turn, with Texas kept as a donor, and p is one plus the number of control states whose estimate is at least as large in absolute value as the Texas estimate, divided by 22. The smallest attainable value is 0.045. Pre-period RMSPE is the root mean squared prediction error of the fit to Texas over the pre-reform years, in the units of the outcome, for the 1999 to 2006 fit. Sources: state tank registries, and county population from the Bureau of Economic Analysis.

Figure 23: Texas entry and synthetic Texas, by outcome



Notes: Entry of new facilities in Texas and in synthetic Texas over 1990 to 2017, one panel per outcome. Synthetic Texas is a weighted average of the 21 control states. The panel labeled entrants per 100 incumbents is the entry rate, entrants as a percent of the previous year's stock of facilities. The synthetic paths are the synthetic difference-in-differences and synthetic control fits for the 1999 to 2017 window, with weights chosen to match Texas over the pre-reform years. The shaded years 1995 to 1998 mark the announcement window, and the dashed line marks the reform. Sources: state tank registries, and county population from the Bureau of Economic Analysis.

A failure to reject is not an estimate of zero. The point estimates are positive in 23 of the 24 cells. With 21 control states, a rejection at the five percent level requires Texas to rank first among the 22 states. Entry is dated by the earliest tank installation in each state registry, and the headline sample leaves out facilities with an undated tank.

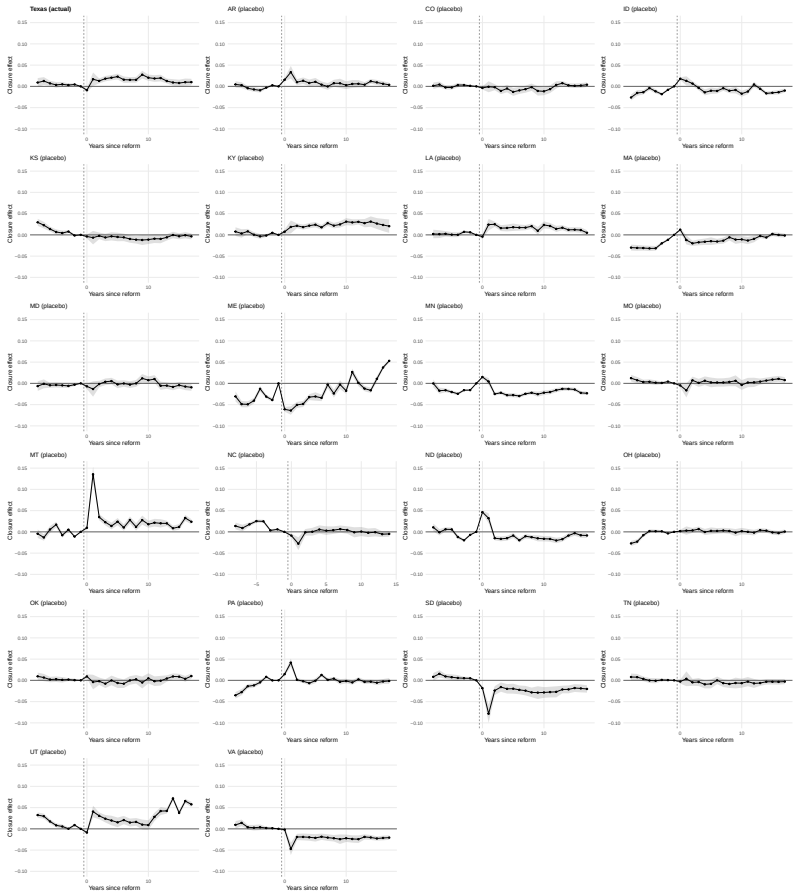
B.9 Inference and robustness

This section collects the checks that support the headline estimate without appearing in the body: the placebo distribution behind the inference and a formal sensitivity analysis for the parallel-trends assumption.

Section 3 sets a placebo distribution beside the cluster-robust standard errors, because the design carries 21 control states and one treated state. Figure 24 plots the event study each placebo run produces, one panel per control state relabeled Texas in turn, on a shared scale

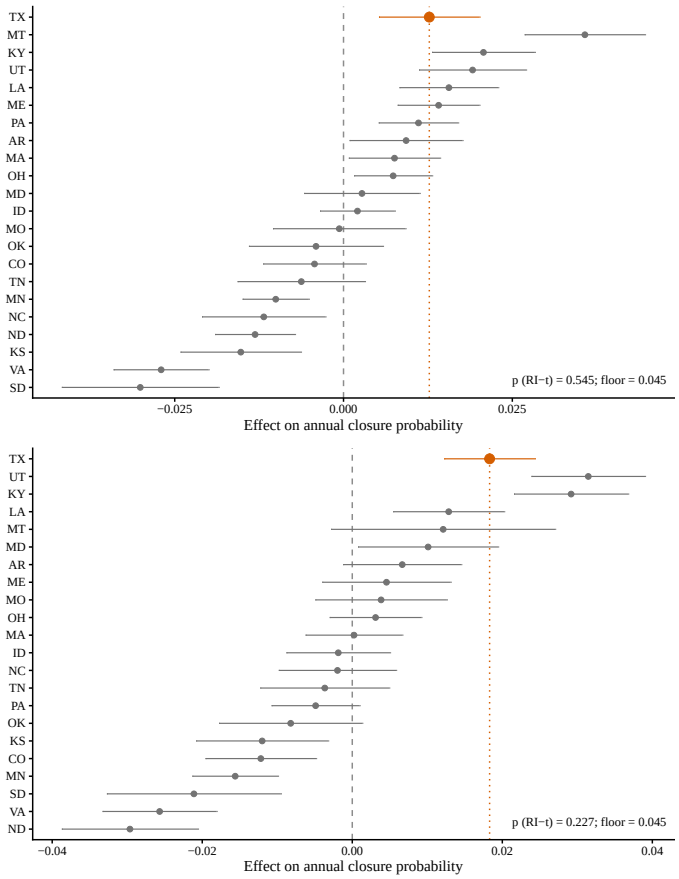
with the true Texas estimate. Figure 25 plots the static estimate from each placebo run beside the Texas estimate, for tank closure and for facility closure, and Figure 26 does the same for the three release-discovery outcomes. With 21 control states the smallest attainable two-sided permutation p-value is 0.045, and 11 of the 21 placebo estimates have a t-statistic at least as large in absolute value as the Texas tank-level estimate's, for a permutation p-value of 0.545.

Figure 24: Placebo event studies, Texas label reassigned to each control state



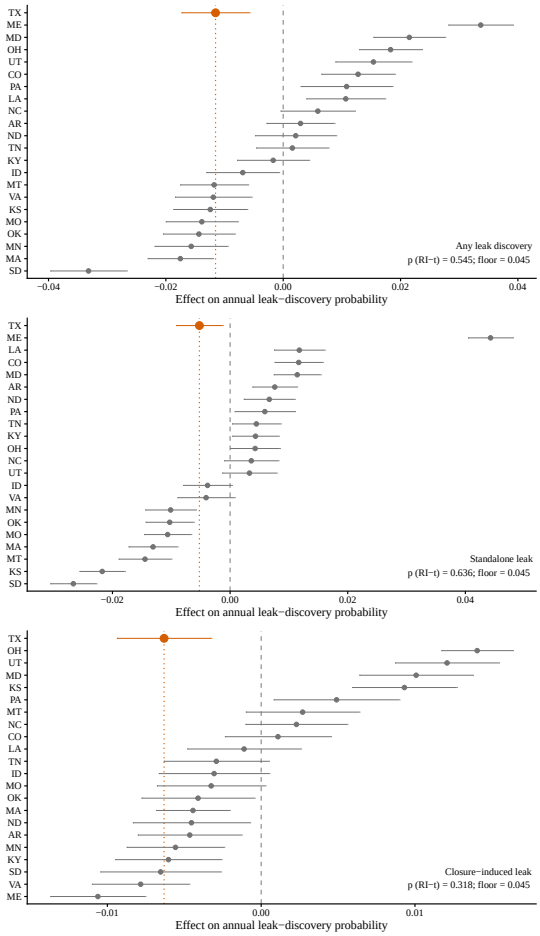
Notes: Coefficients on the interaction of a treatment indicator with an indicator for each year relative to 1998, from Equation 4 estimated on the tank-year panel. The first panel is the actual Texas estimate. In every other panel one control state is relabeled Texas and receives the treatment indicator, and the true Texas stays in the sample as an untreated state. The outcome is an indicator that the tank is permanently closed in the year. Coefficients are in probability units, and 0.01 is one percentage point. The year before the reform is the omitted category. The sample is 7,862,172 tank-years from 1990 to 2015, on tanks active at the December 23, 1998 reform, in Texas and 21 control states. The regression includes facility fixed effects and cell-by-year fixed effects. Shaded bands are 95 percent confidence intervals from standard errors clustered by state. Every panel has the same vertical scale.

Figure 25: Placebo estimates of the reform effect on closure, tanks and facilities



Notes: Each gray marker is the difference-in-differences estimate of the reform effect on the annual closure probability when one control state is assigned the reform in place of Texas. The orange marker is the estimate for Texas, and the dotted vertical line marks it. Estimates are in probability units, and 0.01 is one percentage point. The upper panel uses the tank-year panel, with 7,862,172 tank-years, the tank-closure indicator as the outcome, and facility and cell-by-year fixed effects. The lower panel uses the facility-year panel, with 2,776,307 facility-years, the any-closure indicator as the outcome, facility and portfolio-composition-by-year fixed effects, and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines. Both samples cover 1990 to 2015 and are restricted to units active at the December 23, 1998 reform. Texas stays in the sample as an untreated state in each placebo run. Horizontal bars are 95 percent confidence intervals from standard errors clustered by state, and the dashed line marks zero. Each panel prints a randomization-inference p-value, labeled RI-t, equal to one plus the number of control states whose t-statistic is at least as large in absolute value as the Texas t-statistic, divided by 22. The smallest attainable value, the floor, is 0.045.

Figure 26: Placebo estimates of the reform effect on release discovery

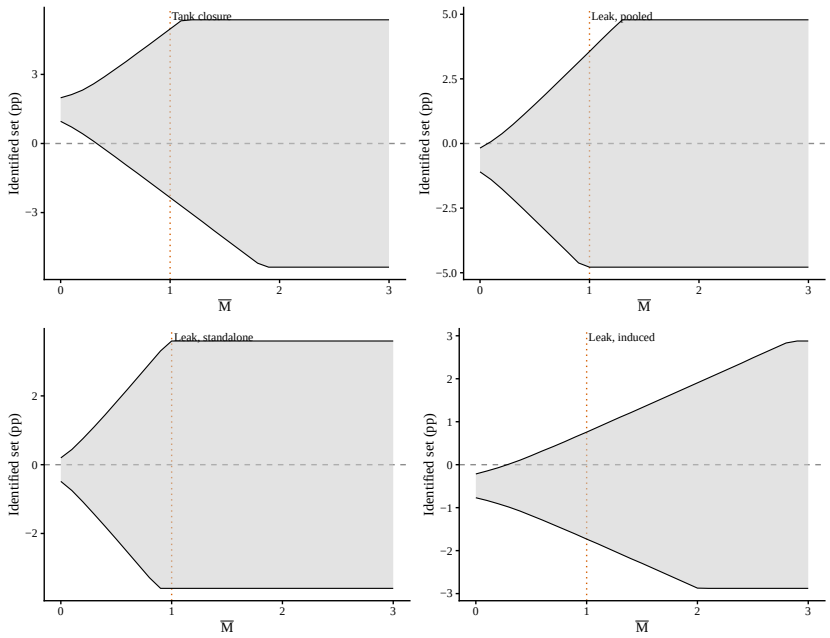


Notes: Each gray marker is the difference-in-differences estimate of the reform effect on the annual probability of the leak-discovery outcome named in the panel when one control state is assigned the reform in place of Texas. The orange marker is the estimate for Texas, and the dotted vertical line marks it. Estimates are in probability units, and 0.01 is one percentage point. The panels are any leak discovery, standalone leaks, and closure-induced leaks, as in Table 12. The unit of observation is the facility-year over 1990 to 2015, on facilities active at the December 23, 1998 reform. Each regression includes facility fixed effects, portfolio-composition-by-year fixed effects, and three indicators for the federal release-detection, spill-and-overfill, and tank-integrity deadlines. The true Texas is removed from the sample in each placebo run. Horizontal bars are 95 percent confidence intervals from standard errors clustered by state, and the dashed line marks zero. Each panel prints a randomization-inference p-value, labeled RI-t, equal to one plus the number of control states whose t-statistic is at least as large in absolute value as the Texas t-statistic, divided by 22. The smallest attainable value, the floor, is 0.045.

The parallel-trends assumption of Section 3 is not directly testable, and Figure 27 reports how large a violation could be before it would overturn the result, following the sensitivity

analysis of Rambachan and Roth (2023). Table 15 reports the same exercise as a breakdown value under two restrictions on the possible violation, relative magnitudes and smoothness, for tank closure and for each leak-discovery margin. The figure plots the 95 percent robust confidence set under the relative-magnitudes restriction as the bound on the violation relaxes.

Figure 27: Sensitivity of the average post-reform effect to a violation of parallel trends



Notes: Each panel plots the 95 percent robust confidence set of Rambachan and Roth (2023) for the average post-reform event-study coefficient, in percentage points, under the relative-magnitudes restriction. The horizontal axis is $\bar{M}$, the bound on any post-reform violation of parallel trends as a multiple of the largest violation already visible before the reform, relaxed from zero. The panels are tank closure and the pooled, standalone, and closure-induced margins of leak discovery. The shaded band lies between the lower and upper bounds of the set. The dotted vertical line marks $\bar{M}$ equal to one, and the dashed horizontal line marks zero. Each event study is aggregated into pre-reform and post-reform bins before the analysis, and the aggregation leaves the average post-reform coefficient unchanged. The confidence sets use the conditional least-favorable method. Sample, fixed effects, and clustering follow Figure 3 and Figure 5.

Table 15: Breakdown values for the parallel-trends sensitivity analysis

	Avg. post effect (s.e.)	RM breakdown $\bar{M}$	SD breakdown M
Tank closure	1.469 (0.269)	0.40	0.005
Leak discovery, pooled	-0.634 (0.239)	0.10	0.000
Leak discovery, standalone	-0.143 (0.180)	0.00	0.000
Leak discovery, closure-induced	-0.492 (0.144)	0.30	0.000

Notes: Rambachan and Roth (2023) sensitivity analysis applied to the event studies of Figure 3 and Figure 5, on the same sample, fixed effects, and state clustering. Each event study is aggregated into pre-reform and post-reform bins before the analysis, and the aggregation leaves the average post-reform coefficient unchanged. The first column is the average post-reform event-study coefficient, in percentage points, with its standard error in parentheses. The relative-magnitudes breakdown is the smallest $\bar{M}$ on the grid where the 95 percent robust confidence set contains zero, and $\bar{M}$ bounds any post-reform violation of parallel trends as a multiple of the largest violation already visible before the reform. An $\bar{M}$ of one permits a post-reform violation exactly as large as the worst pre-reform one, and a breakdown above one means the result survives a violation larger than anything in the pre-period. The smoothness breakdown is the analogous smallest bound M on the per-period deviation from a linear pre-trend where the set contains zero, in probability units, where 0.01 is one percentage point. Larger is more robust, and a breakdown of zero means the confidence set contains zero even under no violation at all.

B.10 Enforcement intensity

Section 3 raises one threat that the regulatory record cannot dispose of: Texas may have inspected its tanks harder after the reform than the control states inspected theirs, and closed tanks on that account rather than on the price. Two measures of enforcement exist for every state, how often inspectors looked and what they found. This section reports both, states what they show, and then says what the pair can and cannot tell the design. Both are level comparisons over the years after the reform. Neither is a test of a change at the reform, and nothing here is a treatment effect.

The measures. EPA publishes, for every state and fiscal year, the number of on-site inspections conducted under the Energy Policy Act and the percentage of the facilities inspected in the preceding twelve months that were found in compliance with the federal technical standards. Inspection coverage is the first count per thousand active tank systems, from fiscal 2008 through fiscal 2024. It is per tank rather than per facility, because the source publishes no state facility count. The failure rate is the complement of the second, the share of inspected facilities found out of compliance. Two standards make up the failure series. Significant operational compliance, SOC, runs from fiscal 2004 through fiscal 2021, and Texas first reports it in fiscal 2005. The technical compliance rate, TCR, replaces it from fiscal 2019 through fiscal 2024. The two are defined differently and need not share a base. States migrated from one to the other one at a time between fiscal 2019 and fiscal 2021, and in those years each standard rests on a changing subset of states. I never average,

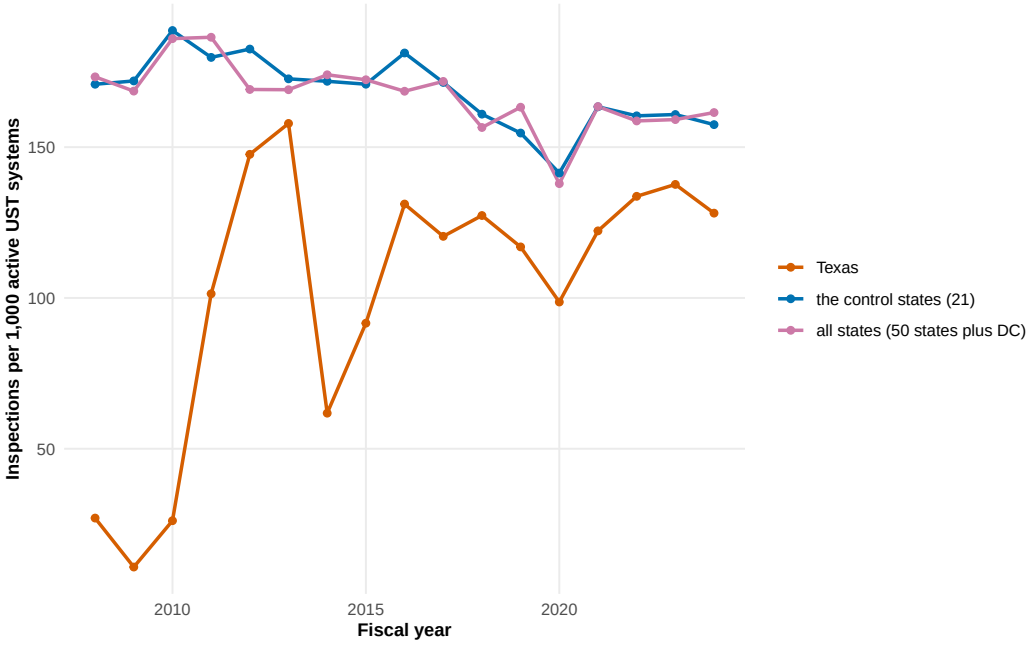
splice, or draw the two as one line.

Texas inspected less often and found less when it did. Over fiscal 2008 to 2024, Texas conducted 102.4 inspections per thousand active tanks a year, against 168.5 in the twenty-one control states of the closure design and 167.1 in all other states. Ranked from the fewest inspections per tank upward, Texas stands at 1 of 22 among the control states and itself and at 3 of 51 among the fifty states and the District of Columbia. Over the SOC years, 21.1 percent of inspected Texas facilities were found out of compliance, against 32.8 percent in the control states and 31.5 percent in all other states, gaps of -11.8 and -10.4 points. Over the TCR years the Texas rate is 15.2 percent against 42.7 and 42.7 percent, a gap of -27.5 points against either group. Ranked from the lowest failure rate upward, Texas stands at 3 of 22 under SOC and 2 of 22 under TCR among the control states and itself, and at 10 of 51 and 7 of 51 among the fifty states and the District of Columbia. Figure 28 and Figure 29 plot the two series by fiscal year. Texas reported few inspections in fiscal 2008 to 2010, and it sits below the control mean in every year of the coverage series. Table 16 and Table 17 report the gaps as regressions of the state-year measure on a Texas indicator, against all other states in Panel A and against the control states in Panel B. For the failure rate the rows are the pooled series and each standard alone, and the second column adds one effect for each standard in each fiscal year. On the pooled row those effects confine every comparison to states reporting the same standard in the same year, and the pooled gap can be read in that column alone. For coverage, with no split by standard, the second column adds fiscal-year effects. There are no state effects, because a Texas indicator that does not vary over time would be absorbed by them. Standard errors are clustered by state, and no significance test accompanies them, because one treated state and a descriptive comparison offer no sharp null. The ranks are rankings of realized state means over each panel's window, not statistics of the regressions. Both series begin after the December 1998 reform, and no reform line is drawn.

What the comparison can say. The closures a state's enforcement forces in a year are the product of three things: the share of its stock its inspectors reach, the share of the inspected facilities found in violation, and the share of violations that end in a closure. Per facility, $E_{st}/N_{st} = \rho_{st}\, y_{st}\, \pi_{st}$. The first two factors are the measures above, with coverage standing in for the inspected share per tank rather than per facility, and the third is observed nowhere. Both measured factors are lower in Texas. For the product to have been higher in Texas than in the control states, Texas would have had to close tanks per violation found at least 2.5 times as often as the controls over the SOC years and 4.6 times over the TCR years, at the inspection rates each reported. Had Texas inspected as large a share of its stock as the control states, the thresholds would be 1.5 and 2.8. That pair is the more conservative one, because a few control states inspect far more often than the rest and pull the control mean up.

What the estimator picks up is not that product but its change. Equation 3 compares the change in Texas closures at the reform with the change in the control states, and enforcement biases it only if the Texas-minus-control enforcement gap moved at the reform. The failure series begins in fiscal 2005 and the coverage series in fiscal 2008, six and nine years after the reform, and the pre-reform gap is never observed. The comparison therefore bounds how

Figure 28: Inspection coverage, Texas against the comparison states

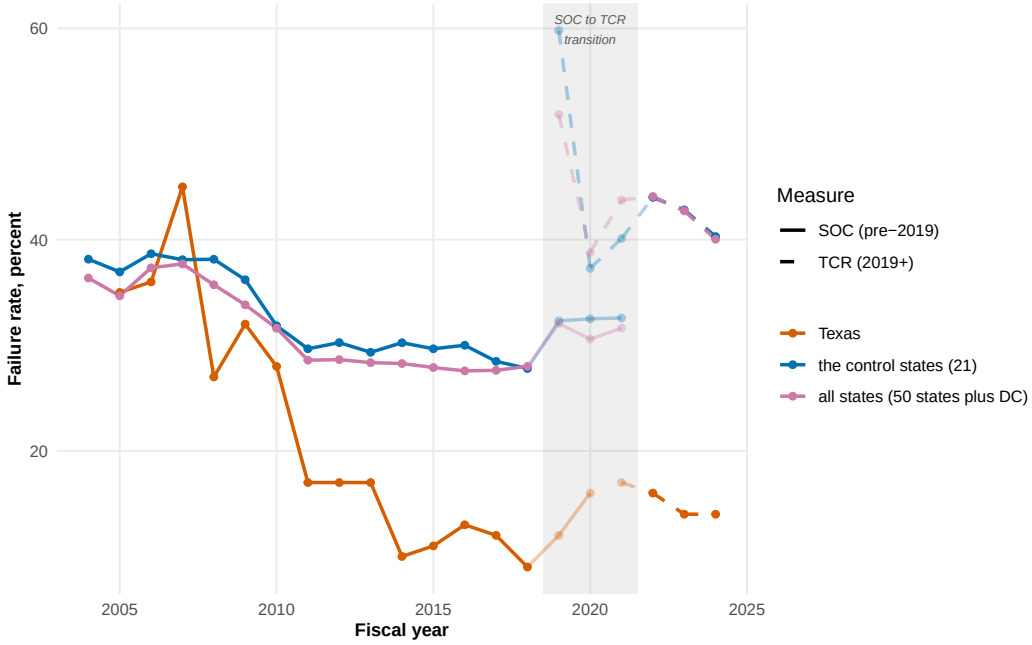


Notes: On-site inspections conducted under the Energy Policy Act per 1,000 active tank systems, by fiscal year. The measure is per tank rather than per facility, because the source publishes no state facility count. The statute's one inspection per facility every three years is therefore not drawn as a reference line. The series begins in fiscal 2008, the first year the inspection count is reported. The control states are the 21 states of the closure design, and all states are the forty-nine remaining states plus the District of Columbia. Both comparison lines are unweighted state averages and exclude Texas. Nothing in the figure is a treatment effect. Source: Environmental Protection Agency, underground storage tank performance measure reports.

large the enforcement channel can have been in the years it covers, and it cannot sign the bias. A Texas program that enforced far less than the controls before the reform and less so after it would bias the estimate upward while showing fewer inspections and fewer violations in every reported year. The years the bound does not reach include 1999, when the federal upgrade deadline moved closures in every state. The coefficients for 2001 to 2004 in Figure 3 are of the same size as those in the covered years.

Two features of the measures decide how far they describe the stock rather than the inspectors. The inspected set is fixed by statute rather than chosen. Since fiscal 2008 each year's inspections are the roughly one third of a state's facilities whose turn in the three-year cycle fell due, and three consecutive years reach every facility (EPA 2014). A year's failure rate departs from the stock's only through the order a state works its cycle, through re-inspection of facilities written up, and through repairs made between visits, and the last two push the measured rate above the stock's. The lower Texas failure rate then fits two readings, and neither is established here. Texas inspectors may have held facilities to the same standard and met a sounder stock, though that stock is itself a survivor of the reform and its condition

Figure 29: Enforcement failure rate, Texas against the comparison states



Notes: Failure rate among the facilities inspected in the preceding twelve months, 100 minus the published combined compliance rate, by fiscal year, in percent. Federal law requires every facility to be inspected at least once every three years from 2007. Each year therefore covers about one third of the stock, and any three consecutive years cover every facility in the state. Significant operational compliance (SOC, solid lines) and the technical compliance rate (TCR, dashed lines) are different standards on possibly different denominators, drawn as separate series and never spliced. The shaded years, fiscal 2019 to 2021, are the reporting transition, when states migrated individually and each line rests on a changing subset of states. The control states are the 21 states of the closure design, and all states are the forty-nine remaining states plus the District of Columbia. Both comparison lines are unweighted state averages and exclude Texas. The series begins after the Texas reform, no reform line is drawn, and nothing in the figure is a treatment effect. Source: Environmental Protection Agency, underground storage tank performance measure reports.

is not independent evidence about enforcement. Texas inspectors may instead have been less demanding, and laxity lowers the violations written up while raising the severity of those that are. The product the bias turns on is not signed by either reading. Fewer inspections per tank, the margin the threat turns on most directly, needs no such reading and runs against the threat on its own.

What remains open. The measures have no pre-reform observation, and the state's own enforcement records before 1999 are what would turn the bound into a measurement. Coverage is per tank rather than per facility. The statute's requirement is per facility, a state whose facilities hold more tanks shows lower per-tank coverage at the same per-facility frequency, and the source publishes no state facility count to convert one to the other. Thirteen

Table 16: Inspection coverage, Texas against the comparison states

	Raw	Year FE
Panel A. Texas versus all states (50 states plus DC)		
Inspection coverage	-64.76	-64.66
	(10.72)	(10.80)
$N = 859$, states = 51, fiscal years = 17		
Panel B. Texas versus the control states (21)		
Inspection coverage	-66.11	-65.93
	(20.02)	(20.42)
$N = 368$, states = 22, fiscal years = 17		
Control mean, excluding compartment reporters: 170.7 per 1,000 active tanks		
Year FE	No	Yes

Notes: State by fiscal year, fiscal 2008 to 2024. The outcome is on-site inspections per 1,000 active tank systems, counted per tank rather than per facility, because the source publishes no state facility count. Panel A compares Texas against all states, Panel B against the control states of the closure design. Cells report the coefficient on a Texas indicator, in inspections per 1,000 active tanks, with the standard error clustered by state in parentheses beneath. The second column adds fiscal-year effects. No state effects enter, because a time-invariant Texas indicator would be absorbed by them. Ranked from the fewest inspections per tank upward, Texas stands at 1 of 22 in Panel B and at 3 of 51 in Panel A. These are rankings of realized state means over the window and are not statistics of the regressions. The control mean excluding the states that report tank compartments rather than tanks is shown inside the table, because those states' inflated denominators understate their coverage. Nothing in the table is a treatment effect, and no significance test is reported. Source: Environmental Protection Agency, underground storage tank performance measure reports.

states report tank compartments rather than tanks, several of them control states, and their inflated denominators understate their coverage. Excluding them raises the control mean to 170.7 and widens the gap. Texas reported few inspections in fiscal 2008 to 2010, the first three years of the coverage series, and its coverage from fiscal 2011 sits closer to the control mean while remaining below it in every year. The Texas SOC mean pools fiscal 2005 to 2007, before the statutory cycle was complete. The state means are unweighted while the closure design weights tank-years within cells. The closures forced per violation are not observed.

Table 17: Enforcement failure rate, Texas against the comparison states

	Raw	Standard × year FE
Panel A. Texas versus all states (50 states plus DC)		
All measures (pooled SOC and TCR)	-13.87	-13.54
	(1.47)	(1.48)
N = 1,018, states = 51, fiscal years = 21		
SOC	-10.41	-10.08
	(1.50)	(1.51)
N = 811, states = 51, fiscal years = 18		
TCR	-27.48	-27.40
	(2.70)	(2.70)
N = 207, states = 51, fiscal years = 6		
Panel B. Texas versus the control states (21)		
All measures (pooled SOC and TCR)	-14.82	-14.44
	(2.26)	(2.32)
N = 451, states = 22, fiscal years = 21		
SOC	-11.75	-11.44
	(2.41)	(2.44)
N = 364, states = 22, fiscal years = 18		
TCR	-27.47	-26.60
	(3.60)	(4.02)
N = 87, states = 22, fiscal years = 6		
Standard × year FE	No	Yes

Notes: State by fiscal year. The outcome is the failure rate among facilities inspected in the preceding twelve months, 100 minus the published combined compliance rate, in percent. Panel A compares Texas against all states, Panel B against the control states of the closure design. Cells report the coefficient on a Texas indicator, in percentage points, with the standard error clustered by state in parentheses beneath. The second column adds one effect per standard and fiscal year, and on the pooled row that confines each comparison to states reporting the same standard in the same year. No state effects enter. SOC, significant operational compliance, spans fiscal 2004 to 2021 and TCR, the technical compliance rate, fiscal 2019 to 2024. The fiscal 2019 to 2021 TCR cells rest on few states. Ranks run from the lowest realized mean failure rate upward. A low rank is a low failure rate. Texas stands at 3 of 22 under SOC and 2 of 22 under TCR in Panel B, and at 10 of 51 and 7 of 51 in Panel A. The ranks are of realized state means over each panel's window, not statistics of the regressions. The series begins after the Texas reform, nothing in the table is a treatment effect, and no significance test is reported. Source: Environmental Protection Agency, underground storage tank performance measure reports.

C The Claims-Implied Premium

This appendix asks whether the premiums Texas carriers file track the expected cost of the risk they insure. The benchmark is the claims-implied premium, or fair premium, the expected annual cleanup cost of a facility's leak risk. I build it first, set the filed premiums against it second, and state last what the comparison cannot show.

The cost object here is not the one Section 2.3 quotes. That section prices the risk the structural model consumes, a national age-by-wall hazard multiplied by the claims-fitted severity of the cell, measured at the facility. The fair premium below differs on three counts. It uses each facility's own fitted hazard in place of its cell average. It measures severity as the amount the fund pays after the operator's deductible rather than the gross cost of the cleanup. It then splits the facility figure across tanks. The two objects answer different questions and need not agree in level.

C.1 The fair premium

The fair premium is the expected cost of a facility's leak risk in a year, the product of how often it leaks and what a leak costs,

$$P_i^{\text{fair}} \;=\; \hat{h}_i \, \hat{S}_i .$$

The frequency $\hat{h}_i$ is the facility's fitted first-release probability from Section 2.3. The severity $\hat{S}_i$ is its fitted cleanup cost net of the deductible the facility bears, making it the share the state fund pays. Both are conditional means rather than realized amounts. Their product is what a break-even insurer would need to collect from facilities like this one, not what any single facility went on to cost. A facility charged less than its own expected cost has the remainder paid by someone else, in this case by the gasoline taxpayers who finance the fund and by the safer facilities in the pool. Both objects are measured at the facility, and the frequency has to be, because a release is recorded against the facility and is never attributable to a particular tank.

The severity needs its own data, because a cleanup cost is observed only when a facility files a claim. I collected trust-fund claims records from six states: Colorado, Louisiana, New Mexico, Pennsylvania, Tennessee, and Utah. All six enter the model fit, since the fit needs every cleanup it can get. Texas has no fund and files no claims. Its severity is taken out of sample as the average state effect across the fit states excluding Louisiana, an offset of -0.33 on the log scale. A claim is the amount a fund paid to clean up one confirmed release. Each claim links to the facility that filed it through the state's site registration number, attaching to the claim the same traits the hazard model uses: tank age, wall construction, fuel, and capacity. The result is 12,168 claims, each one cleanup paired with the traits of the facility it came from.

Harmonizing the cost field takes one decision per state. I measure the fund-payable cleanup cost per release, net of the facility's deductible, deflated to 2023 dollars by the CPI-U, BLS series CPIAUCSL, rebased to the 2023 annual average. Colorado, New Mexico, and Tennessee report the closed, realized cost of completed cleanups. Colorado's field is the

amount reimbursed to the operator, and Tennessee's is the paid amount on closed claims only. Pennsylvania reports the ultimate incurred loss. That measure is paid cost plus case reserves, and it retains large claims while they remain open. The state's most expensive releases are therefore not truncated. The states also record the operator's share differently, and I take each at the point where its own records already net it out rather than imposing one rule on all. Each of the three above is fund-paid by construction. New Mexico is the one state whose recorded amount is gross of the operator's share, and I subtract its $10,000 deductible directly. Louisiana reports a recommended amount across applications, and I divide by the application count to reach one release. Utah reports the amount its petroleum storage tank fund paid on each release, and I use it as reported. Appendix A documents each state's source field.

From these claims I fit a model mapping a facility's traits to its expected cleanup cost. Every facility then carries a fitted severity whether or not it ever filed a claim. Two features of cleanup costs determine the estimator. Many releases cost the fund nothing, because the cleanup falls below the deductible, and the claims that do pay have a heavy right tail. Poisson pseudo-maximum-likelihood handles both (Santos Silva and Tenreyro 2006). A regression of log cost cannot use the zeros, since the log of zero is undefined, and discarding them would select on the outcome. Poisson pseudo-maximum-likelihood retains them. It also estimates the dollar mean $\mathrm{E}[C \mid x]$ directly rather than the mean of log cost, avoiding a retransformation that is biased when the error variance moves with the covariates, and it stays consistent under heteroskedasticity.

The features that carry the fit govern how much contaminated material a release puts in the ground and how hard it is to reach. An older tank has been leaking longer before anyone notices, a single wall puts fuel directly against soil, and a larger tank holds more of it. Their interactions matter for the same reason, because a large old single-walled tank is not the sum of three separate penalties. The model is

$$\mathrm{E}[C_i \mid x_i] = \exp(\beta_0 + g(x_i) + \alpha_{s(i)}),$$

where C_i is the fund-payable cost of claim i in 2023 dollars, x_i holds the facility's traits and every two-way interaction among tank age band, single-walled status, tank count, capacity, and the two fuel indicators, $g(\cdot)$ is the function of them an elastic net selects, and $\alpha_{s(i)}$ is a state effect.

Its construction mirrors the hazard model at every step, with a log link and a dollar outcome in place of a logit and an indicator. It uses the same trait set, the same elastic-net selection tuned by cross-validation, and the same out-of-sample scoring. Folds are drawn by facility and stratified by state, and no facility spans training and holdout. Selection does more work here than in the hazard model, because the trait set is rich and the claims sample is three orders of magnitude smaller. Cross-validation again lands at the lasso corner, $\alpha = 1$, and retains 62 features. A tank inherits its facility's composition features alongside its own age and wall.

Pennsylvania's fund supplies an out-of-model check. Its actuarial review implies about $162,000 per claim and a yearly claim rate near 2.5 to 3 percent (AON, 2022 USTIF Performance Review). The model here fits a median covered severity of $158,600 and a mean

hazard of 2.3 percent for the same state. Both rest on the same claims. The match therefore confirms that the pipeline reproduces the fund's numbers rather than validating them independently.

Risk rating prices tanks while the fair premium is measured at the facility, and the comparison with filed premiums therefore needs a premium for each tank. I split a facility's premium across its tanks in proportion to each tank's share of the facility's expected loss,

$$P_{ij}^{\text{fair}} \;=\; P_i^{\text{fair}}\, w_{ij}, \qquad w_{ij} \;=\; \frac{r_{ij}\,\widehat{S}_{ij}}{\sum_k r_{ik}\,\widehat{S}_{ik}}.$$

The relativity r_{ij} is the national age-by-wall hazard entering only as a within-facility ratio, and $\widehat{S}_{ij}$ is the tank's fitted cleanup cost. The split conserves the facility premium by construction, $\sum_j P_{ij}^{\text{fair}} = P_i^{\text{fair}}$, and therefore redistributes risk across a facility's tanks without re-leveling it. This is ordinary ratemaking practice. A rate is equitable when "differences in rates reflect material differences in expected cost" (Actuarial Standards Board, ASOP No. 12, §3.2.1), and the fair premium is that benchmark.

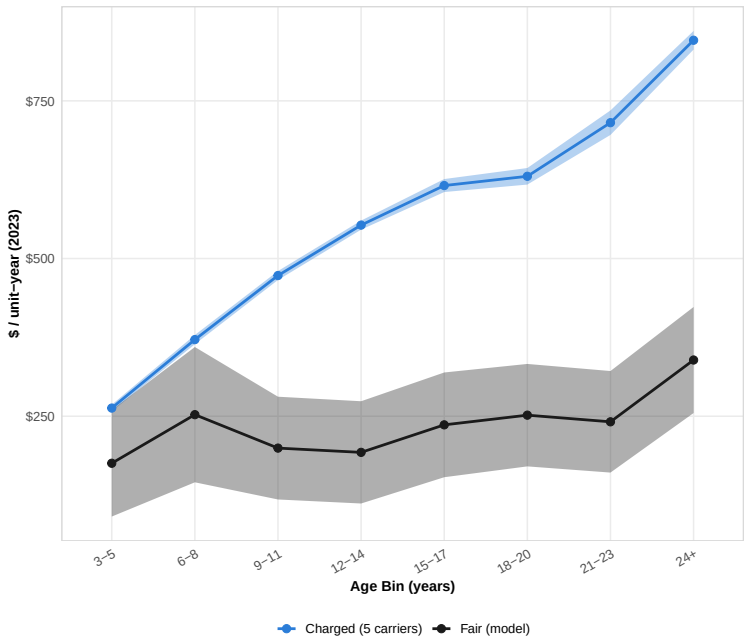
Both first-stage models carry estimation error, and I carry it into the premium. The two are fit on independent samples, the hazard on the facility-year panel and the cost model on the claims, and I treat their errors as independent, combining them by the delta method, $\operatorname{Var}(h\widehat{S}) = \widehat{S}^2 \operatorname{Var}(h) + h^2 \operatorname{Var}(\widehat{S})$. The hazard variance is the analytic cluster-robust interval the risk model produces at the age-by-wall cell level, and each facility inherits its cell's hazard uncertainty. The cost variance is a facility-clustered bootstrap. The reported band is $h\widehat{S} \pm 1.96\,\widehat{\text{SE}}$, floored at zero.

C.2 Filed premiums against the fair premium

The comparison that adds the filed minimum policy premium to the fair premium is Figure 2 in Section 2.4.

Figure 30 shows the same comparison without the minimum-premium adjustment. On the raw comparison the charged premium runs 2.48 times the fair premium pooled, and most of that gap is the minimum policy premium, the floor below which no carrier will write. Weighted across carriers and deflated to 2023 dollars, that minimum is $563 per policy-year, or $219 per tank-year split across a facility's tanks, and it binds for 15.0 percent of Texas facilities. Netting the floor out as the fixed cost of writing a policy puts the implied ratio at 1.42 times the fair premium at the tank and 1.44 at the facility, ranging from 0.67 to 1.55 across age bins. Which of the two comparisons is the right one depends on whether the floor is read as an expense load or as a price, and the filings state it as a floor rather than as a stated fee. Treating it as an additive fixed cost is therefore an interpretation. The comparison also holds the contract fixed, at coverage above the standard base deductible up to the federal minimum one-million-dollar per-occurrence limit, because facilities' actual deductible and coverage-limit choices are unobserved.

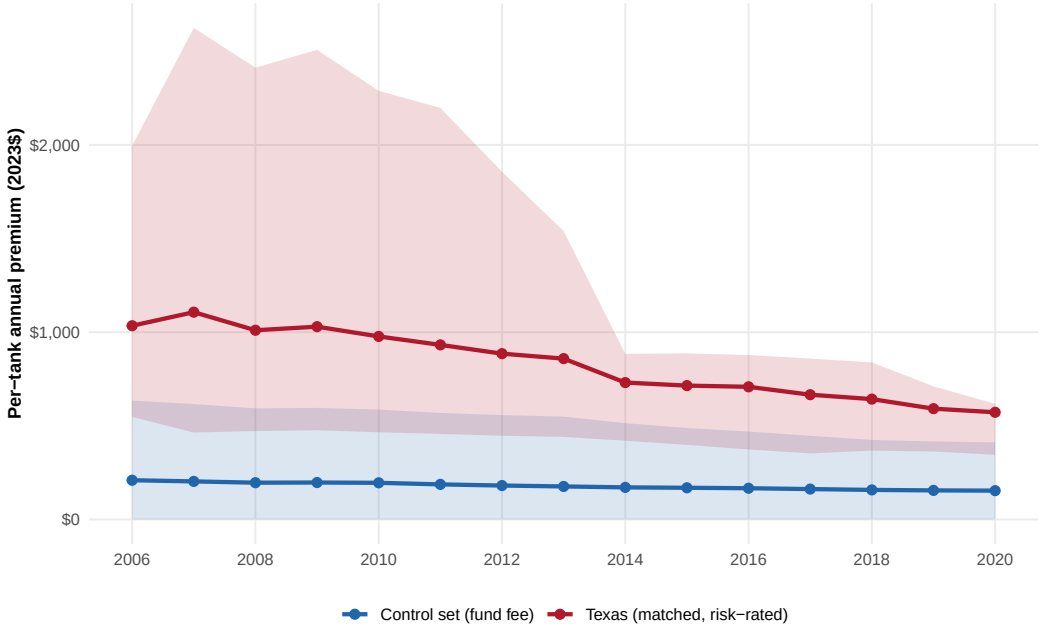
Figure 30: Filed Texas premiums against the fair premium, without the minimum-premium adjustment



Notes: Mean charged premium (blue line) and mean fair premium (black line) per tank-year, by three-year bin of tank age, for Texas tanks in 2015, in 2023 dollars. The figure plots 17,718 tanks. The charged premium applies the filed rating manual of each of five carriers to the insured tanks of the facility. The charged premium is for the standard contract, with coverage between the base deductible and the federal minimum limit of one million dollars per occurrence. The fair premium is the tank's share of its facility's expected cleanup cost, a release rate times a fitted cleanup cost. The fair premium in this figure carries no minimum policy premium, and Figure 2 in Section 2.4 reports the adjusted comparison. Shaded bands are 95 percent intervals. The fair premium band carries the estimation error of the release-rate and cleanup-cost models, and the charged-premium band carries the sampling error of the mean.

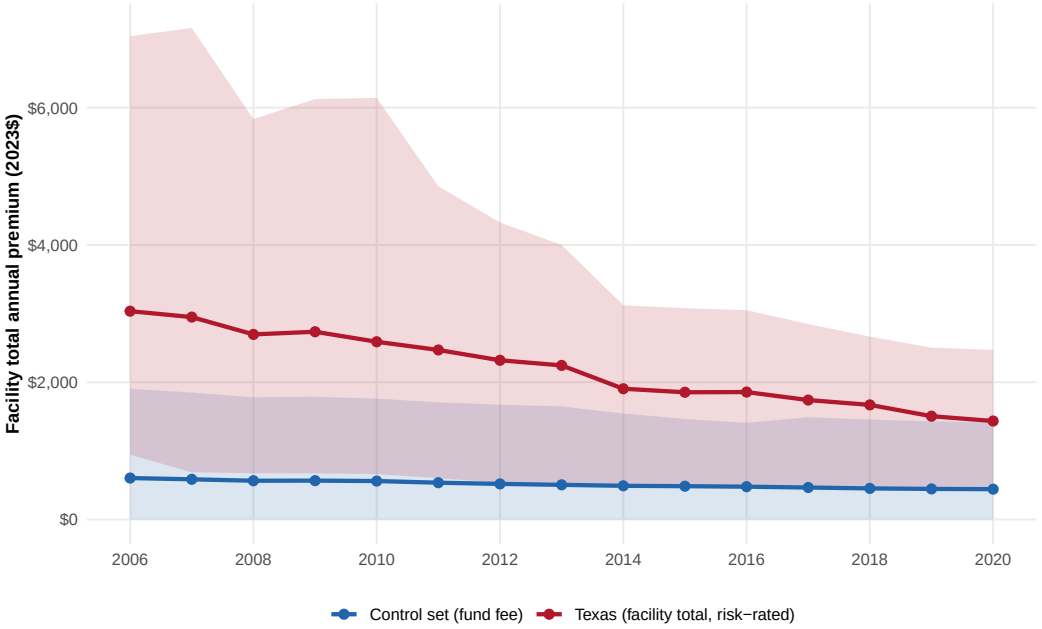
The price path over the risk-rated era shows the reform working through composition. In every year from 2006 the matched Texas sample pays far more per tank than the control states pay in fund fees, and that gap is the price of risk the reform introduced (Figure 31). The Texas per-tank premium falls across the period, from $1,035 in 2006 to $573 in 2020, while the control fee drifts down more gently, from $210 to $154. The Texas decline is composition rather than a rate cut. As the reform retires the oldest and priciest tanks, the tanks that remain are younger and cheaper to insure, and the average surviving premium therefore falls while its spread across facilities narrows. What a facility pays in total falls harder still, from $3,035 to $1,435, because a facility both sheds tanks, from about 2.93 to 2.51 apiece, and pays less on the tanks it keeps (Figure 32). The level gap is the repricing of risk that Section 3 exploits, and the decline over the period is a first look at the closure response that design measures directly.

Figure 31: Per-tank price of coverage, matched Texas sample against the control fee



Notes: Mean annual price of coverage per tank by year, 2006 to 2020, in 2023 dollars. The Texas series (red line) covers the insured tanks of the matched estimation sample, priced under the filed rating manuals of five carriers. Each tank receives the mean premium that its carrier's manual yields for tanks of the same wall construction and five-year age bin in that year. The control series (blue line) is the per-tank fund fee in force, averaged over facilities in the control states of the matched sample. Shaded bands span the 10th to 90th percentiles across tanks for Texas and across facilities for the control states.

Figure 32: Annual premium per facility, Texas against the control fee



Notes: Annual price of coverage per facility by year, 2006 to 2020, in 2023 dollars. The Texas series (red line) covers every Texas facility that one of five carriers prices. Each insured tank receives the mean premium that its carrier's filed rating manual yields for tanks of the same wall construction and five-year age bin in that year, and the series sums those premiums over the facility's tanks. The control series (blue line) is the per-tank fund fee in force times the facility's active tank count, in the control states of the matched sample. Lines are means across facilities, and shaded bands span the 10th to 90th percentiles. The Texas figures are filed premiums and are not amounts recorded on policies.

C.3 What the comparison cannot show

The level of the fair premium rests on pairing a leak frequency from the release database with a claim severity from the fund database, two different event populations. Texas files no claims, and its severity is therefore predicted from the claims states. The severity is a fitted mean over a heavily right-skewed distribution, with a pooled median claim of $106,000 against a mean of $348,000. A pure premium is correctly built on the mean, and that is why these premiums look large beside a typical cleanup.

None of this measures welfare. The comparison shows how filed premiums sit against expected cost, and over- or under-charging risk is welfare-ambiguous on its own. Whether a premium that tracks cost raises welfare depends on responses the comparison cannot see. Firms may respond to the premium or ignore it. They may pass it into pump prices, where consumers respond in turn. Nothing in a static comparison settles any of that. Section 5 prices those responses directly.

D Structural Model Details

D.1 Constructing measured revenue

The model needs a measure of what a facility earns, and the data do not provide one directly. I observe no facility's own gallons sold and no facility's own prices. What I do observe is every facility's storage capacity and the public price series that bracket what a gallon of gasoline earns its seller, and I build revenue from those, with one market assumption connecting them: a facility's throughput moves with its storage capacity. The construction is deliberately coarse, since facilities in the same bin earn different revenues for reasons capacity cannot capture, and I return to what that restriction costs below.

The inputs are public: the retail gasoline price and the state gasoline tax, by state and year, come from the EIA State Energy Data System and the FHWA tax-rate series. The wholesale price refiners charge resellers comes from the EIA refiner sales-for-resale prices, reported by refining region. The federal gasoline tax is \$0.184 a gallon, flat over the sample. Facility size comes from the tank registry, where I sum each facility's tank capacity and sort facilities into the four bins of Section 2.2. Every input is an administrative price or a registry capacity, which makes the measure available for all facilities in all years, with none of it drawing on a facility's own accounts. The series cover the 18 study states from 1999 to 2020.

One state's early-period margin is bounded, not measured. North Carolina sits in the Lower Atlantic, which the Gulf supplies through the Colonial Pipeline, but EIA reports wholesale prices only for the whole East Coast region, which is pulled up by the Northeast. With no sub-region wholesale price published, North Carolina's margin before 2014 is floored at a small positive value. It is one cell of fifty-four, and the state fixed effect absorbs the rest.

Because most of the retail price of gasoline is the cost of the fuel itself, the retail price is a poor measure of what a station clears. The right measure is the margin: the retail price net of the wholesale price, the state tax, and the federal tax,

$$m_{g,y} = P^{\text{ret}}_{g,y} - P^{\text{whl}}_{r(g),y} - \tau^{\text{st}}_{g,y} - \tau^{\text{fed}},$$

where g indexes the state, y the year, $r(g)$ the state's refining region, and $\tau^{\text{fed}} = \$0.184$. The retail price and the state tax vary by state. The wholesale price varies by region. The margin $m_{g,y}$ is what a station keeps on a gallon. I hold it in nominal dollars, to match the nominal premium and deductible the facility pays.

Revenue is that margin scaled to the fuel a facility moves, and throughput scales with tank size, and I therefore proxy a facility's gallons by its capacity bin rather than by volumes I do not observe. Each tank's capacity is capped at 60,000 gallons before summing, which keeps a few very large facilities from stretching their bin's mean, and each facility then carries the mean capped capacity of its bin, $\bar{c}_B$. The margin is averaged within each of three rate eras, 1999 to 2013, 2014 to 2018, and 2019 to 2020, written $\bar{m}_{g,e}$, and a facility therefore forecasts a steady local margin rather than a yearly swing. Measured operating revenue is the product, in model units of $\Omega = \$10{,}000$,

$$R_{B,g,e} = \frac{\bar{c}_B \cdot \bar{m}_{g,e}}{\Omega}.$$

A facility operating in capacity bin B, state g, and era e earns $R_{B,g,e}$. Revenue is thus a small table with one entry per size bin, state, and era, read off the facility's current bin.

This construction imposes one behavioral assumption: operating revenue is exogenous to the facility's dynamic problem, a payoff attached to the facility's state, not a quantity the firm manages. First, revenue is pinned down by three observables, the size bin, the state, and the era, and the only lever a firm has on its revenue is therefore which capacity bin it occupies, and it moves between bins only through the capital choices the model already prices, downsizing and replacement. It has no other lever on revenue: throughput, effort, and retail markup are outside the model. Second, two facilities in the same bin, state, and era earn the identical $R_{B,g,e}$, which assumes away idiosyncratic profitability. There is no unobserved station that quietly earns more in a way that would jointly raise its sales and its survival, which is the endogeneity the Sweeting two-step is built to shut down. Third, because I average the margin within an era, the firm treats the local margin as the permanent flow it keeps, and within-era gasoline-price swings therefore do not drive the timing of exit or replacement. The cost of these restrictions is that any true within-cell revenue heterogeneity is loaded onto the choice shocks and onto ψ rather than modeled. The return is that revenue enters the payoff as a clean, exogenous state, which is exactly what lets ψ be identified from size-by-margin variation rather than from the firm's own conduct.

D.2 Computing the value function

The linear system $(I - \beta M)V = R$ has one row per state, 298,448 in each of the 104 environments, and the matrix is never formed. The solve iterates the contraction $V \mapsto R + \beta MV$, whose convergence rate is bounded by $\beta = 0.95$, with the matrix-vector product computed in compiled code that applies the action maps, the aging kernel, and the choice weights cell by cell rather than through a stored matrix. Anderson acceleration of depth five replaces each iterate with a least-squares combination of the last five, and it cuts the iteration count substantially relative to plain successive approximation. Each solve runs to an infinity-norm residual below 10^{-9} and halts with an error rather than returning an unconverged value function. The estimator of Section 5 repeats this solve at every parameter evaluation, once per environment, and the counterfactuals of Section 6 repeat it under each policy.

The estimation converges. The nested pseudo-likelihood loop stops at iteration 13, when the parameter vector and the choice probabilities each change by less than 10^{-5} between successive iterations. The choice-probability update is damped at 0.6. The inner maximization at each iteration runs to a tighter tolerance than the outer loop, and the measured step therefore reflects the estimate and not the optimizer's own precision.

D.3 Model objects and fit

The dynamic model treats a tank's age as a Markov process. Each year a surviving tank either advances one five-year band or stays, with a probability fixed by the band it currently occupies. Table 18 reports the empirical transition rates behind that process. A tank stays in its own band with probability close to four in five in every band from the first through

the seventh, and it advances with the complementary probability. Band eight holds tanks past thirty-five years. No tank observed there ever leaves, and the kernel makes the band absorbing by construction. This kernel is the facility's own forecast of its fleet's aging, estimated once from the registry and imposed on every environment the model solves.

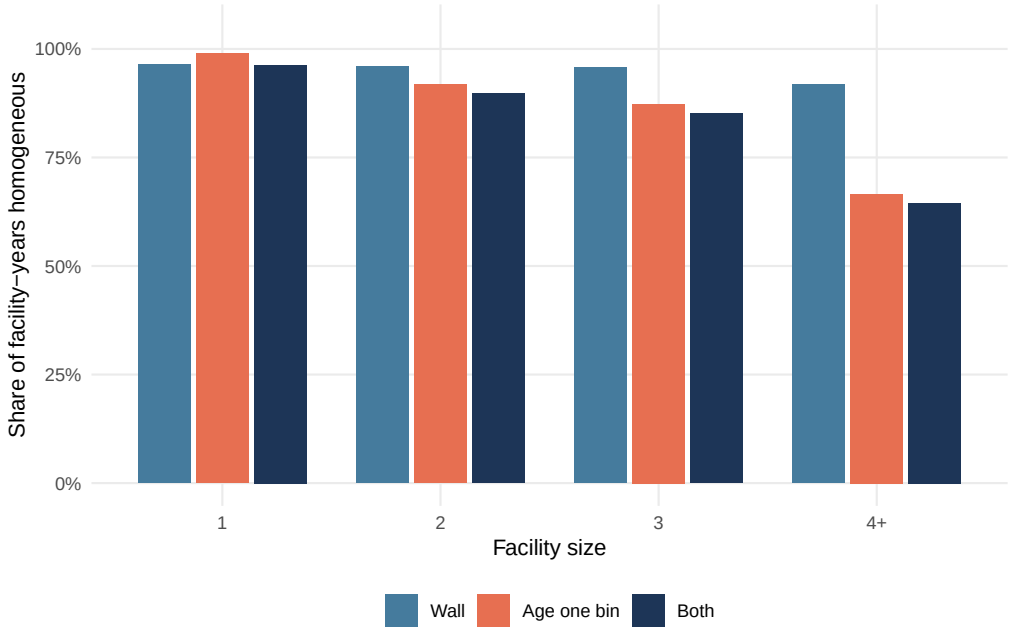
Table 18: Age-band transition kernel

Age band	Stay	Advance
1	0.770	0.230
2	0.769	0.231
3	0.776	0.224
4	0.795	0.205
5	0.812	0.188
6	0.814	0.186
7	0.831	0.169
8	1.000	0.000

Notes: Empirical one-year transition probabilities between tank age bands. The unit of observation is a tank-year. The sample is the 6,949,981 tank-years of the tank-year panel from 1999 to 2020 for tanks that are not closed in the year and have a recorded age. Age is the calendar year less the installation year. Band 1 is ages 0 to 4, each higher band covers the next five years, and band 8 is 35 years and older. Each tank-year is assigned the band of its age and the band of its age plus one year. Stay is the share of a band's tank-years whose two bands are the same, and Advance is the share whose second band is the next one up. A tank moves at most one band in a year, and Stay and Advance sum to one. Band 8 has no higher band, and its Stay is one by construction.

The portfolio model tracks a facility through a small set of tank cells rather than through each tank's individual history, and that simplification needs its own justification. Figure 33 reports how often a facility's active tanks actually differ from one another, by wall type and by installation vintage, at each facility size. Tanks installed at the same station are typically installed together and age together. The median facility carries a zero-year spread in tank age at every size, and most facilities operate tanks of a single wall type. The state space the model solves is therefore not an approximation forced by computation. It matches how stations actually build and replace their tanks.

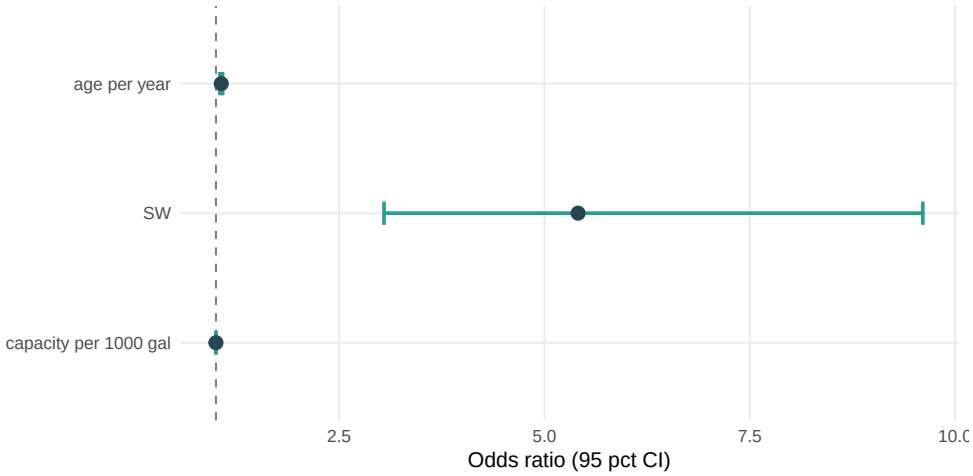
Figure 33: Within-facility tank homogeneity by size



Notes: Bars show, by facility size, the percent of facility-years with homogeneous tanks. The bars labeled Wall give the percent with all tanks of one wall type, single-walled or double-walled. The bars labeled Age one bin give the percent with the oldest and the youngest tank in the same five-year age band. The bars labeled Both give the percent that meet both conditions. The age bands are ages 0 to 4, 5 to 9, and so on up to 35 years and older. Facility size is the number of tanks at the start of the year: one, two, three, or four or more. The sample is the 2,193,723 facility-years with a recorded tank age in Texas from 2006 to 2020 and in the control states from 1999 to 2020.

The model assigns which tank a facility removes by a fixed rule: the oldest single-walled tank first, then the oldest double-walled tank. Figure 34 and Table 19 ask whether that rule describes what operators actually do, using a conditional logit that compares each removed tank only against the other tanks standing at the same facility in the same year. Wall type and age carry the removal decision. A single-walled tank faces far higher removal odds than a double-walled one, and each additional year of age raises the odds further. Capacity carries no weight at all. Operators remove the tank the post-reform price schedule targets. That is exactly the tank the model's rule names.

Figure 34: Removal odds ratios, within facility



Notes: Odds ratios from the conditional logit of Table 19. Each facility-year with exactly one removal compares the removed tank with the other tanks standing at the facility that year. Each point is the odds ratio of removal for one more year of tank age, for a single-walled tank against any other tank, and for 1,000 more gallons of capacity. Horizontal lines are 95 percent confidence intervals from standard errors clustered by state, and the dashed vertical line at one marks equal odds of removal. The sample is the 27,591 facility-years of the tank-year panel, 1985 to 2020, with two or more tanks and exactly one removal, and the 104,838 tank-years of their tanks.

Table 19: Which tank is removed: within-facility comparison

Closed tank attribute	OR	SE	p
age per year	1.065	0.008	0.000
SW	5.410	0.293	0.000
capacity per 1000 gal	1.000	0.000	0.861

Notes: Conditional logit of which tank a facility removes in a year with exactly one removal. The unit of observation is a tank-year. Each facility-year is one choice set with the removed tank as the outcome and the other tanks standing at the facility as the alternatives. Only differences among the tanks of one facility in one year identify the coefficients, and facility characteristics such as location, demand, owner, and size drop out. The sample is the 27,591 facility-years of the tank-year panel, 1985 to 2020, with two or more tanks and exactly one removal, and the 104,838 tank-years of their tanks. Age is in years, SW is an indicator that the tank is single-walled with every other tank as the comparison, and capacity is in thousands of gallons. OR is the odds ratio, and a value above one means that a tank with more of the variable is more likely to be the one removed. SE is the standard error of the log odds ratio, clustered by state, and p is the two-sided p-value from that standard error.

Table 20: Removal-rule fidelity and the cost of misclassification

Panel A: How often the removal rule names the right tank

Action	Size	N	Match	Wrong wall
Downsize	1	62	1.000	0.000
Downsize	2	1,231	0.784	0.041
Downsize	3	1,706	0.790	0.038
Downsize	4+	5,180	0.709	0.053
Replace	1	580	1.000	0.000
Replace	2	1,195	0.986	0.003
Replace	3	2,737	0.986	0.004
Replace	4+	2,835	0.930	0.014

Panel B: Premium and hazard error on the misclassified events

Action	Size	Misses	Mean premium err	p90 premium err	Mean hazard err pp
Downsize	2	266	58	124	0.210
Downsize	3	359	53	124	0.125
Downsize	4+	1,505	76	159	0.084
Replace	2	17	65	138	0.305
Replace	3	39	55	124	0.168
Replace	4+	198	82	182	0.107

Notes: The structural model assigns which tank leaves a facility by rule: the oldest single-walled tank first, then the oldest double-walled tank. The sample is the 15,526 facility-years from 1999 to 2020 with at least one tank removed and no exit. A downsize installs no tank in the year and a replacement installs at least one. Size is the number of tanks at the start of the year, and 4+ means four or more. Panel A scores the rule against the tanks the facility removed. N is the number of events. Match is the share of events whose removed tanks fall in exactly the wall-and-age-band cells the rule names. Wrong wall is the share of events with a double-walled tank removed and a single-walled tank kept. Panel B covers the events the rule misses, and Misses is their number. Premium errors are absolute differences, in nominal dollars per year, between the premium on the tanks removed and the premium on the tanks the rule removes, with each tank priced on the Texas rate schedule for its wall type and age band in the filing era of the event year. Hazard errors are absolute differences, in percentage points, between the annual leak hazard of the facility's remaining tanks after the rule's removal and after the actual removal, evaluated at their average age band and majority wall type. Mean is the average over a row's events, p90 is the ninetieth percentile, err means error, and pp means percentage points. Source: insurer rate filings for the Texas rate schedule.

A rule that matches the average pattern can still misname the tank in any single event, and the model needs to know how costly that is. Table 20 scores the rule event by event against every observed downsize and replacement. The rule names the correct tank in 84.6 percent

of 15,526 events, the number Section 5 reports for assumption A1. Its errors concentrate in one narrow cell, replacements at four-or-more-tank facilities that shed a single tank, and even there the errors are mostly about which age band the tank falls in rather than its wall type. Panel B prices what those errors cost the model. A misclassified tank carries a small error in its premium and its hazard, and neither error moves the states the model tracks by an amount that would change a facility's choice.

The model reproduces the aggregate action shares and fits less well within Texas. Table 21 reports the observed and predicted share of facility-years in each action, for the full estimation sample and separately for Texas and the control states. Within Texas the model predicts more downsizing and less exit than the data show. Section 6 discusses that pattern.

Table 21: Model fit: predicted versus observed action shares

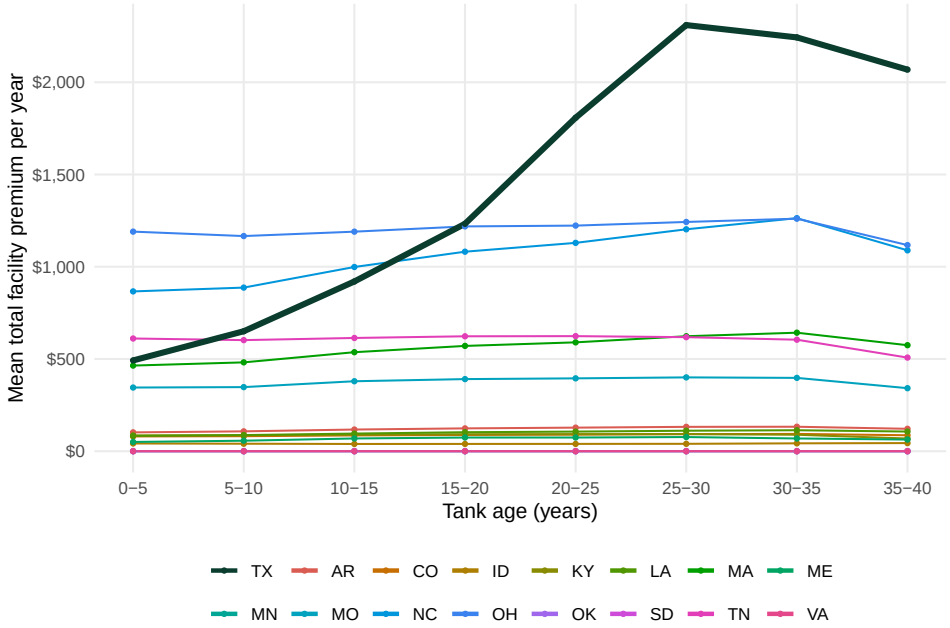
	Observed	Predicted
All facility-years		
Maintain	0.978	0.974
Downsize	0.003	0.006
Replace	0.003	0.004
Exit	0.016	0.016
Texas, risk-rated premiums		
Maintain	0.987	0.982
Downsize	0.001	0.011
Replace	0.004	0.005
Exit	0.007	0.001
Control states, flat fees		
Maintain	0.977	0.973
Downsize	0.003	0.005
Replace	0.003	0.004
Exit	0.017	0.018

Notes: Each cell is the share of facility-years in which the group takes the action, and the four shares sum to one within a group up to rounding. Maintain is a year with no removal and no install, downsize a removal of at least one tank with no install, replace a removal with at least one install in the same year, and exit the closure of the last tank. Facility-years that install a tank and remove none are not in the estimation sample. The sample is the estimation sample of the structural model, 1,654,626 facility-years, Texas from 2006 to 2020 and 14 control states from 1999 to 2020. Observed shares are the raw frequencies in that sample. Predicted shares are the model's choice probabilities at the estimated parameters, averaged over the same facility-years. The shares enter the estimation only through the likelihood and are not moments matched by construction.

D.4 Identification figures

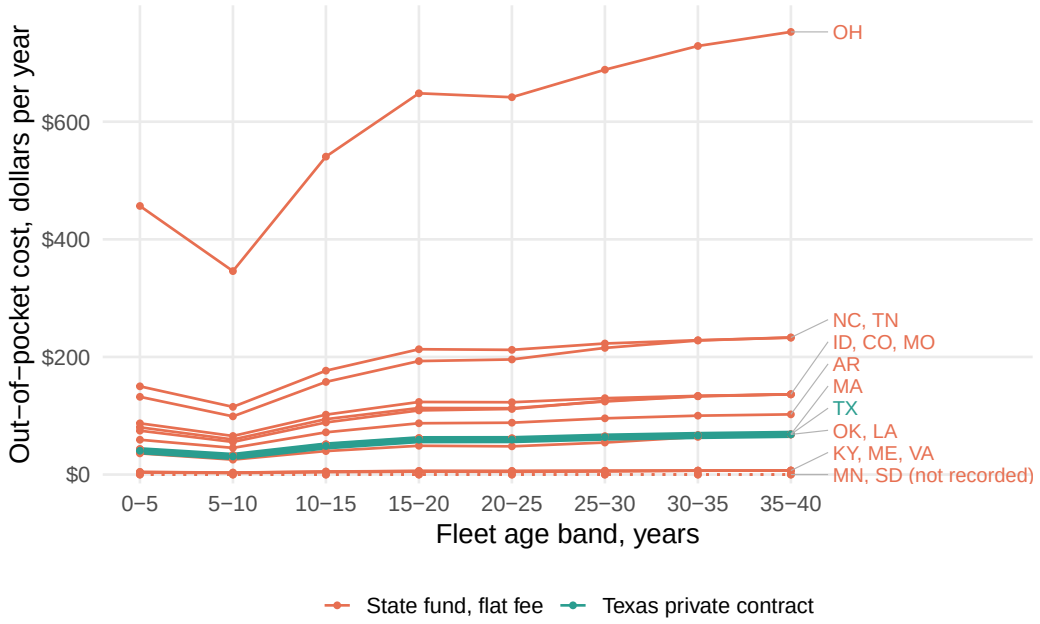
The identification argument of Section 5 rests on two pieces of price variation, and both are first-stage inputs fixed before any dynamic parameter is estimated. Figure 35 plots the premium side, the filed premium a facility faces as its fleet ages, under the Texas risk-based schedules and under the control funds' flat fees. Figure 36 plots the out-of-pocket side, the expected cost a facility carries itself after a release, its annual leak hazard times its state's deductible.

Figure 35: Mean facility premium by fleet age, Texas against the flat-fee control states



Notes: Mean total facility premium per year, in nominal dollars, by the facility's average tank age in five-year bands. Each point is the mean over the facility-years of one state in one age band, with facility sizes pooled, and each line is one state. The heavy dark line is Texas. A Texas facility is priced on the filed rate schedule of the carrier it held that year, or at the priceable-market mean when no schedule can be reconstructed for its carrier (Section A.6). The other lines are control states, where the premium is the state fund's per-tank fee, averaged over the sample years, times the facility's tank count. The fee does not vary with tank age, and a line at zero is a state with a per-tank fee of zero in the data. The cells shown hold 1,853,068 facility-years with six or fewer tanks at the start of the year, in Texas from 2006 to 2020 and in the control states from 1999 to 2020. Kansas and Maryland and the Texas facility-years that pay no premium to a carrier are excluded. Cells with fewer than thirty facility-years are suppressed. Sources: insurer rate filings for Texas and state fund fee schedules for the control states.

Figure 36: Expected out-of-pocket cost by fleet age, by state



Notes: Mean expected out-of-pocket cost per facility-year, in nominal dollars, by the facility's average tank age in five-year bands. Each point is the mean over the facility-years of one state in one age band, with facility sizes pooled, and each line is one state, labeled at its right end. The cost is the facility's annual leak hazard times its state's deductible, with the hazard evaluated at the facility's average age band and majority wall type. A line's level scales with the deductible, and its slope across age bands comes from the hazard. The heavy teal line is Texas and the orange lines are control states with a state fund. The Texas deductible is the reference deductible in the carriers' filings and not a measured one. Dotted lines at zero are states with no positive deductible on record. The cells shown hold 2,099,962 facility-years with six or fewer tanks at the start of the year, in Texas and the control states from 1999 to 2020. Kansas and Maryland are excluded. Cells with fewer than thirty facility-years are suppressed. Sources: insurer rate filings for the Texas deductible and state fund statutes and rules for the control-state deductibles.

D.5 Welfare accounting

Every term of the welfare identity of Section 6 is a present value per facility in the observed stock. The subscript g names a facility's pricing environment, a fund state or a Texas carrier-year, and the weight $\mu_g(s)$ is the share of the observed facility-years that sit in environment g at state s. The weights sum to one over environments and states. A counterfactual changes the contract an environment carries and leaves the number of facilities that begin in each state unchanged. It changes what they do afterward and how many facilities open.

Each term follows a flow along a facility's own path. For a flow $f(a, s)$ paid when a facility in state s takes action a, let $\bar{f}_g(s) = \sum_a P_g(a \mid s)\, f(a, s)$ be its expected value at state s and M_g the transition matrix the choice probabilities induce, with exit absorbing. The present

value of the flow over the observed stock is

$$\mathrm{PV}_g(f) = \sum_s \mu_g(s)\,[(I - \beta M_g)^{-1}\,\bar{f}_g](s).$$

A facility that exits stops paying premiums and stops leaking, and the present value counts neither after it leaves. The expected discounted number of leaks is $\mathrm{Leaks}_g = \mathrm{PV}_g(h)$, with $h_a(s)$ the post-action hazard.

Producer surplus is the firm's value in money,

$$PS = \frac{1}{\gamma} \sum_g \sum_s \mu_g(s)\, V_g(s).$$

Because revenue is exogenous, PS moves across counterfactuals only through how long facilities operate, the fleets they hold, and the costs they bear. A policy raises PS by letting facilities keep their revenue streams at lower cost, and the accounting is silent on demand-side responses. The insurer's position is premium revenue net of claims,

$$\mathrm{Insurer} = \sum_g \left[\mathrm{PV}_g(\mathrm{prem}) - \mathrm{PV}_g(h\, S_{aw})\right].$$

The severity S_{aw} is the fitted mean cleanup cost of a leak, estimated from the claims records of the states that file them and transferred to every facility, Texas included, as the same expectation for the same tank cell. It is capped at the one-million-dollar statutory limit and evaluated at the post-action cell, and a leak at a single-walled fleet therefore books a larger payout than one at a double-walled fleet of the same age. The firm bears the tail beyond the cap,

$$\mathrm{FirmAboveCap} = \bar{S}_{>L} \sum_g \mathrm{Leaks}_g, \qquad \bar{S}_{>L} = \mathrm{E}\big[(S - \$1{,}000{,}000)^+\big],$$

and government outlay is nonzero only in the subsidy experiment,

$$\mathrm{Gov} = \frac{1}{\gamma} \sum_g \mathrm{PV}_g(m\, s_{\mathrm{sub}}\, c^{\mathrm{inst}}).$$

The last incumbent term is the damage a leak does beyond the cleanup, External $= E_{\mathrm{ext}} \sum_g \mathrm{Leaks}_g$, charged at the infant-health value of \$25,600 per leak on every contract, the baseline included, and each comparison therefore charges the same damage. The first-best premium prices that term. The other contracts do not, and every leak their facilities retain still imposes it. Under exact internalization the incumbent ledger collapses: premium revenue equals claims plus the above-cap loss plus the external damage, the insurer position equals the last two, and the incumbents' net welfare equals producer surplus less government outlay. That identity holds to machine precision in the first-best run and is a test of the implementation rather than a feature of the results.

The entrant block applies the same flows to the facilities that open. Let D_g be the discounted number of entrants in environment g, the sum over future years of the entry probability times

the pool remaining in that year, and let $\omega_g(s)$ be the distribution of states at which entrants open. An entrant flow is $D_g \sum_s \omega_g(s)\,[(I-\beta M_g)^{-1}\bar{f}_g](s)$, and the insurer, above-cap, and external terms of the entrant block are built from it exactly as the incumbent terms are. Entrant producer surplus is the value of a new facility and the entry cost is what entrants pay to open,

$$PS^E = \frac{1}{\gamma}\sum_g D_g\,\bar{V}_g, \qquad \text{EntryCost} = \frac{1}{\gamma}\sum_g D_g\,[\,ec + (1-s_{\text{sub}})\,c^{\text{inst}}\,\bar{n}_g\,].$$

Under the subsidy the government also pays $s_{\text{sub}}\,c^{\text{inst}}\,\bar{n}_g$ on each entrant's tanks. Section D.6 gives the closed form for D_g.

The identity rests on two conventions. The money metric divides exactly the payoff-unit terms, producer surplus, the entry cost, and government outlay, by γ and rescales nothing else, because premiums, claims, and the above-cap loss are already dollars. Utility is linear in the coverage-cost dollar, and compensating and equivalent variation coincide. Premiums are a transfer. The firm pays them inside V, the insurer receives them as premium revenue, and only real resource flows remain in W.

Two caveats bound the severity input. Its level is a fitted mean transferred from other states' claims, and because the same S_{aw} enters the baseline and every counterfactual, level error cancels from the parts of each comparison where behavior does not move and scales the parts where it does. Severity never enters a facility's payoff, only the accounting. The second caveat concerns the youngest cells. Leaks in a tank's first decade are mostly installation failures rather than the corrosion the rest of the age gradient measures, and the fitted severity in those cells is correspondingly high. The model books that severity on new tanks. A replacement therefore looks more expensive to insure against than the aging risk alone implies, and the contracts that induce replacement are scored conservatively.

D.6 Entry: the pool, the entry cost, and welfare by pool size

The entry step of Section 5 rests on a pool of potential entrants that is set by rule rather than observed. This section gives the rule, reports the entry cost and how well it fits, and shows how the welfare ranking of Section 6 moves with the size of the pool.

The risk-rated contract ranks first at every pool size. The entry cost rises with the pool, from 12.76 at the baseline pool to 19.63 with no limit on the pool, and the order of the three contracts below the risk-rated one changes with it. One entry cost reproduces the total number of entrants and does not reproduce their distribution across states.

The pool. The number of facilities a state can support in a year is λ times its population. The rate λ is the largest ratio of facilities to residents in the sample, counting the facilities operating at the start of a year and the facilities that open during it. That maximum is one facility for every 549 residents, in Virginia in 1999. The search runs over the control states from 1999 through 2017 and Texas from 2006 through 2017, leaving out Maine and South Dakota, whose registry counts are not reliable enough to define a maximum. The pool of potential entrants in a state and year is the cap less the facilities operating at the start of

the year. South Dakota's recorded stock exceeds its cap in every year. It is excluded from the entry-cost fit, and in every contract its facilities open at the rate observed in the data.

Entry and the stock of operating facilities determine each other over time, because each year's pool is the cap less a stock that past entrants have added to. With entry probability p_g, cap N_g, and discount factor β, the discounted number of entrants in environment g solves that recursion in closed form,

$$D_g = \frac{p_g \beta \left(N_g/(1-\beta) - \hat{s}_g \right)}{1 + p_g \beta \hat{S}_g},$$

where $\hat{s}_g = \sum_s \mu_g(s) \left[(I - \beta M_g)^{-1} \mathbf{1} \right](s)$ is the discounted number of years the incumbents remain open and $\hat{S}_g = \sum_s \omega_g(s) \left[(I - \beta M_g)^{-1} \mathbf{1} \right](s)$ is the discounted number of years an entrant remains open. The first cohort of entrants opens one year after the observed stock is measured.

The entry cost. Table 22 reports the entry cost at the baseline pool and at pools two and five times as large, and at the limit in which the pool never binds, where the binomial likelihood becomes a Poisson one. A larger pool lowers the entry probability that matches a given count of entrants and therefore raises the estimated cost. The standard errors are clustered by state and take the value of a new facility as given.

Table 22: Entry cost by size of the pool of potential entrants

Pool	Entry cost	Standard error	Dollar equivalent
Baseline pool	12.762	(0.884)	\$231,875
Twice the baseline	13.775	(0.860)	\$250,274
Five times the baseline	14.843	(0.852)	\$269,681
No limit on the pool	19.631	(0.847)	\$356,665

Notes: Each row is a separate maximum-likelihood estimate of the entry cost on 259 state-years in 14 states, in which 12,554 facilities open, with standard errors clustered by state in parentheses. Entrants in a state and year are binomial draws from the pool of potential entrants, with a logit entry probability in the value of a new facility net of the install cost of its tanks and of the entry cost. The last row is the Poisson limit of that likelihood. The entry cost is in model units of 10,000 dollars, and its dollar equivalent divides the estimate by γ, the weight on a dollar of coverage cost, and multiplies by 10,000. The entry cost excludes the install cost of the entrant's tanks. The standard errors take the value of a new facility, the install cost, and the pool as given.

The fitted model matches the 12,554 entrants in total. It does not match where they open. It places 12,502 of them in Texas, where 2,675 facilities opened, and 52 in the control states, where 9,879 opened. The value of a new facility is higher in Texas than in any control state, and with one entry cost and a unit shock scale that difference in value becomes a large difference in the entry probability.

Welfare by pool size. Table 23 splits the net welfare of each contract in Table 6 into its incumbent and entrant blocks and reports the total at each pool size. The incumbent block does not depend on the pool. The entrant block is negative under every contract, for the reason Section 6 gives, and it grows in size with the pool. The risk-rated contract ranks first in every column. The order of the other three contracts is not stable: with a pool five times the baseline the removal mandate ranks above the flat fee and the subsidy.

Table 23: Net welfare of incumbents and entrants, and totals by pool size

	Baseline pool, by block		Total net welfare, by pool size			
Scenario	Incumbents	Entrants	Baseline	Twice baseline	Five times baseline	No limit
First best	57,632	-763	56,868	56,677	56,548	55,927
Risk-rated pricing (baseline)	32,109	-8,128	23,981	18,989	14,269	689
Uniform flat fee	32,115	-10,643	21,472	12,643	3,170	-22,384
Replacement subsidy	31,507	-12,554	18,953	6,290	-9,897	-54,750
Age-20 removal mandate	27,760	-11,001	16,759	12,881	9,087	-3,660

Notes: Present value of net welfare in 2023 dollars per facility in the observed stock. Every row charges the same infant-health damage of 25,600 dollars per leak. Incumbents and Entrants are the two blocks of the welfare identity of Section 6 at the baseline pool of potential entrants. They sum to the Baseline column, the net welfare Table 6 reports. The remaining columns re-estimate the entry cost with a pool two and five times the baseline and with no limit on the pool, and re-solve entry under each contract. The incumbent block is the same in every column. At the larger pools the external damage of the entrant block is restated to the infant-health value in proportion to the number of entrants.

What the entry step cannot show. The pool is an assumption, and the entry cost and the size of the entrant block move with it. The standard errors do not carry the sampling error of the value of a new facility. The welfare ledger counts no benefit of a new facility to the drivers it serves and does not count the private shock of the facilities that enter, and the entrant block is therefore a lower bound on the value of entry. A model in which the entry cost or the value of a new facility varies by state would be needed to match where facilities open.